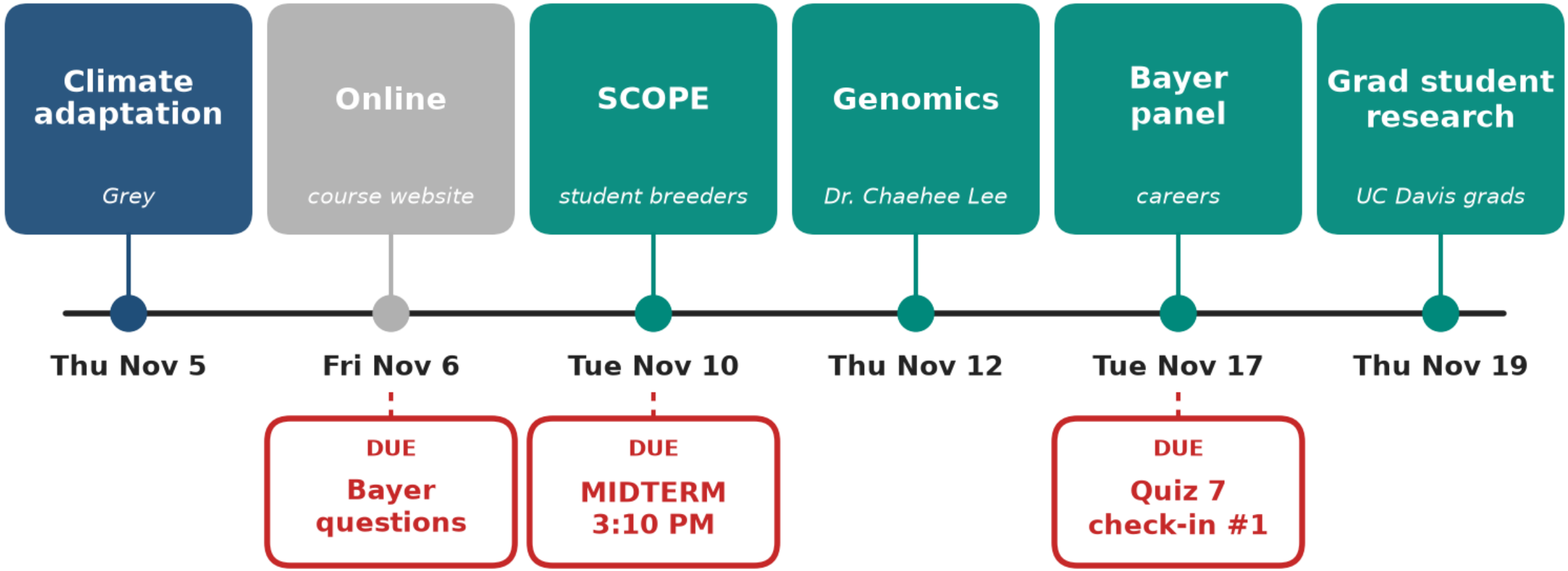
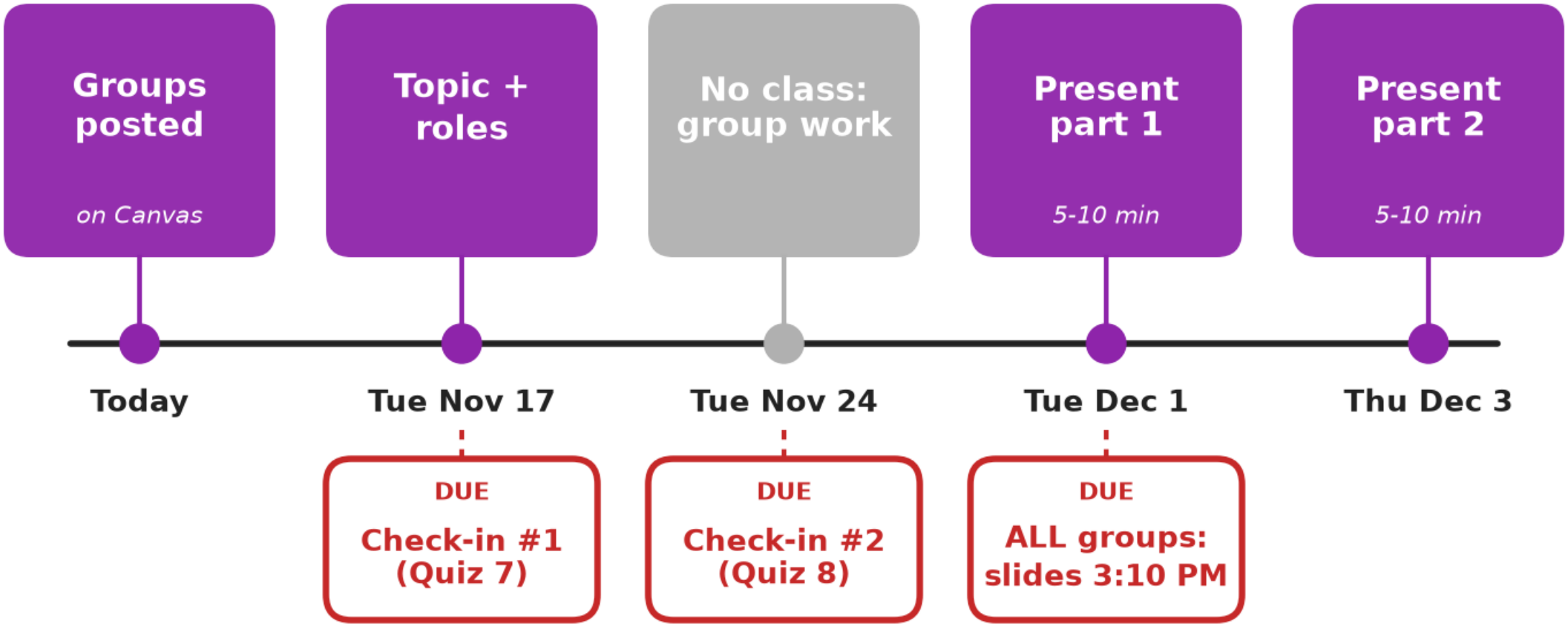
Finding the genes



Coming up: guests + the midterm



Final project: your group, your topic



Meet your group this week and pick a topic together. Thanksgiving Nov 26: no class.


BAYER
Crop Science

Bayer panel: Tue Nov 17

Send me your questions by Friday

1. Submit questions for the panel
2. Read the networking guide

Scan with your phone

greymonroe.github.io/PLS_152/bayer.html

Last Tuesday recap

Kahoot-style warm-up

12 quick questions
Not graded
Fast and correct wins.



Scan with your phone

[Join on your phone or laptop: greymonroe.github.io/live-poll-test/quiz-play.html?quiz=pls152-fg-quiz](greymonroe.github.io/live-poll-test/quiz-play.html?quiz=pls152-fg-quiz)

[host screen](greymonroe.github.io/live-poll-test/quiz-host.html?quiz=pls152-fg-quiz)

Last week

A peak is a region, not a proof

Today

Why the real gene often makes no peak at all

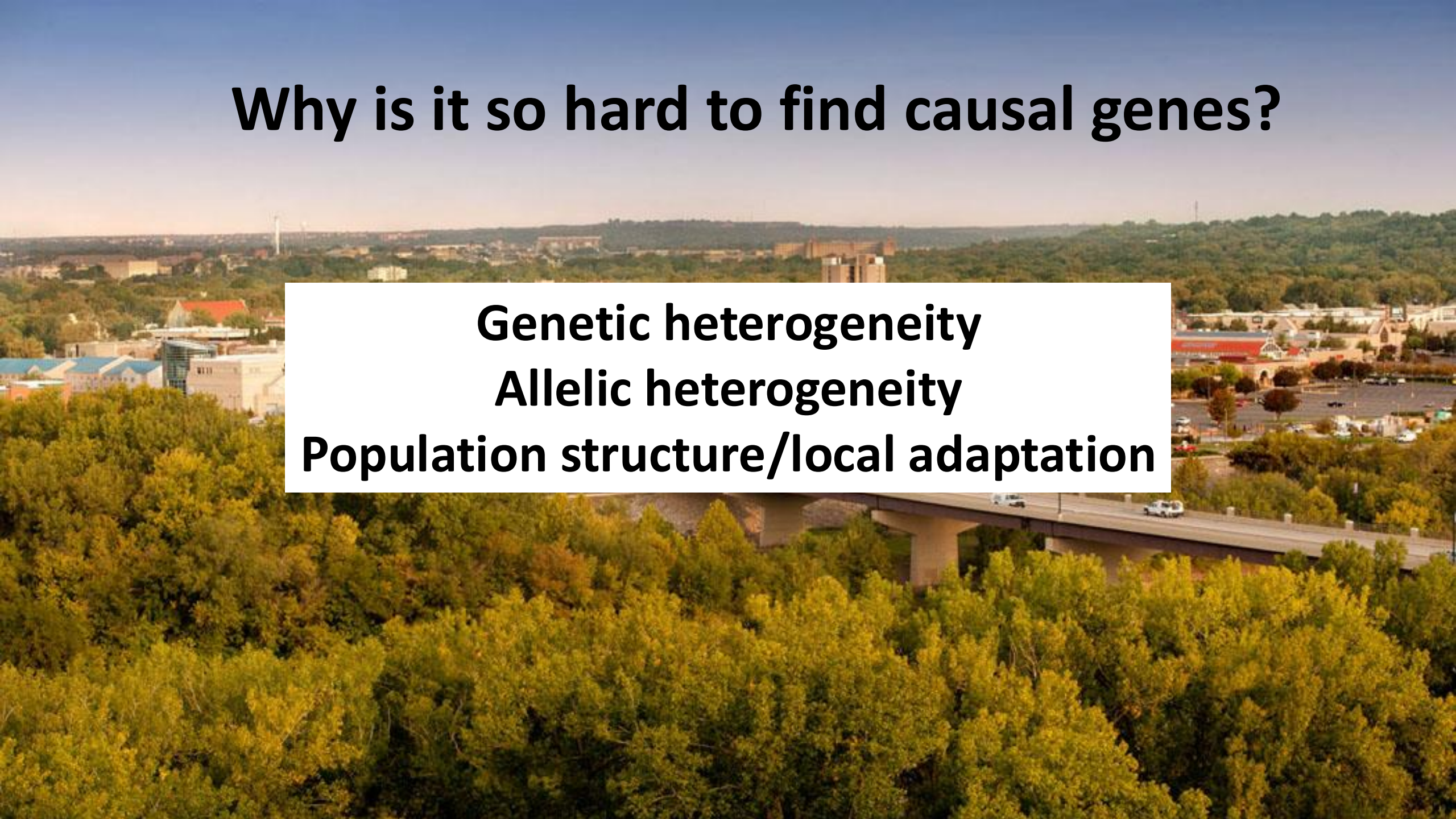
Missing heritability



The case of the missing heritability

When scientists opened up the human genome, they expected to find the genetic components of common traits and diseases. But they were nowhere to be seen. **Brendan Maher** shines a light on six places where the missing loot could be stashed away.

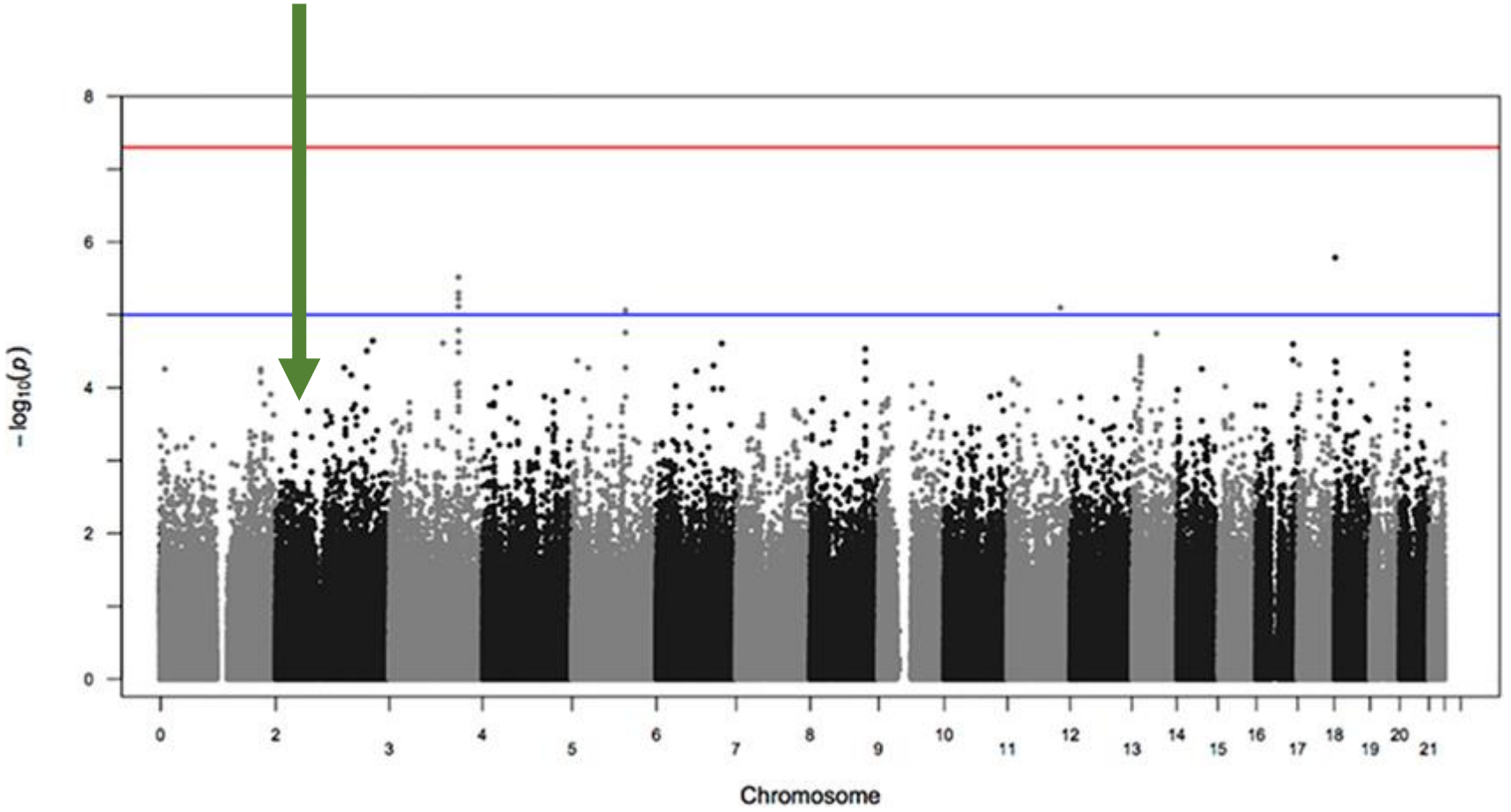
Why is it so hard to find causal genes?



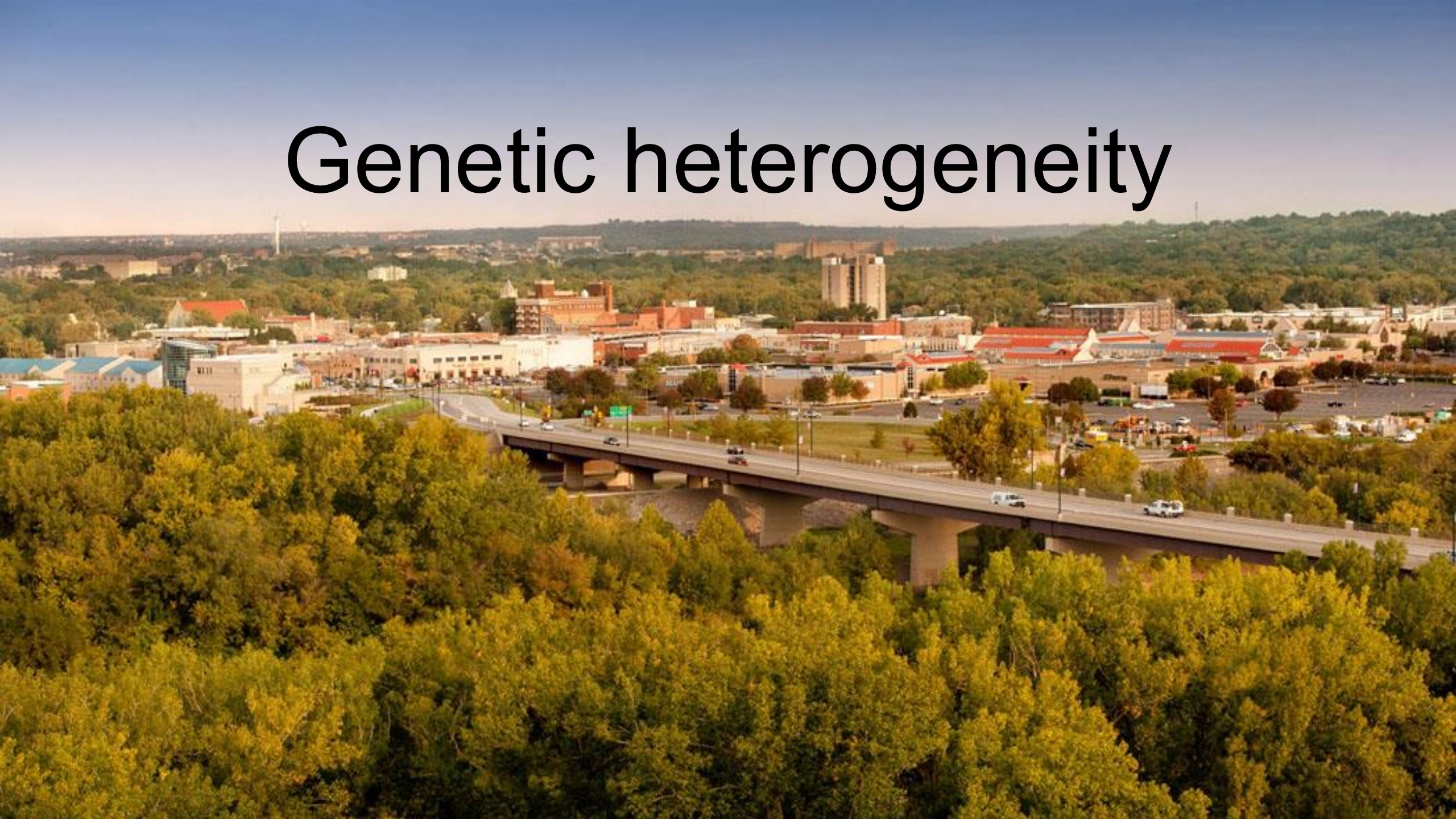
Genetic heterogeneity
Allelic heterogeneity
Population structure/local adaptation

“False negatives”

Example: Causal variant is here but doesn’t show a peak in GWAS. Why?



Genetic heterogeneity



If one gene controlled flowering time





early



late



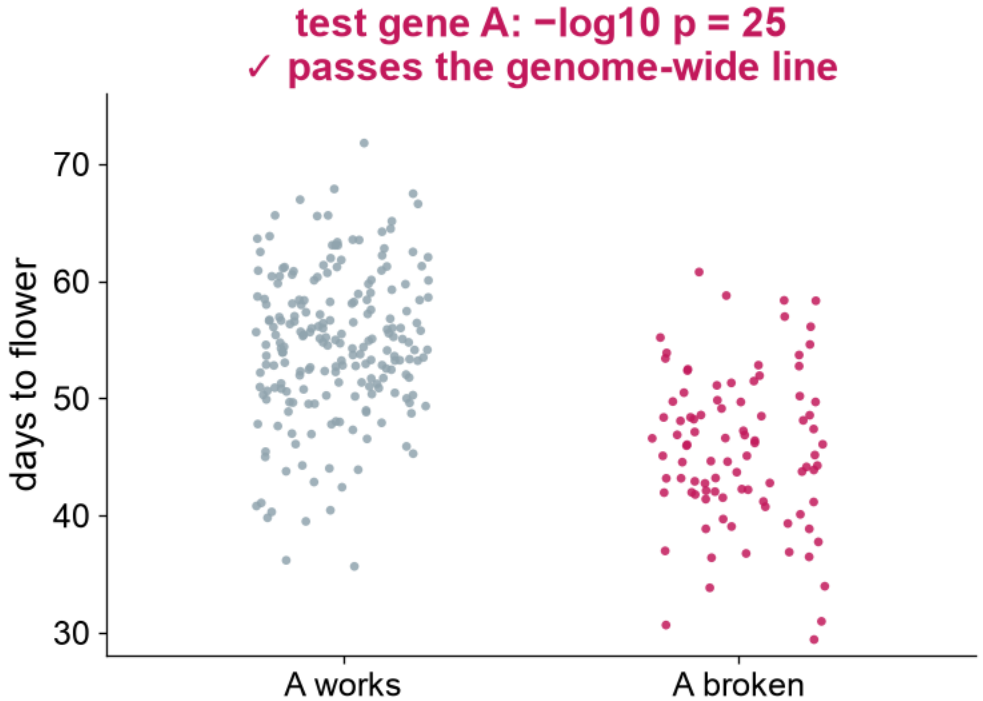
early = already flowering · late = still a rosette

If one gene controlled flowering time



early = already flowering · late = still a rosette

dot color = which gene is broken: **A** none (late)



Reality: different broken genes, same trait



early = already flowering · late = still a rosette

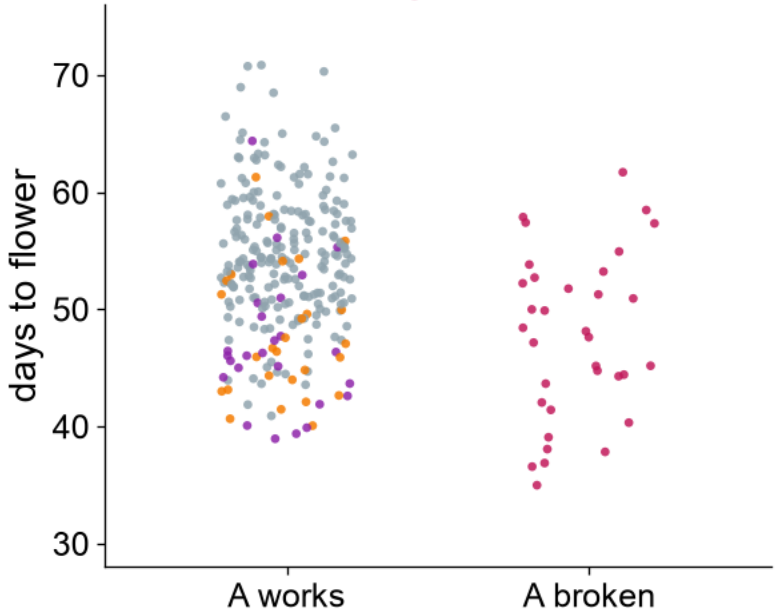
Reality: different broken genes, same trait



early = already flowering · late = still a rosette

dot color = which gene is broken: **A** **B** **C** none (late)

test gene A: −log10 p = 6
✗ misses the genome-wide line



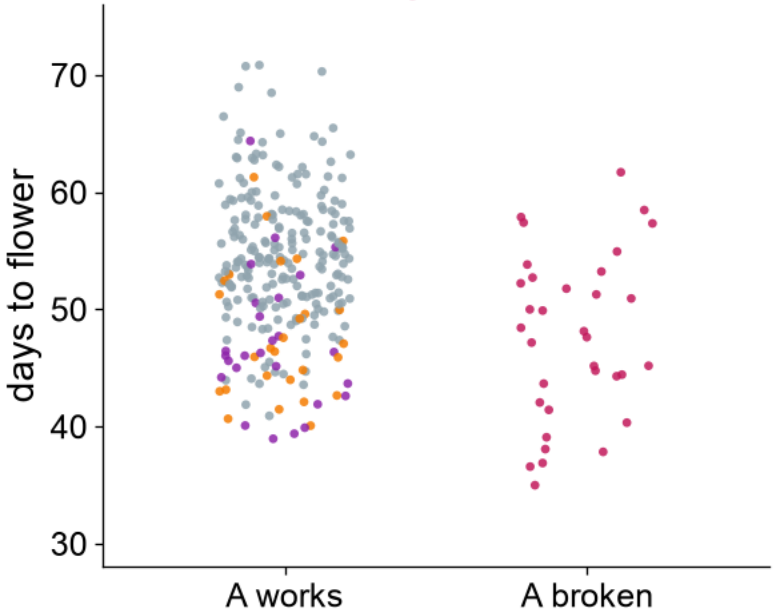
Reality: different broken genes, same trait



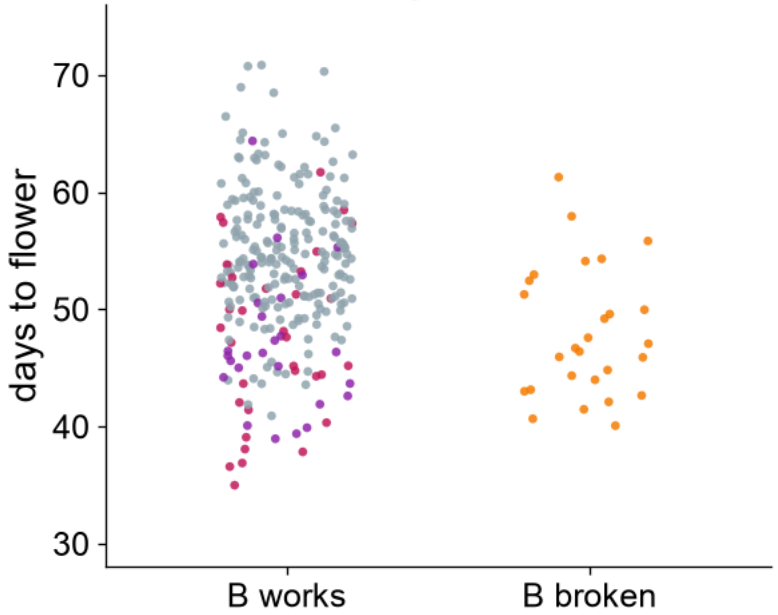
early = already flowering · late = still a rosette

dot color = which gene is broken: **A** **B** **C** none (late)

test gene A: −log10 p = 6
✗ misses the genome-wide line



test gene B: −log10 p = 4
✗ misses the genome-wide line

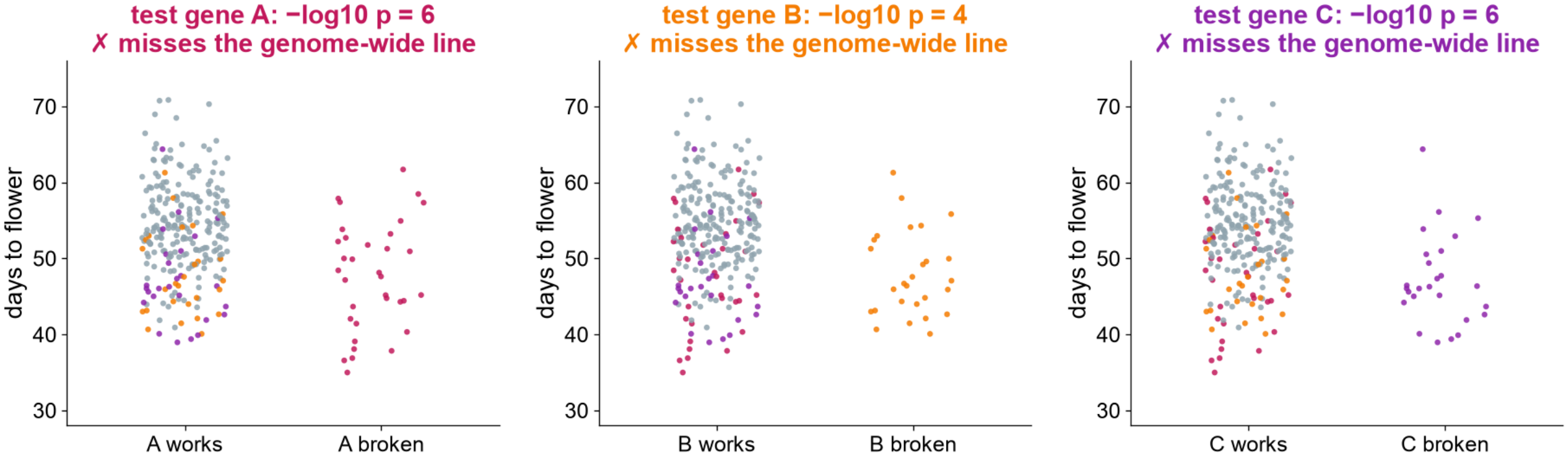


Reality: different broken genes, same trait

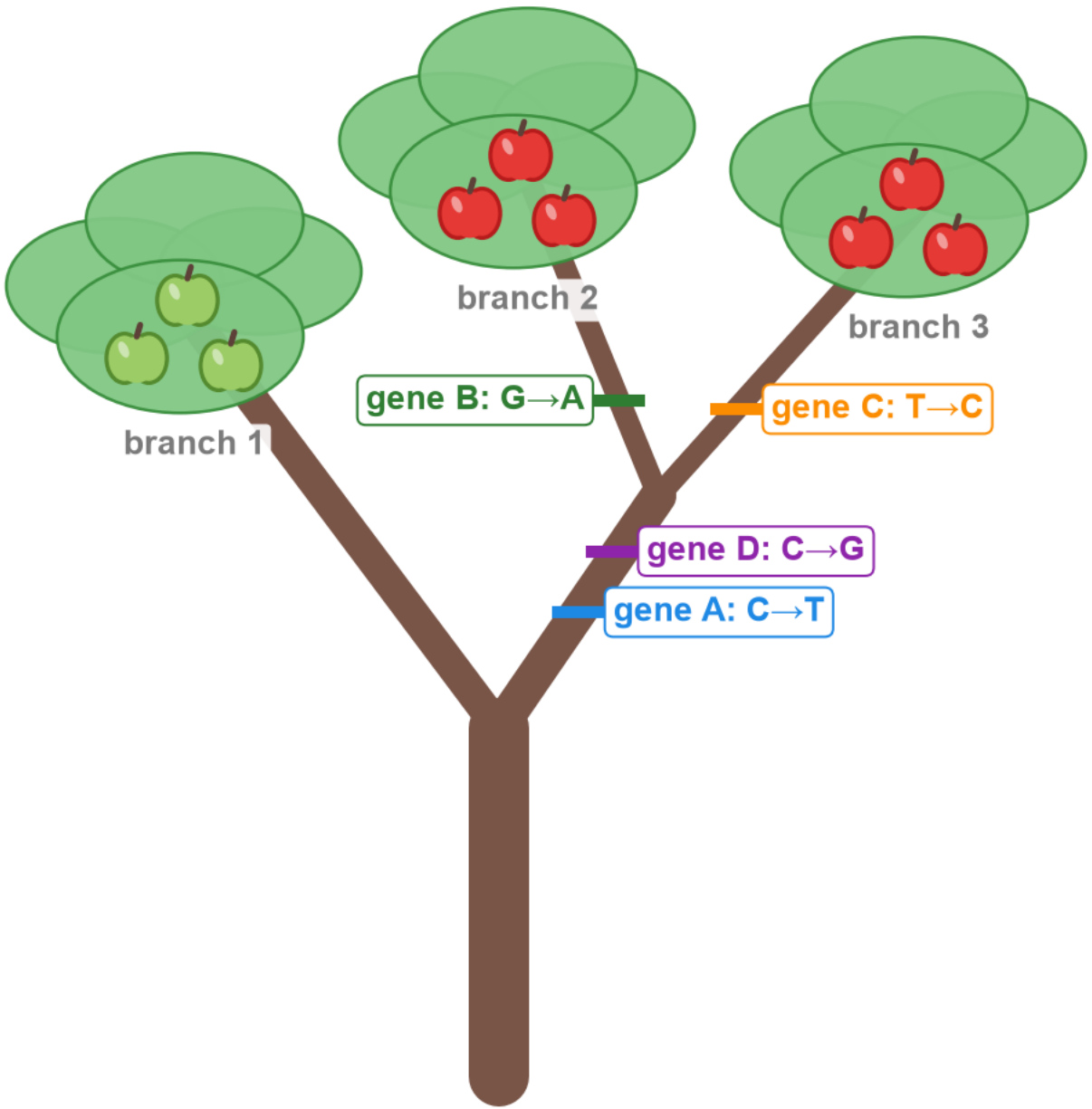


early = already flowering · late = still a rosette

dot color = which gene is broken: **A** **B** **C** none (late)



Which gene makes the apple red?



	gene A	gene B	gene C	gene D
branch 1	GAC**C**TGA	TTA**G**CAA	CAG**T**ACG	AGG**C**TTA
branch 2	GAC**T**TGA	TTA**A**CAA	CAG**T**ACG	AGG**G**TTA
branch 3	GAC**T**TGA	TTA**G**CAA	CAG**C**ACG	AGG**G**TTA

A change on a branch is inherited by every apple further out on that branch

after Korte & Farlow 2013 · toy sequences

Live poll

Multiple choice

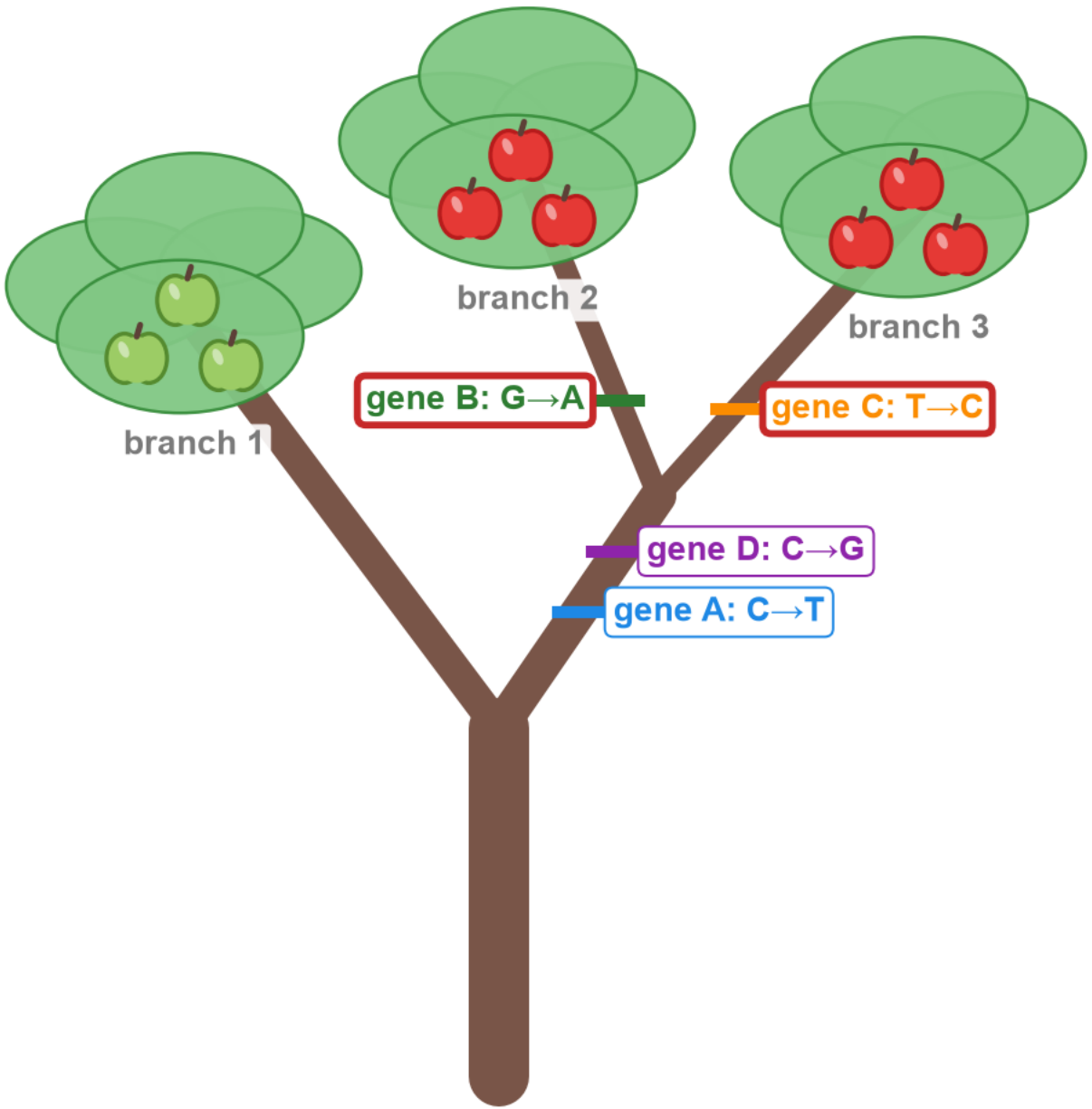
Two red apples, two different mutations (one in Gene B, one in Gene C), and both share Gene A. Which gene causes the red apple phenotype?

Scan with your phone

Join on your phone or laptop: greymonroe.github.io/live-poll-test/submit.html?poll=pls152-ftg-redapple

host screen

Reading 1: different genes, same red

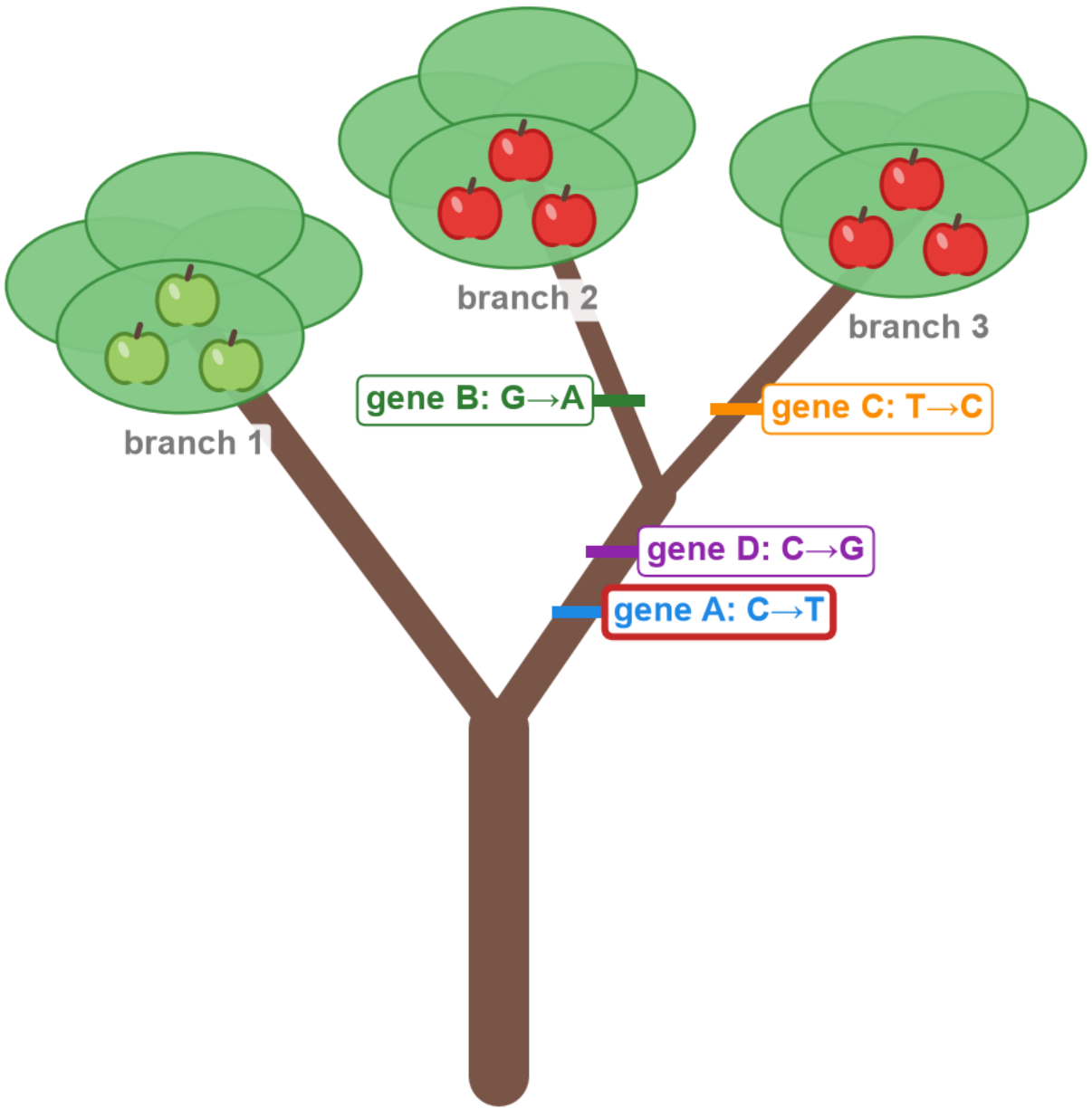


	gene A	gene B	gene C	gene D
branch 1	GAC**C**TGA	TTA**G**CAA	CAG**T**ACG	AGG**C**TTA
branch 2	GAC**T**TGA	TTA**A**CAA	CAG**T**ACG	AGG**G**TTA
branch 3	GAC**T**TGA	TTA**G**CAA	CAG**C**ACG	AGG**G**TTA

B explains branch 2, C explains branch 3: each change alone fits only half the red apples

after Korte & Farlow 2013 · toy sequences

Reading 2: one change shared by both reds

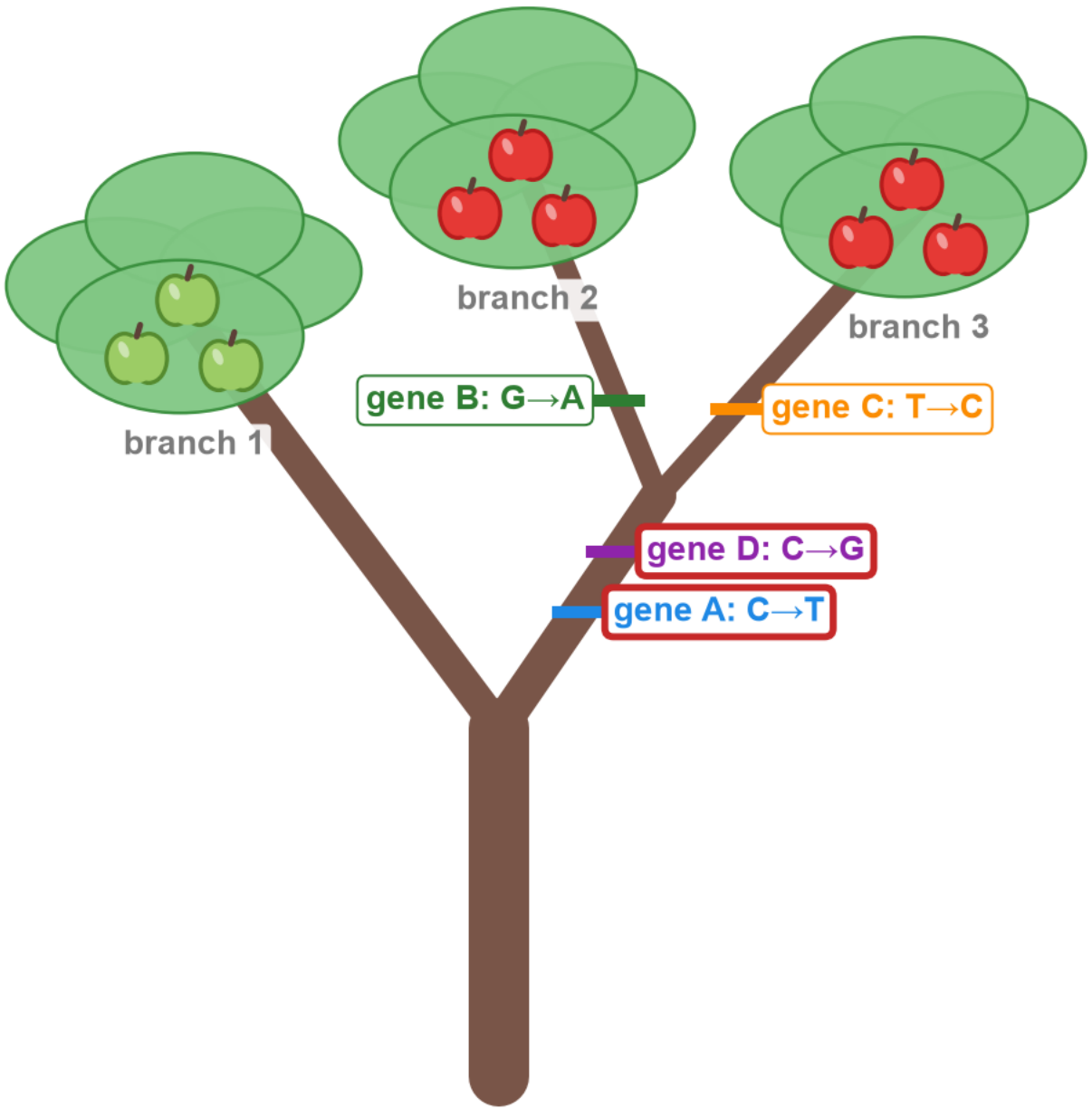


	gene A	gene B	gene C	gene D
branch 1	GAC**C**TGA	TTA**G**CAA	CAG**T**ACG	AGG**C**TTA
branch 2	GAC**T**TGA	TTA**A**CAA	CAG**T**ACG	AGG**G**TTA
branch 3	GAC**T**TGA	TTA**G**CAA	CAG**C**ACG	AGG**G**TTA

Gene A's T fits both red branches perfectly... because both grew from the branch where it happened

after Korte & Farlow 2013 · toy sequences

A or D? Shared ancestry = population structure

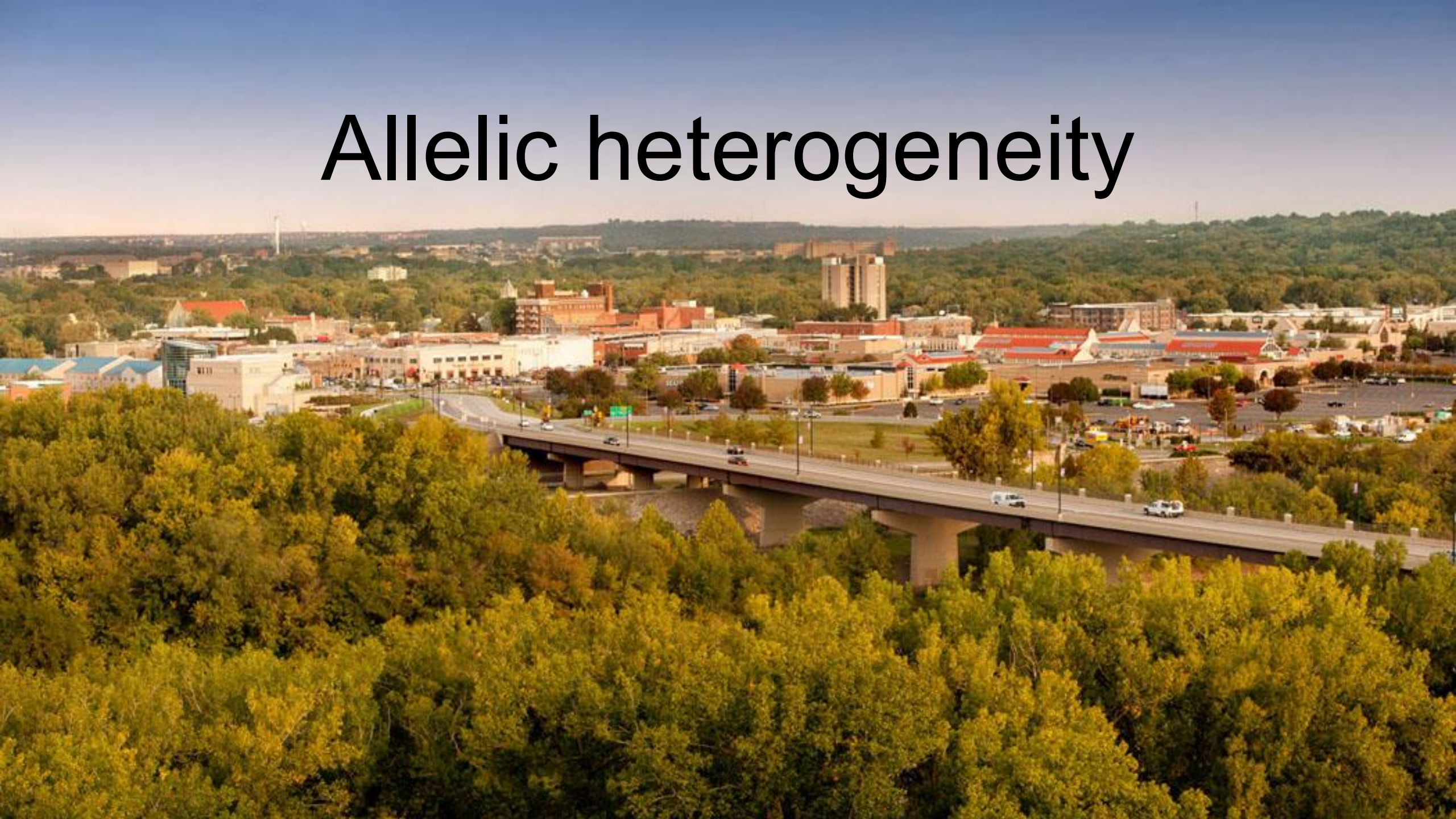


	gene A	gene B	gene C	gene D
branch 1	GACCTGA	TTAGCAA	CAGTACG	AGGCTTA
branch 2	GACTTGA	TTACAA	CAGTACG	AGGGTTA
branch 3	GACTTGA	TTAGCAA	CAGCACG	AGGGTTA

So does gene D: both changes happened on the shared branch. Shared history makes every passenger look causal: population structure

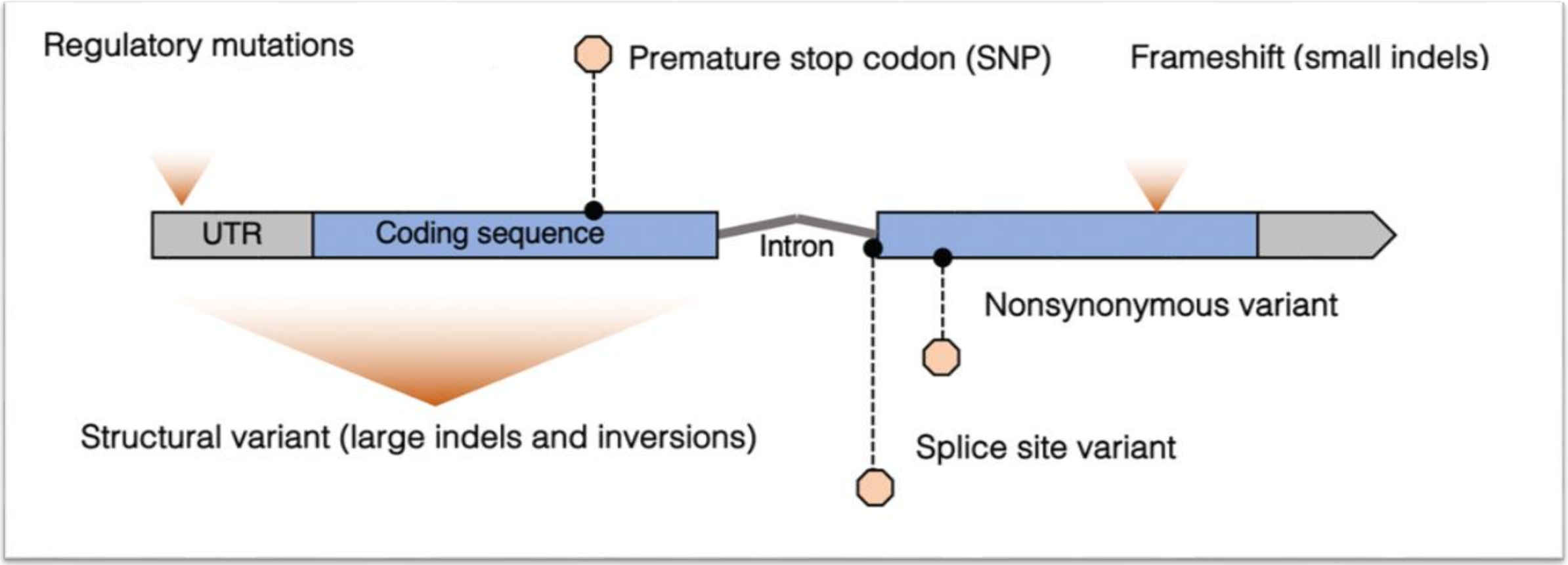
after Korte & Farlow 2013 · toy sequences

Allelic heterogeneity



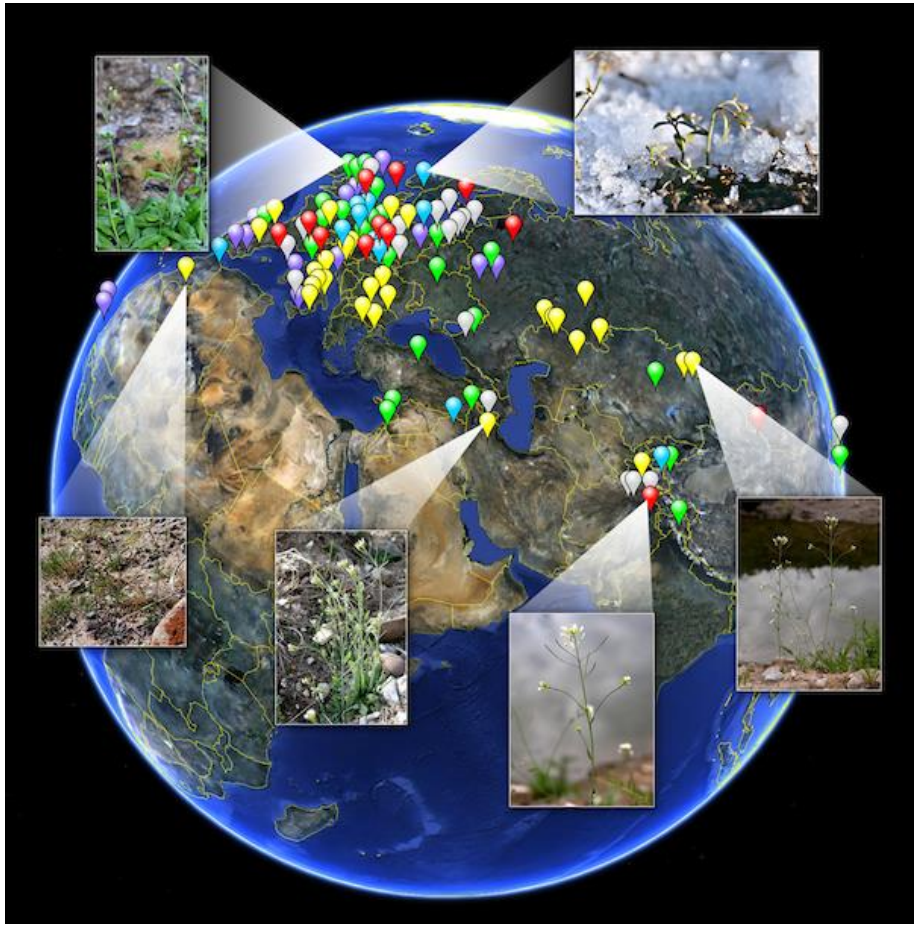
Loss of function (LoF): a common cause of allelic heterogeneity

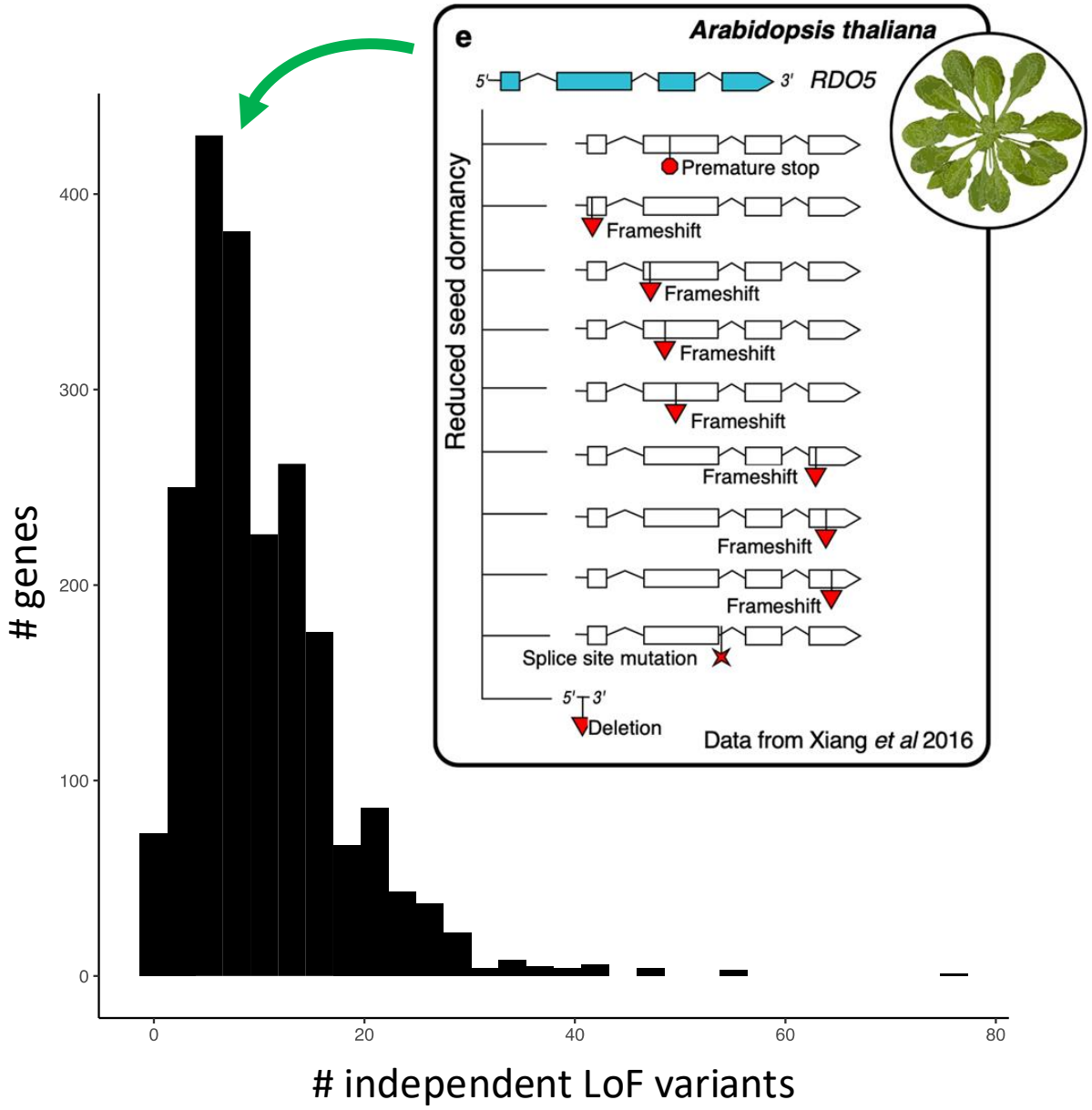
There are many ways to break a gene!



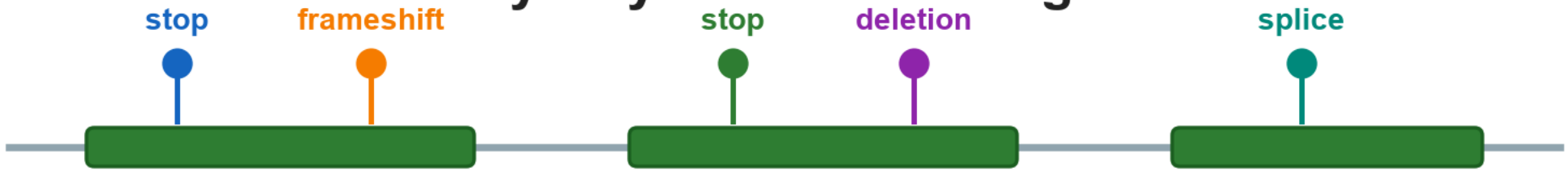
Allelic heterogeneity by LoF in *Arabidopsis*

1,135 genomes sequenced of *Arabidopsis thaliana*



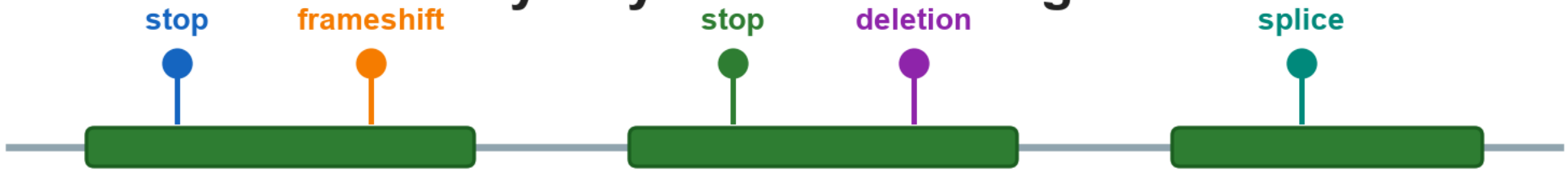


Many ways to break one gene

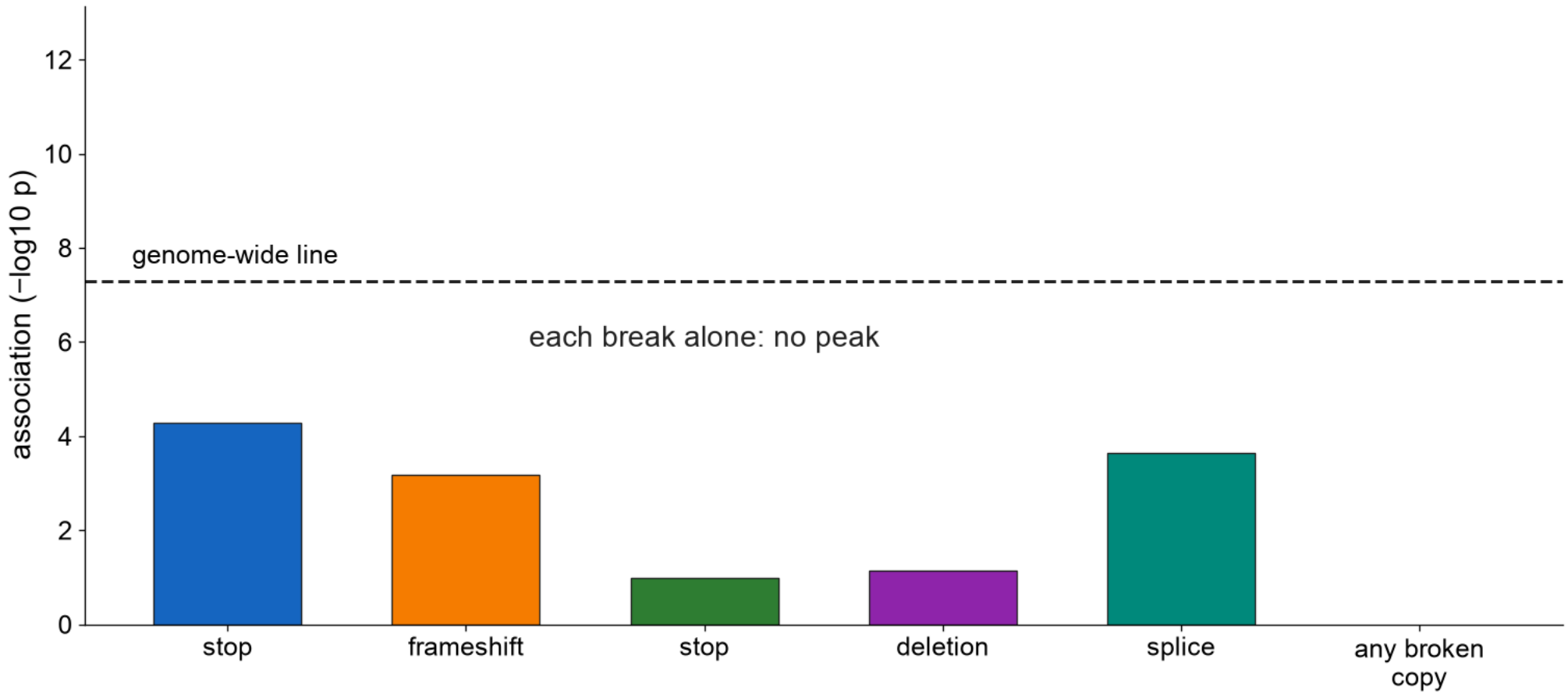


height gene (all five breaks → semi-dwarf) · each break found in only 2% of plants

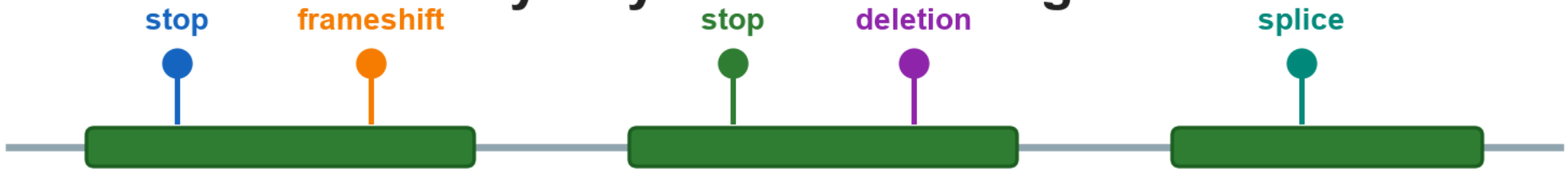
Many ways to break one gene



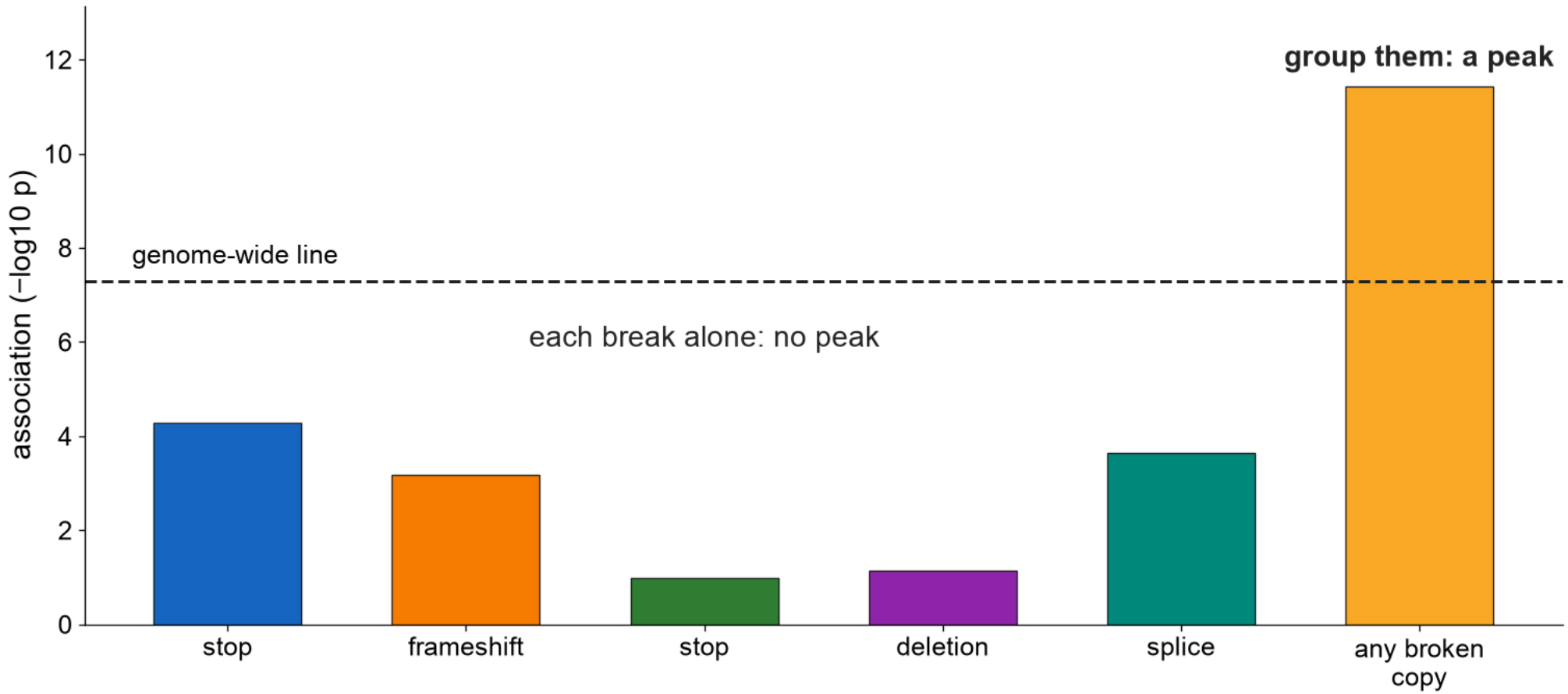
height gene (all five breaks → semi-dwarf) · each break found in only 2% of plants



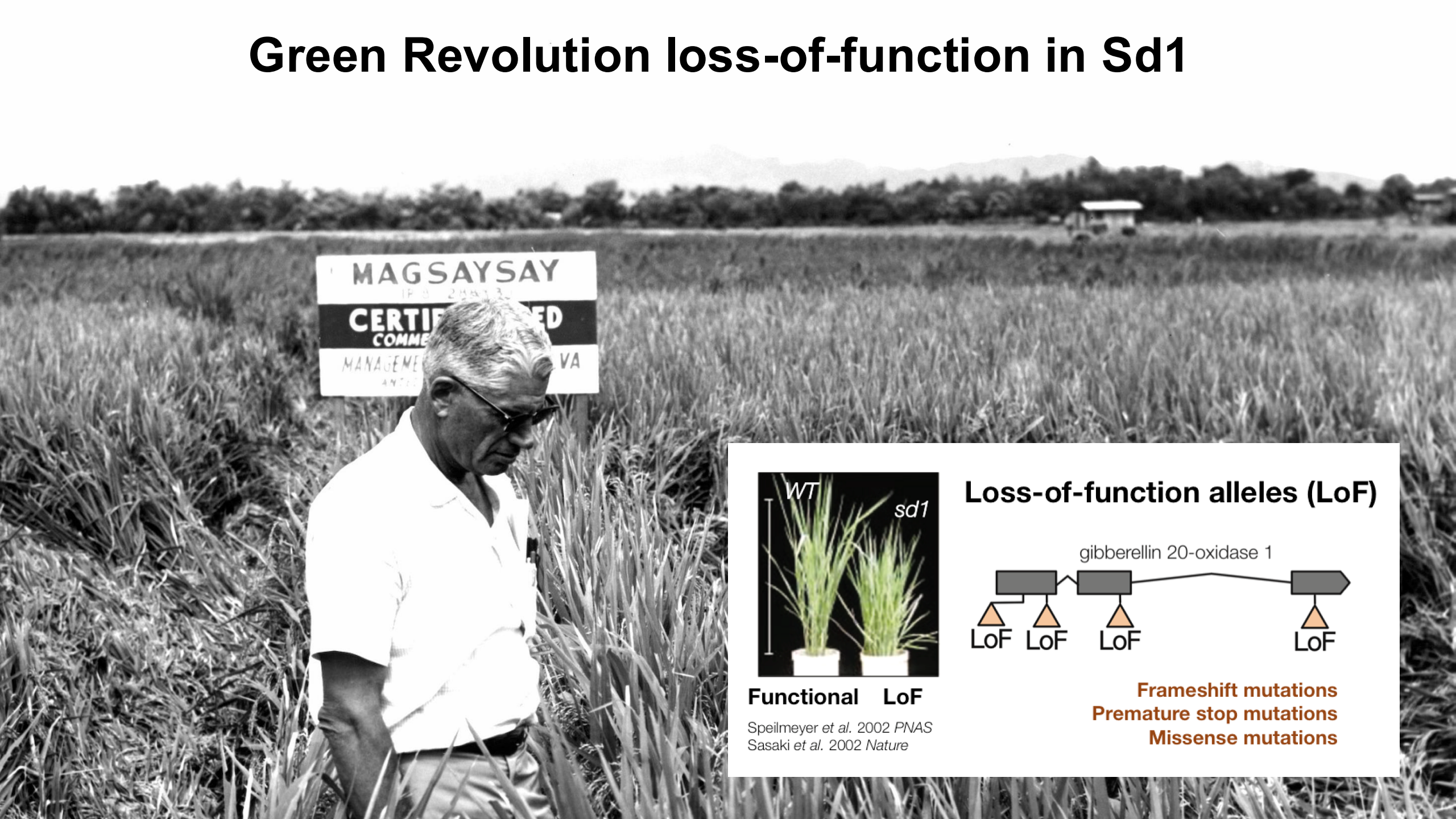
Many ways to break one gene



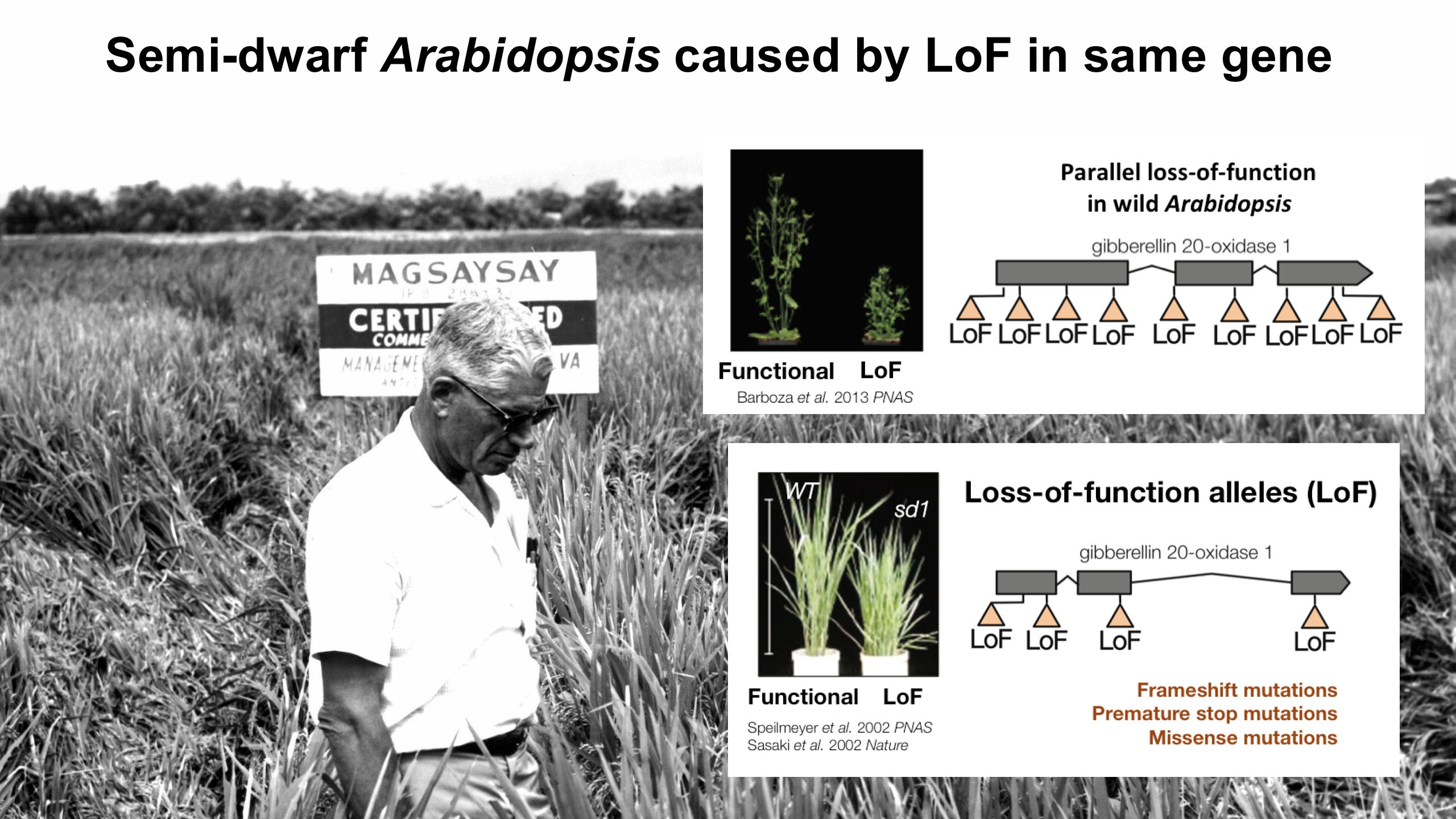
height gene (all five breaks → semi-dwarf) · each break found in only 2% of plants



Green Revolution loss-of-function in Sd1



Semi-dwarf *Arabidopsis* caused by LoF in same gene



Functional LoF

Barboza *et al.* 2013 *PNAS*

Parallel loss-of-function in wild *Arabidopsis*

gibberellin 20-oxidase 1

Functional LoF

Speilmeyer *et al.* 2002 *PNAS*
Sasaki *et al.* 2002 *Nature*

Loss-of-function alleles (LoF)

gibberellin 20-oxidase 1

Frameshift mutations
Premature stop mutations
Missense mutations

Live poll

Multiple choice

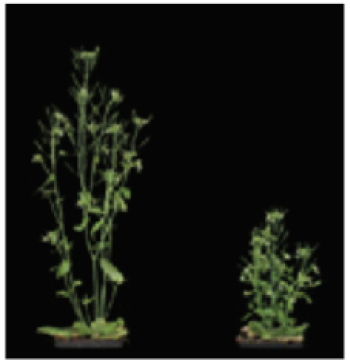
Wild Arabidopsis carries many DIFFERENT broken alleles of the same height gene (GA20-oxidase 1). Will GWAS find a peak there?

Scan with your phone

Join on your phone or laptop: greymonroe.github.io/live-poll-test/submit.html?poll=pls152-ftg-sd1

host screen

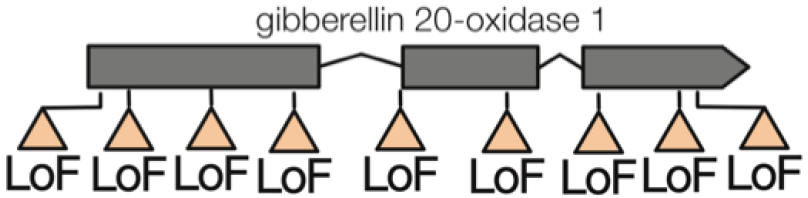
Gene is hidden from GWAS



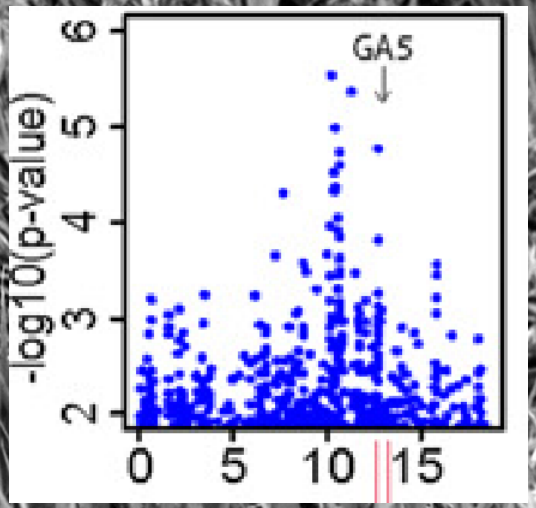
Functional LoF

Barboza *et al.* 2013 *PNAS*

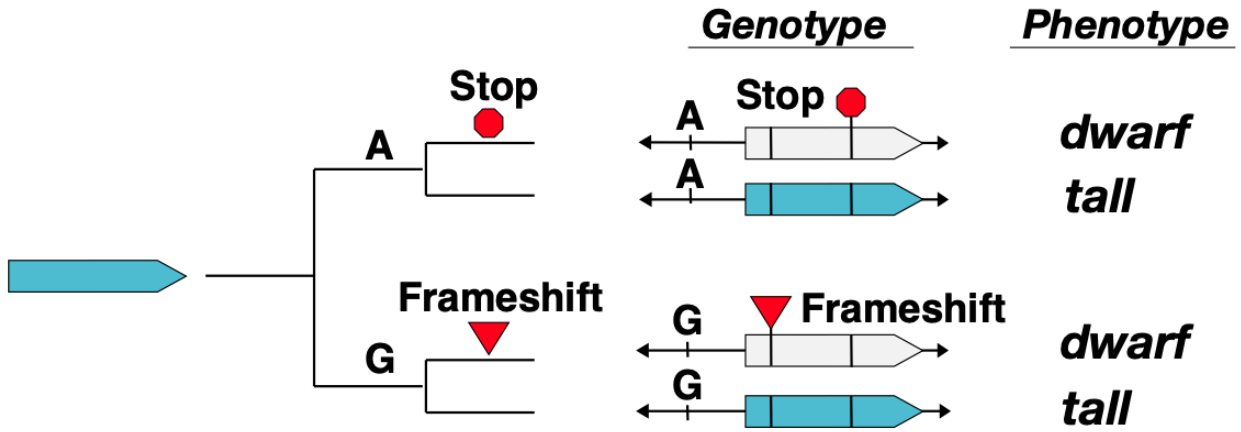
Parallel loss-of-function in wild *Arabidopsis*



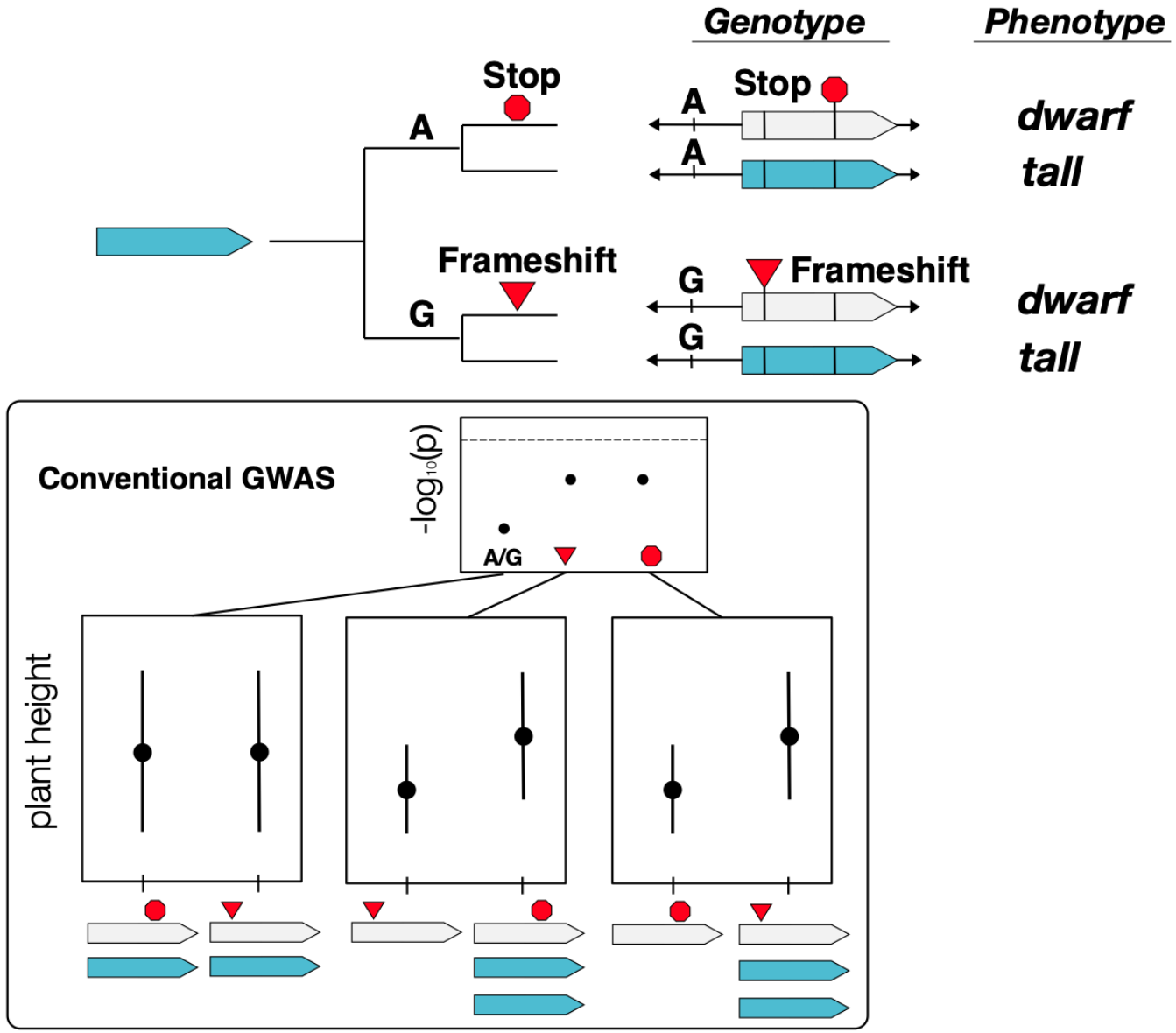
Not significant!



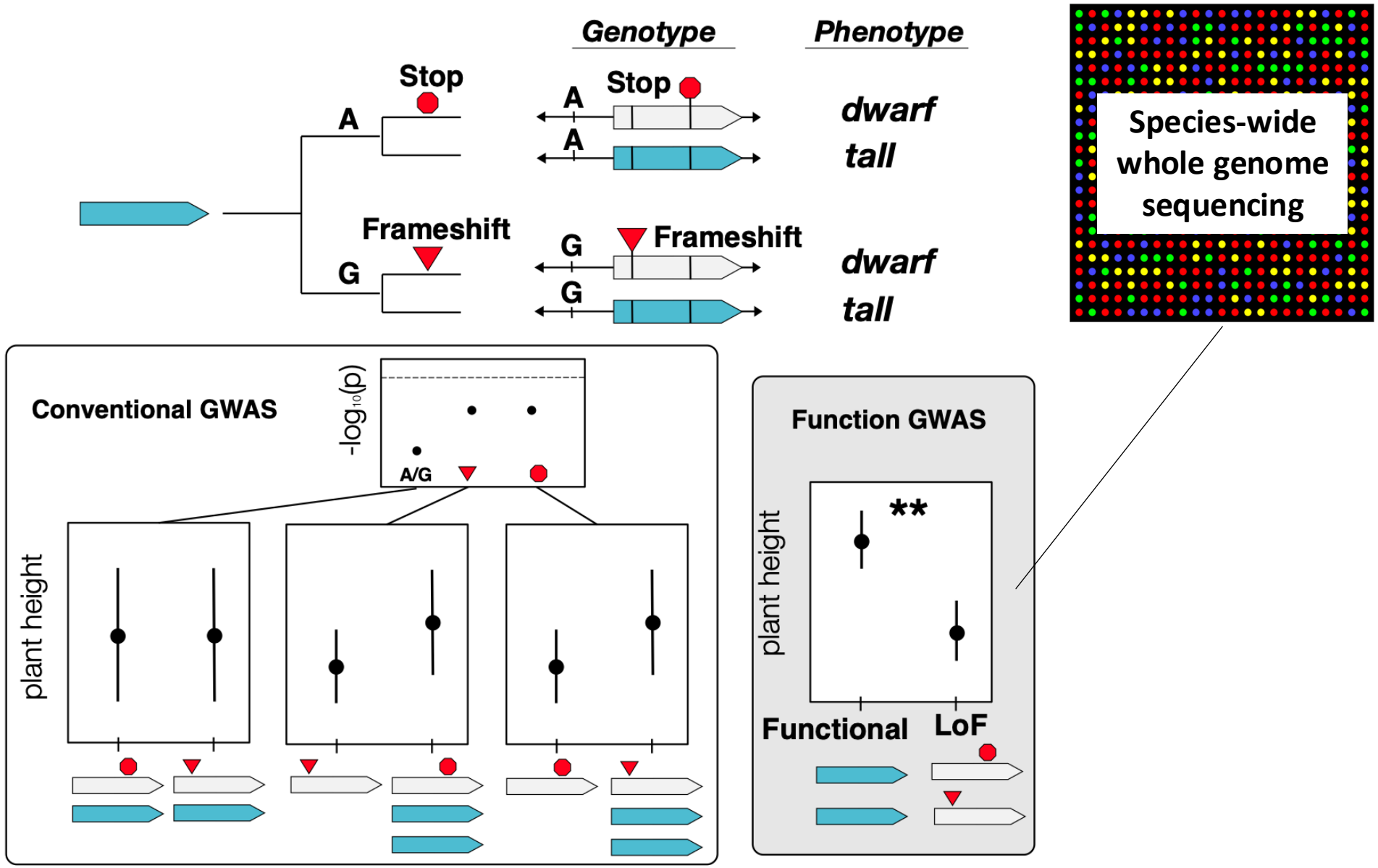
Using information about functional effects to overcome allelic heterogeneity



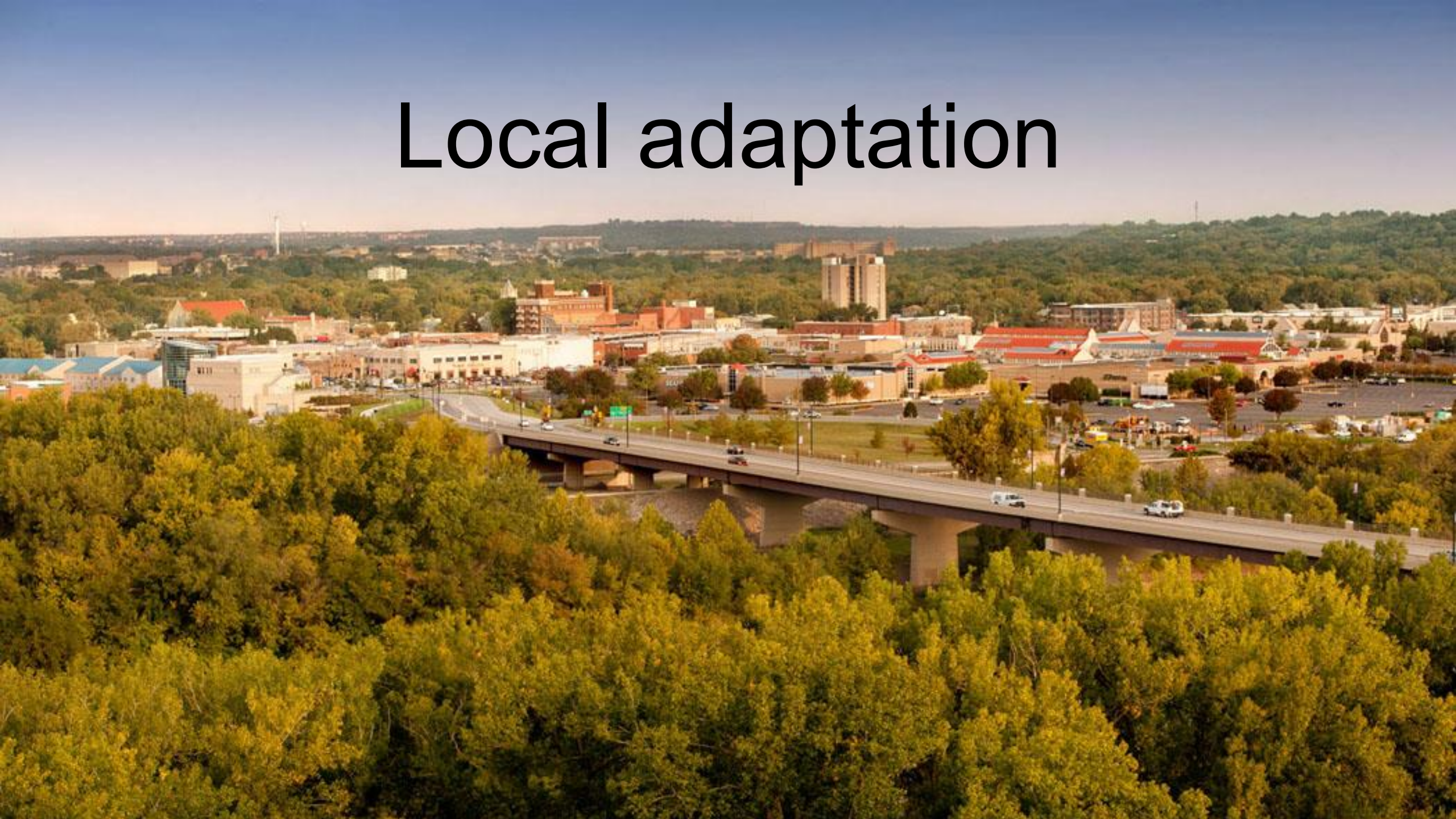
Using information about functional effects to overcome allelic heterogeneity



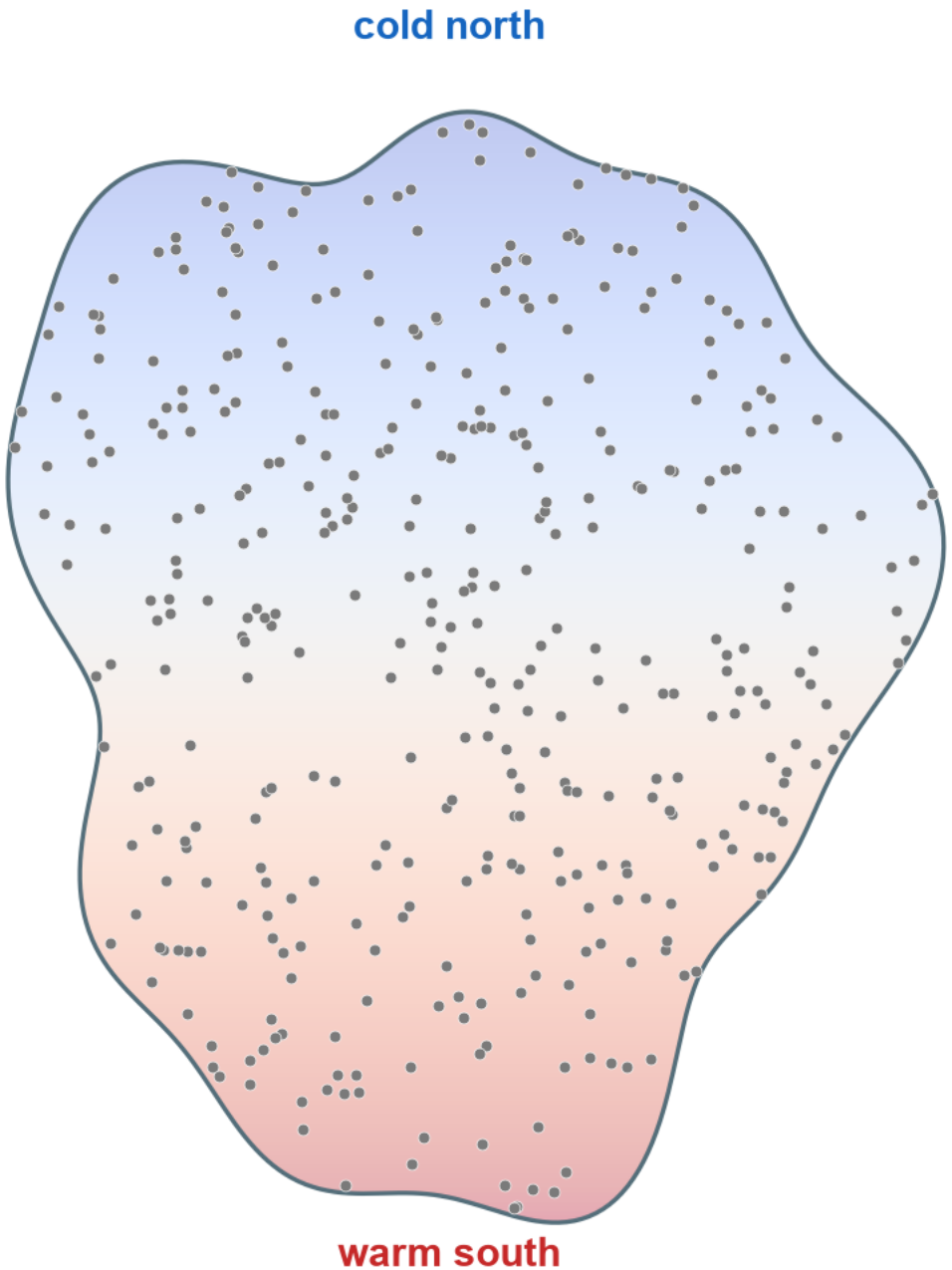
Using information about functional effects to overcome allelic heterogeneity



Local adaptation



Plants collected across a landscape

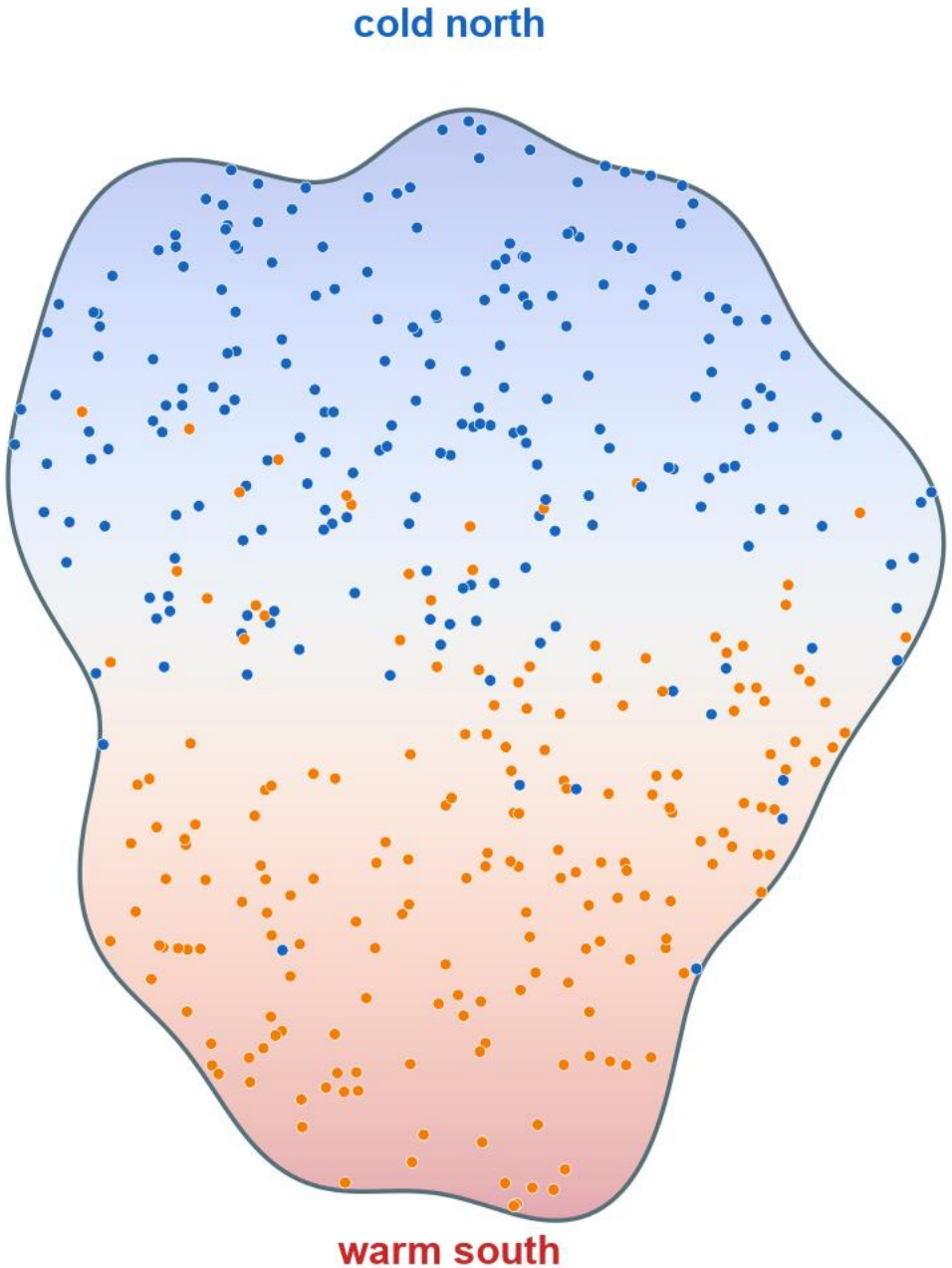


400 wild plants

1 dot = 1 plant, where it was collected

Seeds from all of them are grown together in ONE common garden.

Shared history: who is related to whom

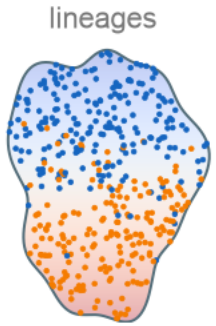


Two lineages spread from
the north and the south,
and met in the middle.

- **northern lineage**
- **southern lineage**

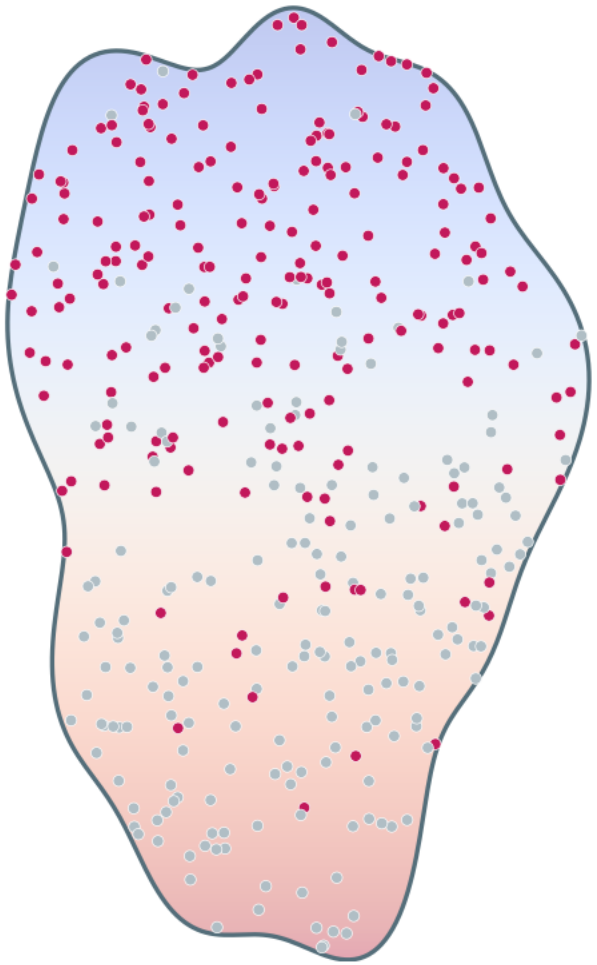
This is population structure.

A neutral SNP follows history



◆ neutral SNP

cold north

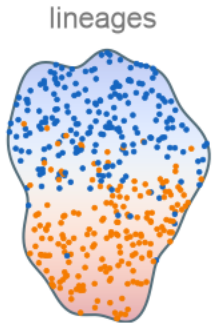


warm south

● T allele ● other allele

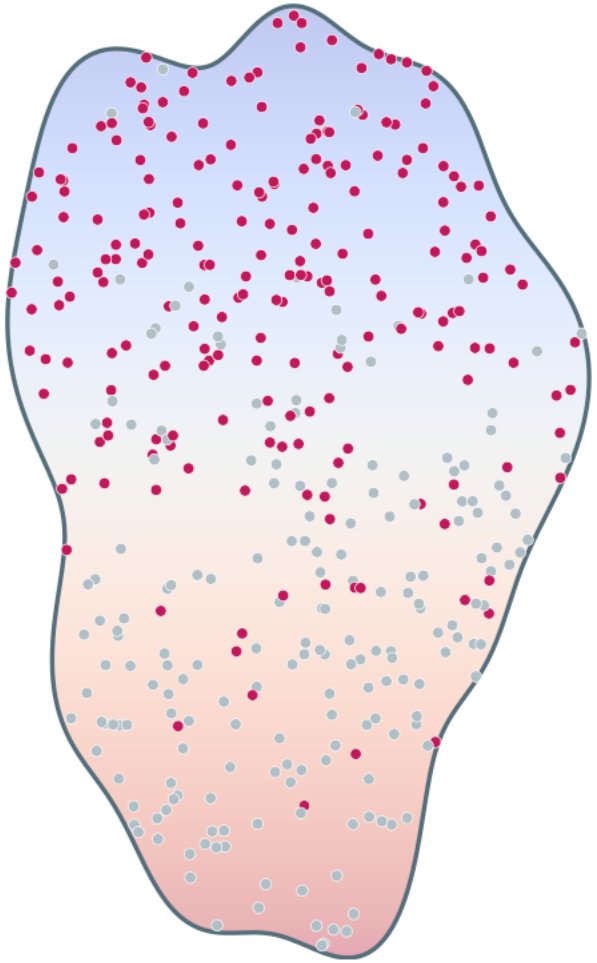
no effect on flowering

A flowering gene follows climate (local adaptation)



◆ neutral SNP

cold north



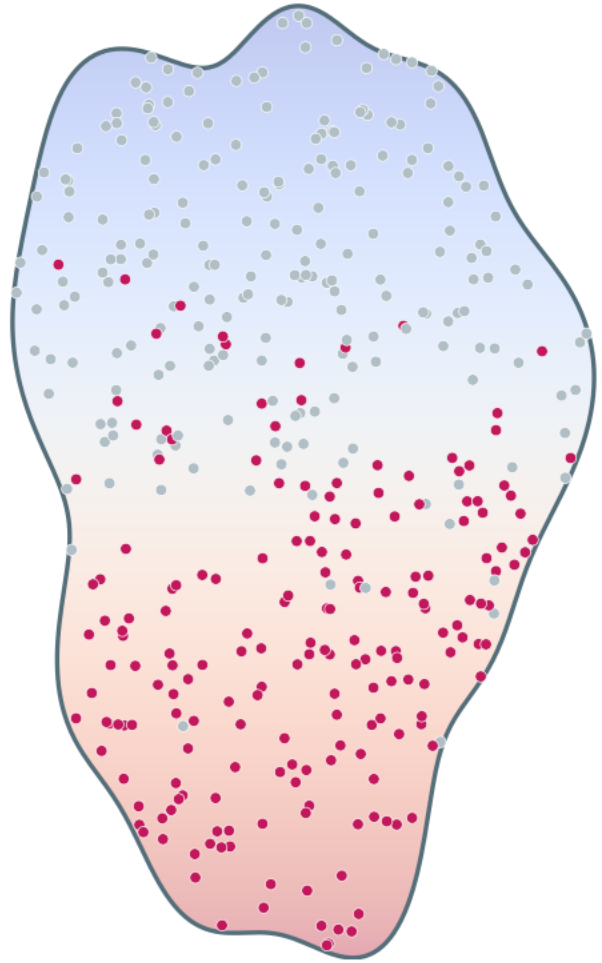
warm south

● T allele ● other allele

no effect on flowering

★ local-adaptation gene

cold north

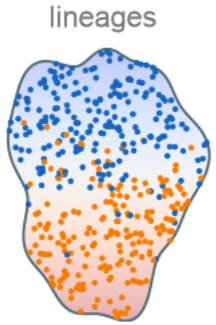


warm south

● early allele ● other allele

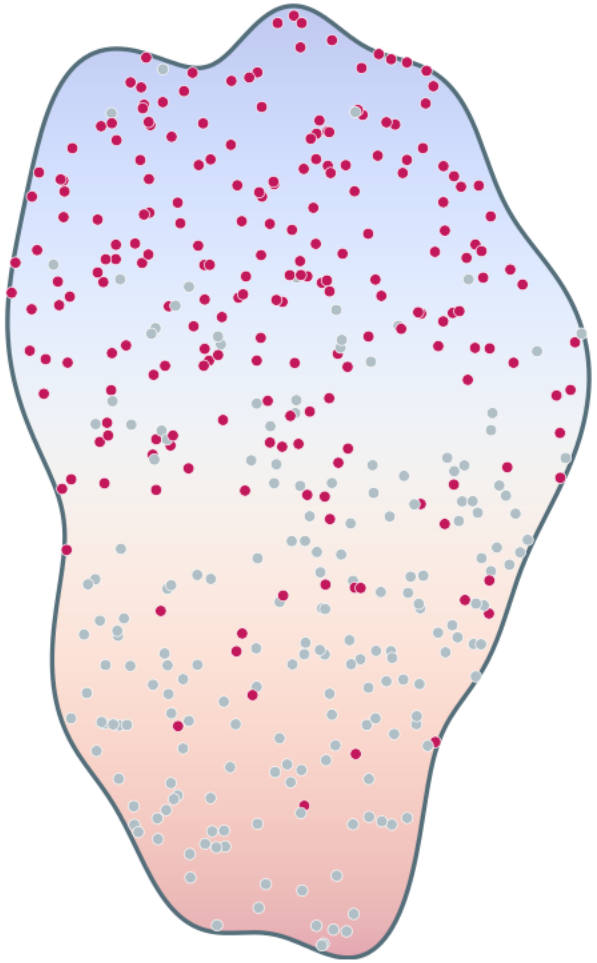
common where it is warm

A second flowering gene, found everywhere



◆ neutral SNP

cold north



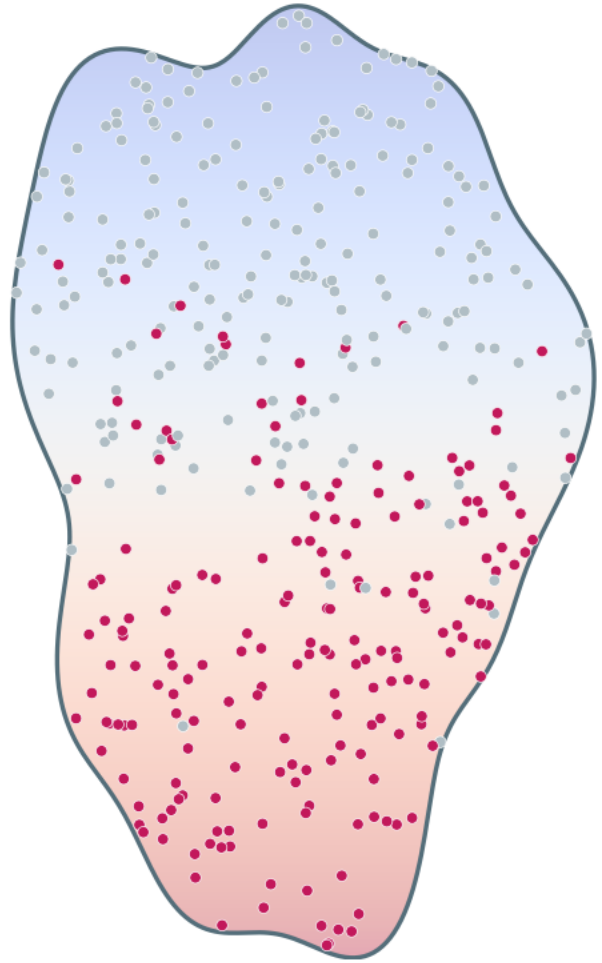
warm south

● T allele ● other allele

no effect on flowering

★ local-adaptation gene

cold north



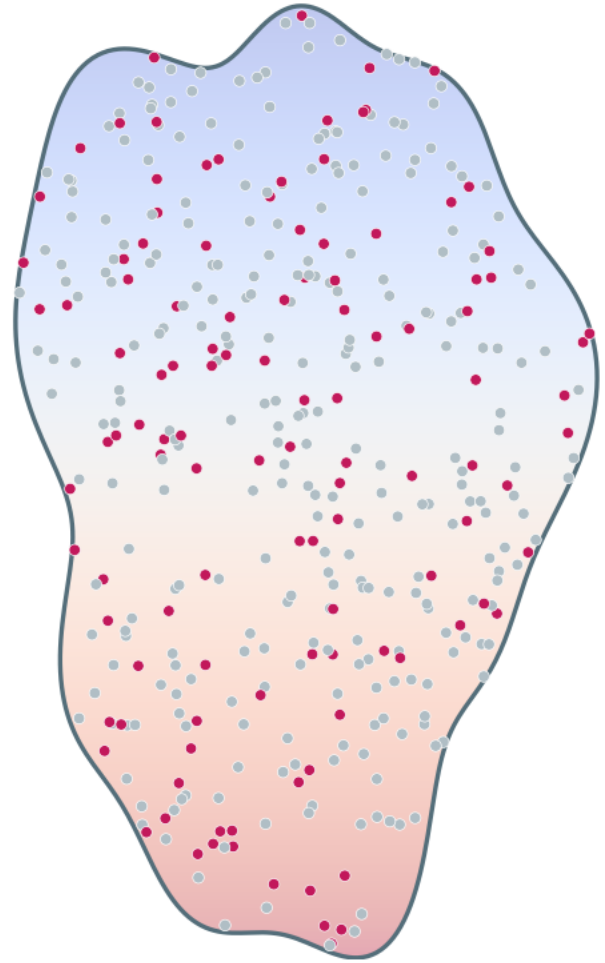
warm south

● early allele ● other allele

common where it is warm

★ gene found everywhere

cold north



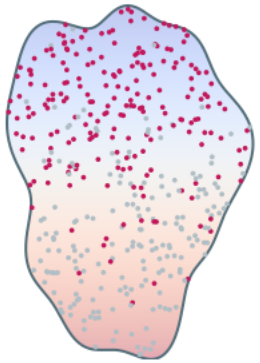
warm south

● early allele ● other allele

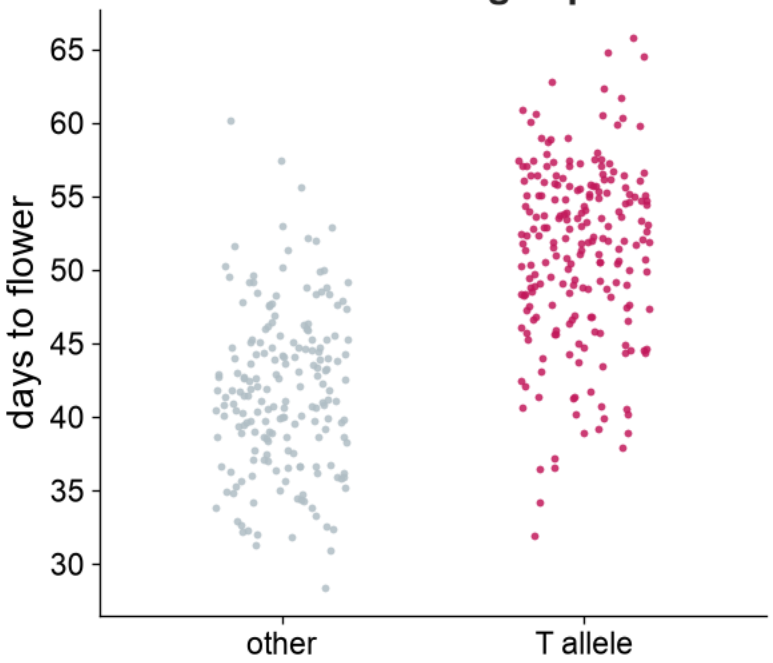
common everywhere

In the common garden: which alleles go with flowering time?

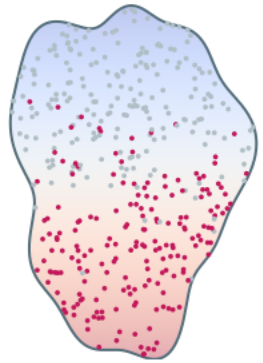
◆ neutral SNP



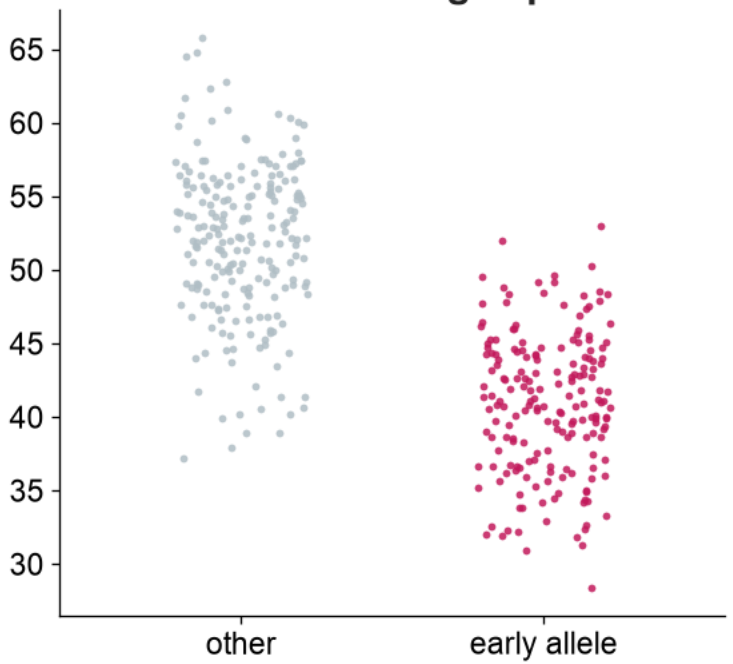
association: −log10 p = 46



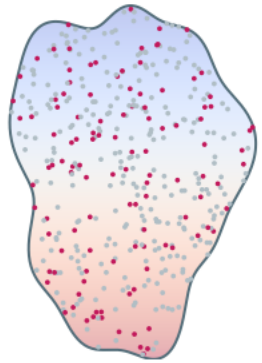
★ local-adaptation gene



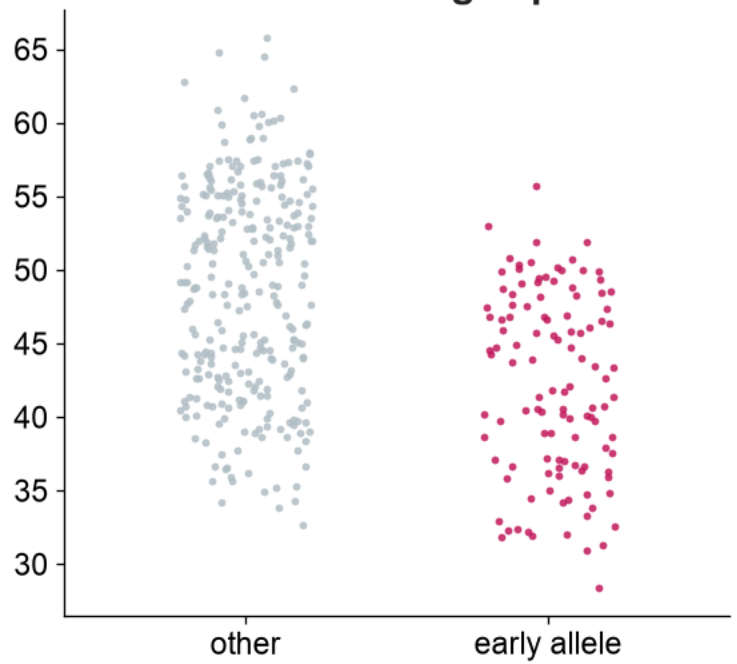
association: −log10 p = 69



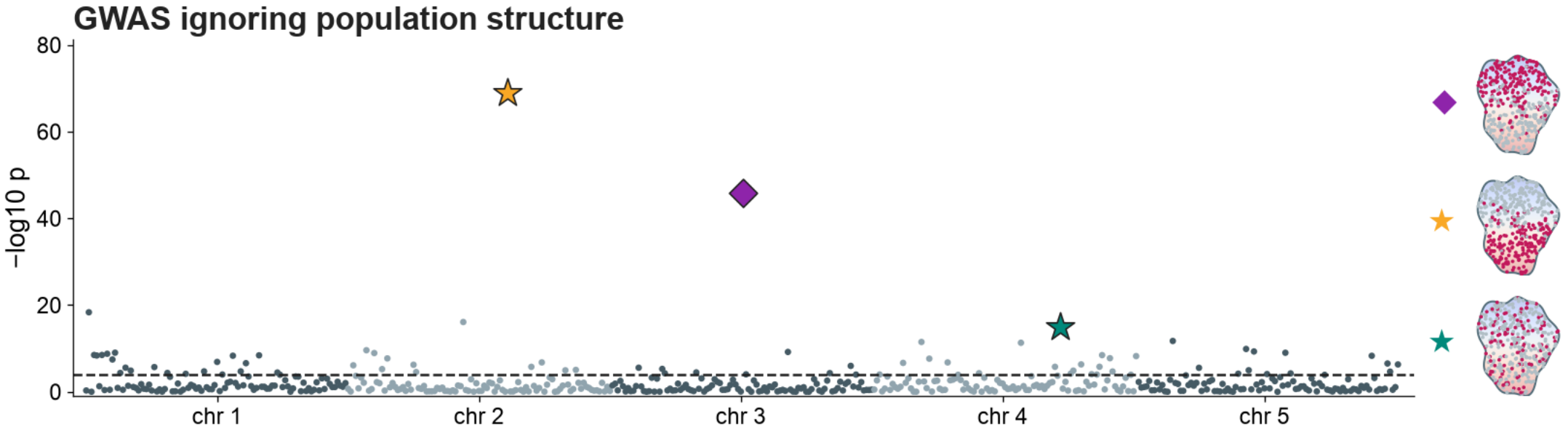
★ gene found everywhere



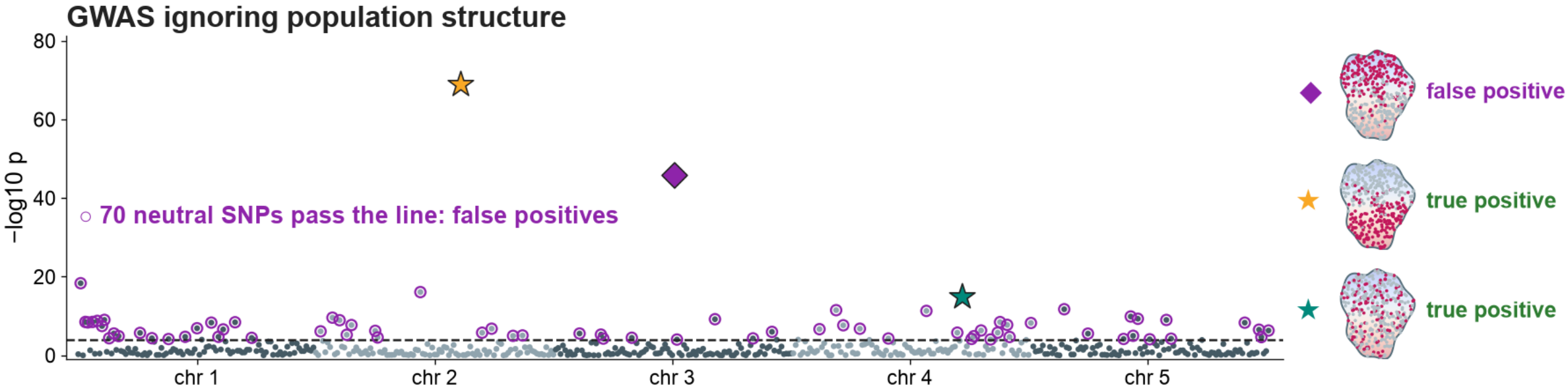
association: −log10 p = 15



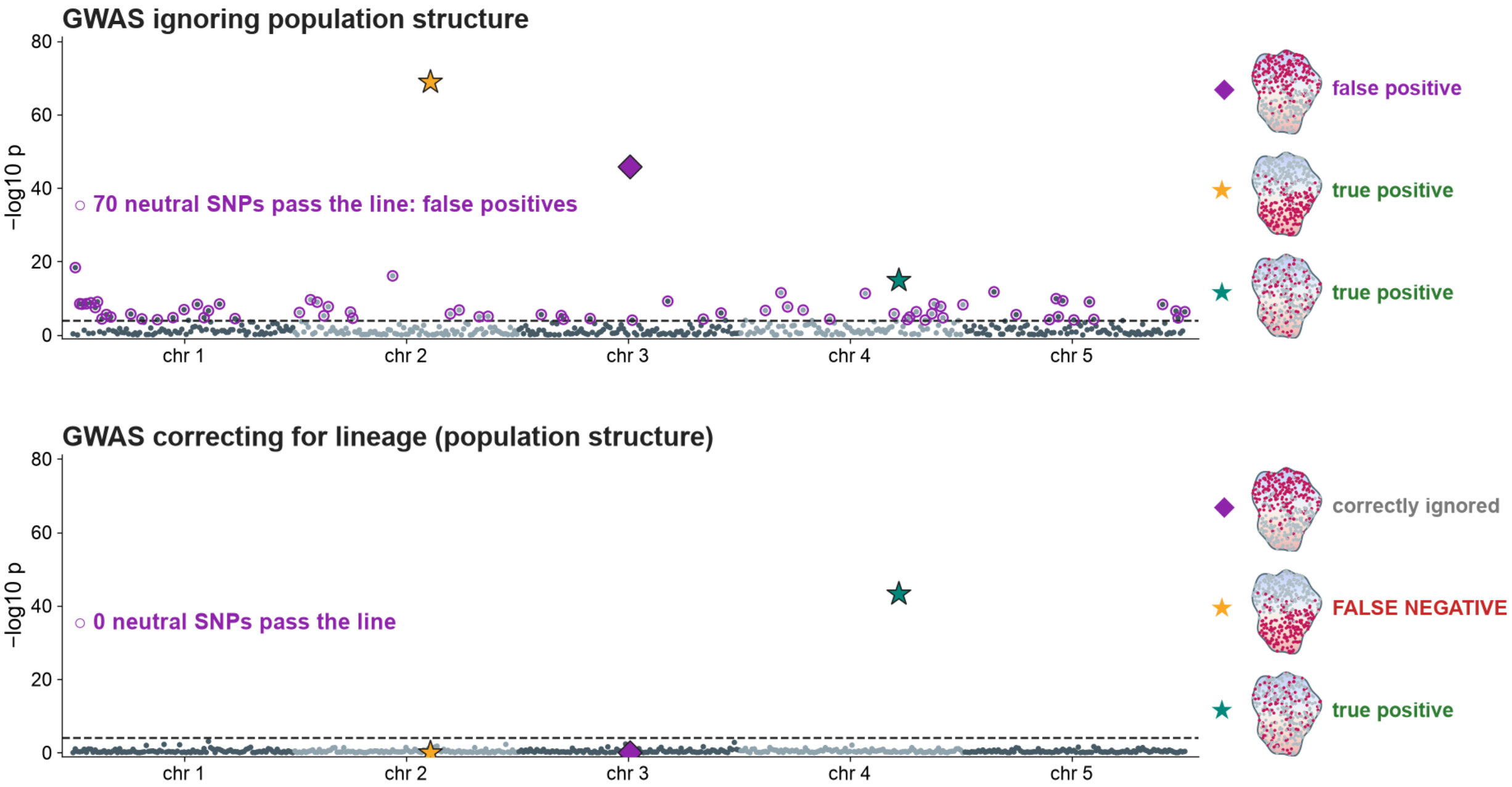
All three line up with flowering time. Only two are causal.



◆ our neutral SNP ★ the two real flowering genes dashed = significance line lineage of each plant assigned from its genome



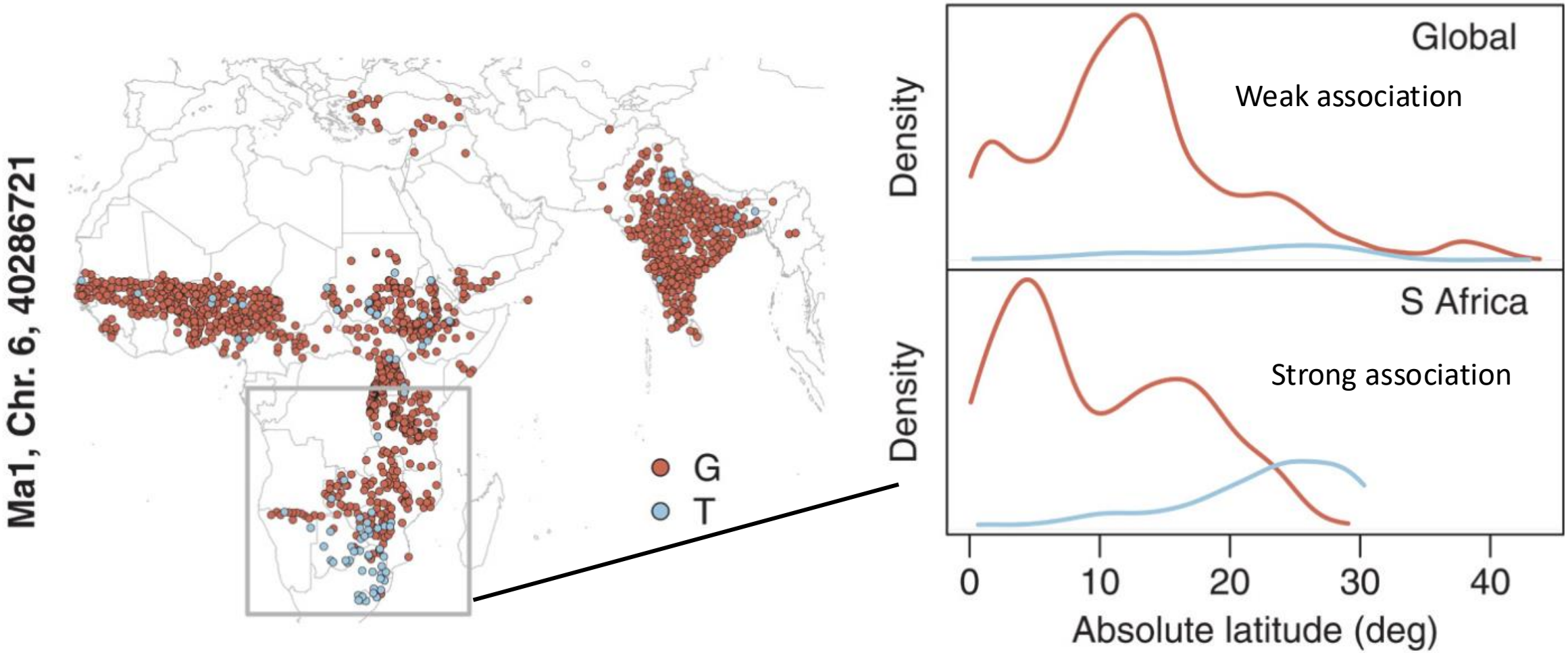
◆ our neutral SNP ★ the two real flowering genes dashed = significance line lineage of each plant assigned from its genome


GWAS ignoring population structure
−log10 p
80
60
40
20
0
○ 70 neutral SNPs pass the line: false positives
chr 1
chr 2
chr 3
chr 4
chr 5
false positive
true positive
true positive
GWAS correcting for lineage (population structure)
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80
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○ 0 neutral SNPs pass the line
chr 1
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chr 4
chr 5
correctly ignored
FALSE NEGATIVE
true positive
◆ our neutral SNP ★ the two real flowering genes dashed = significance line lineage of each plant assigned from its genome



Locally adapted genotypes are genetically distinct at many loci whether causal and neutral.

Case: Gene affecting maturing date in sorghum hidden by controlling for population structure. Solution: studying subset of population.



Live poll

Word cloud

The causal gene is there, but GWAS shows no peak. What can cause this false negative?

Scan with your phone

Join on your phone or laptop: greymonroe.github.io/live-poll-test/submit.html?poll=pls152-ftg-falseneg

host screen

Break



What are the genetics of adaptation to a new environment?

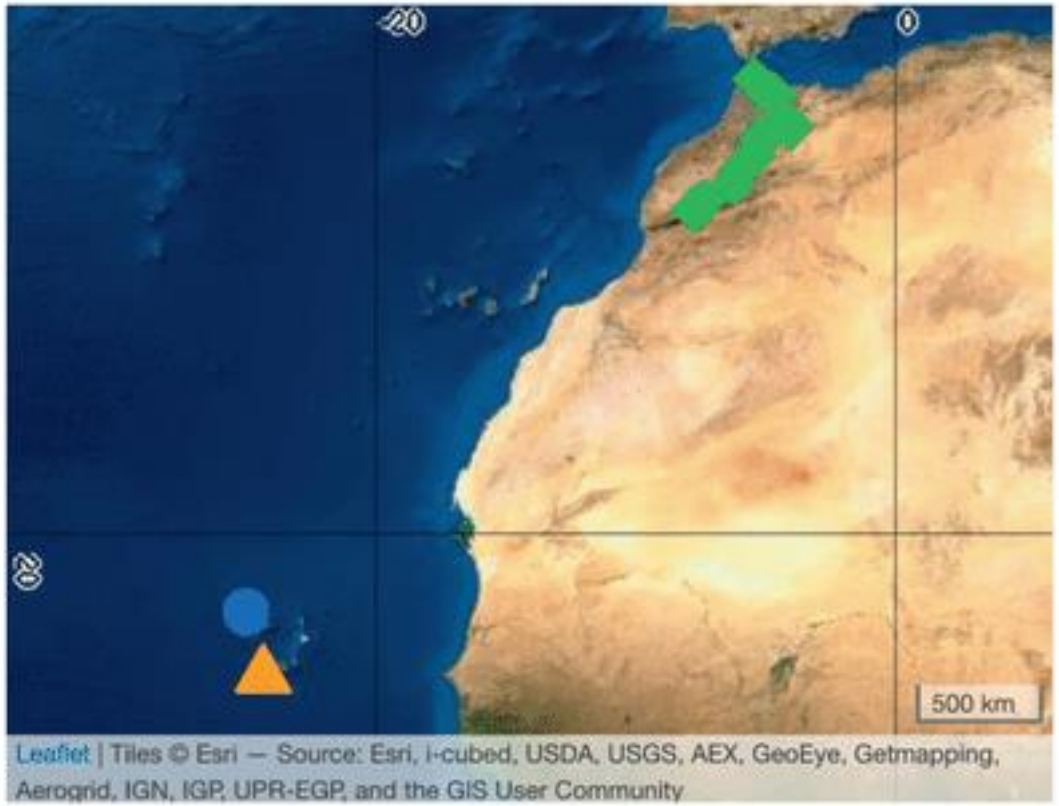
Important for understanding invasive species, how species respond to climate change, even how to select for crops.


Cape Verde Islands, off the coast of West Africa



a

Arabidopsis plants colonized the Cape Verde Islands







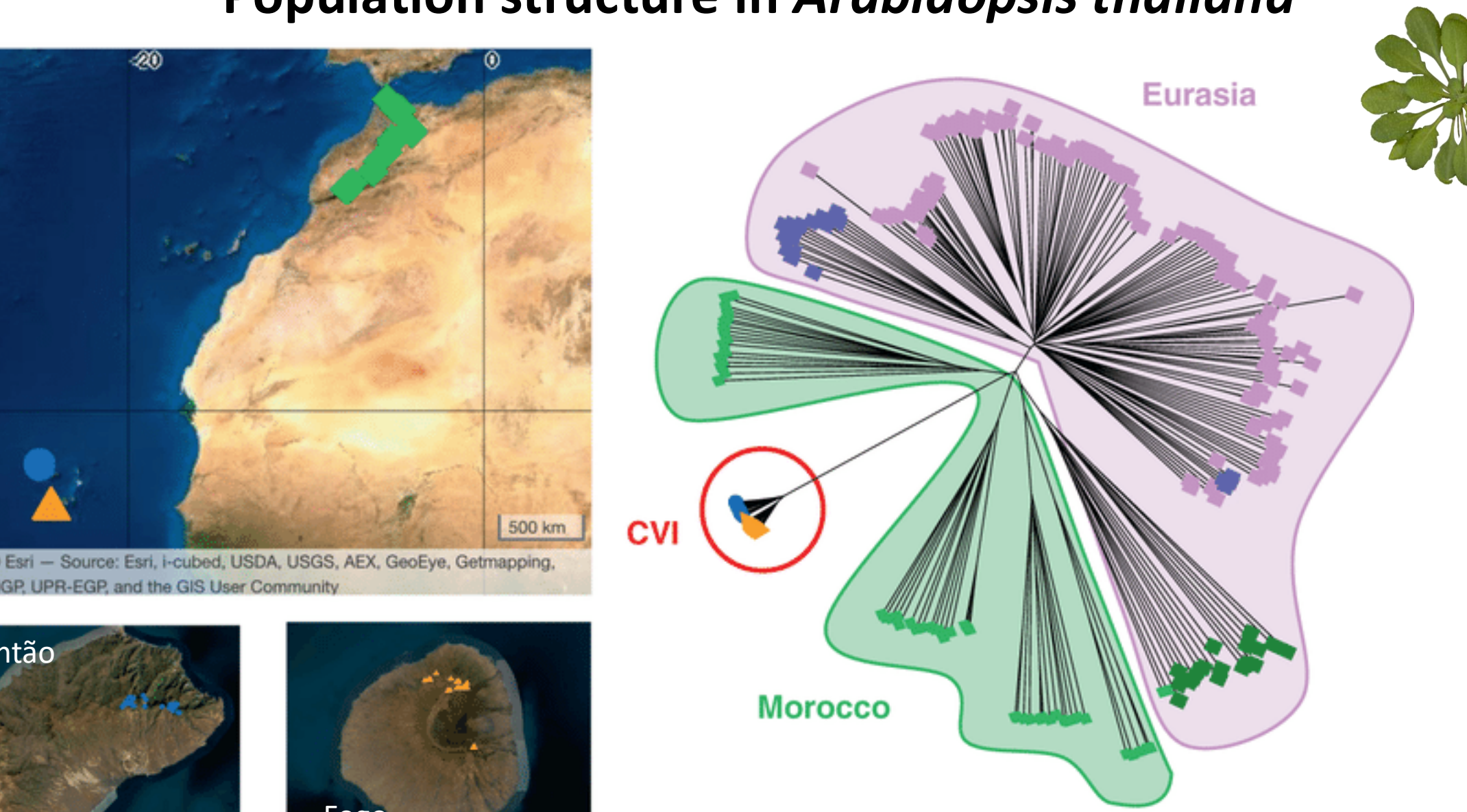
Cape Verde Islands

Fulgioni et al 2022



Are there mutations that helped *Arabidopsis* adapt to this new environment?

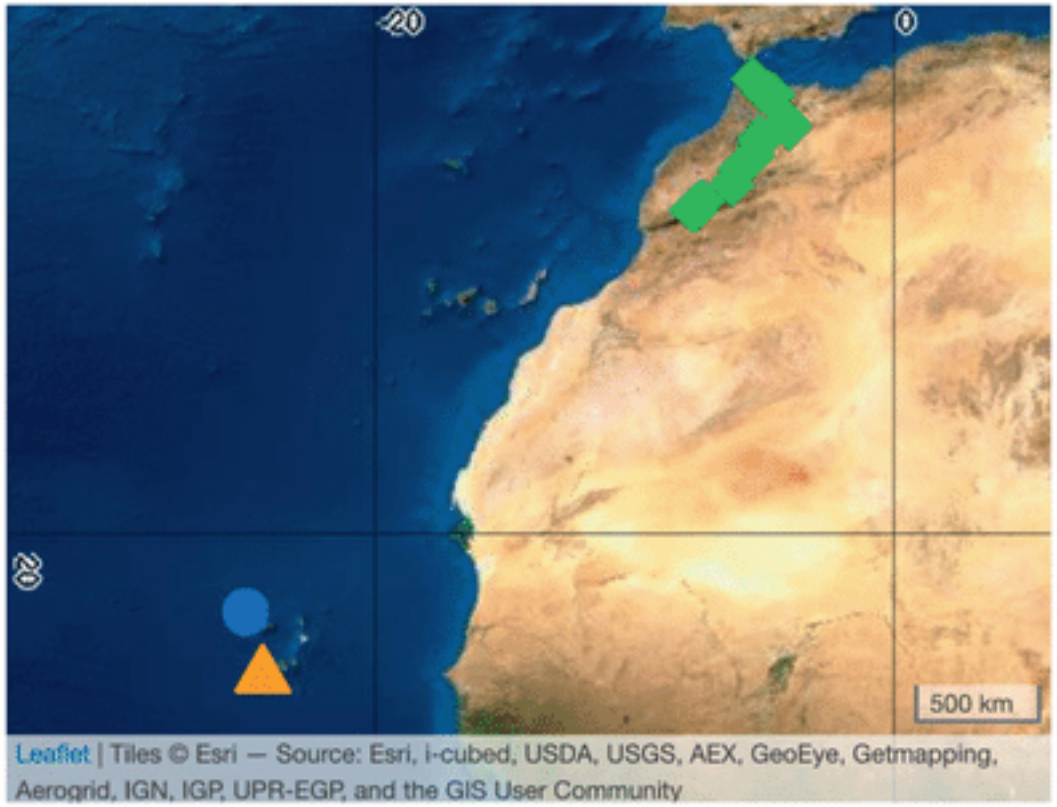
Population structure in *Arabidopsis thaliana*



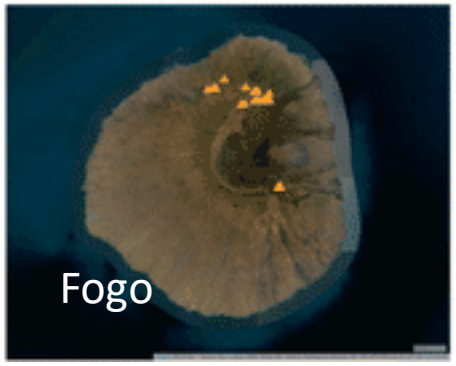
Fulgioni et al 2022

Population structure in *Arabidopsis thaliana*

a



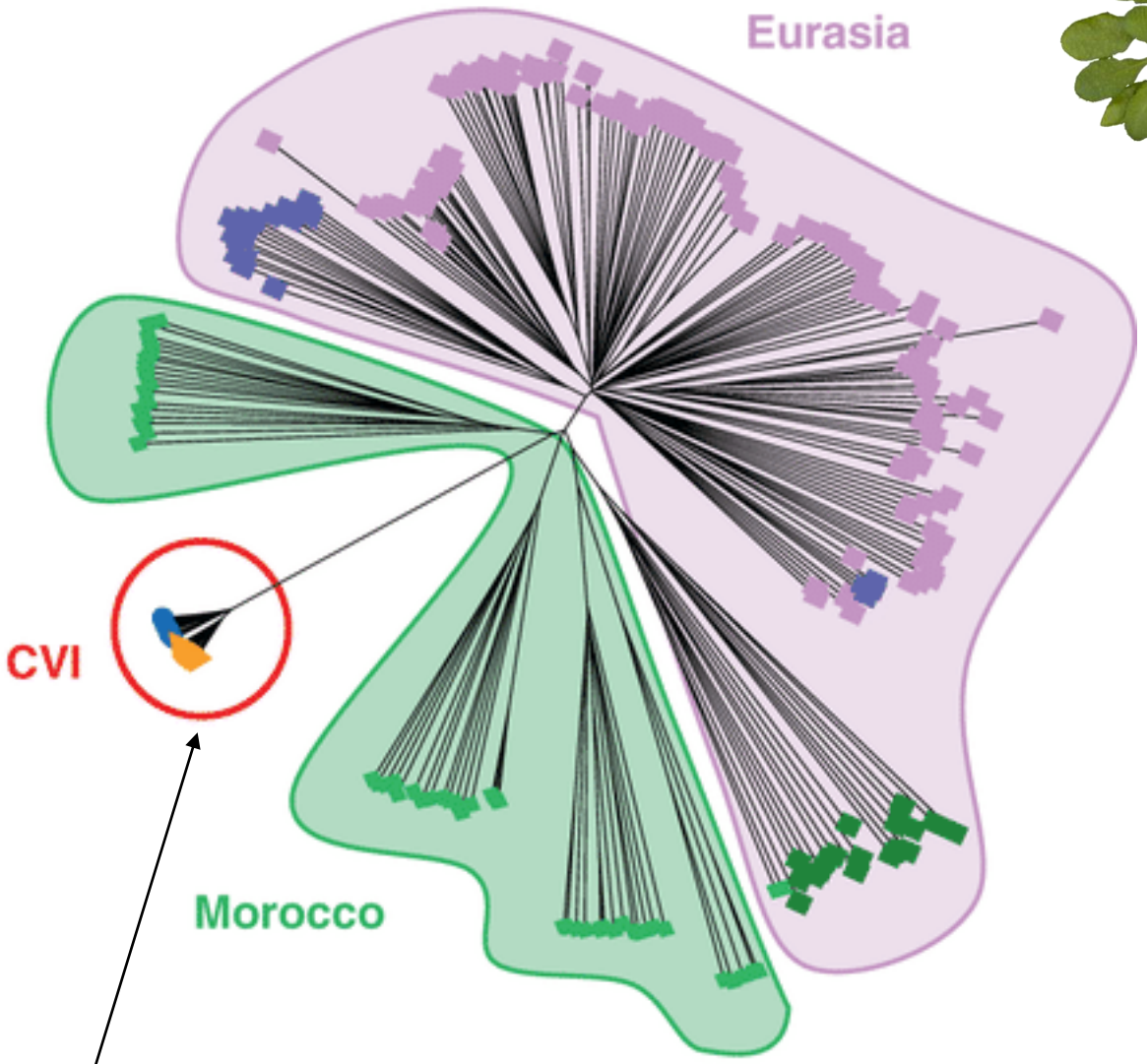




Cape Verde Islands

Fulgioni et al 2022

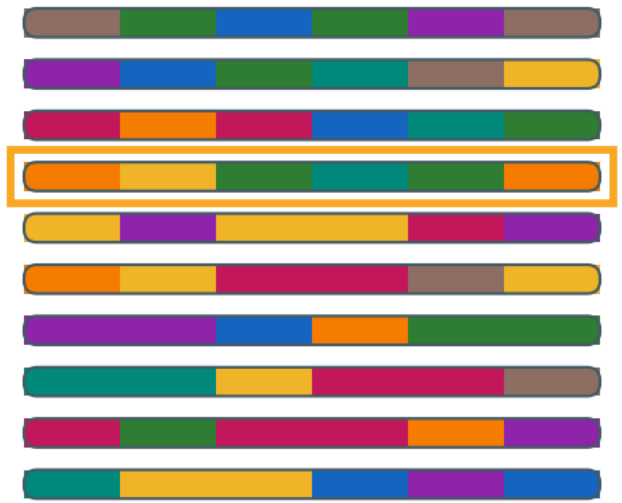




Any mutation unique to the founder of the Cape Verde Island population will be associated to traits involved in adaptation to island habitat

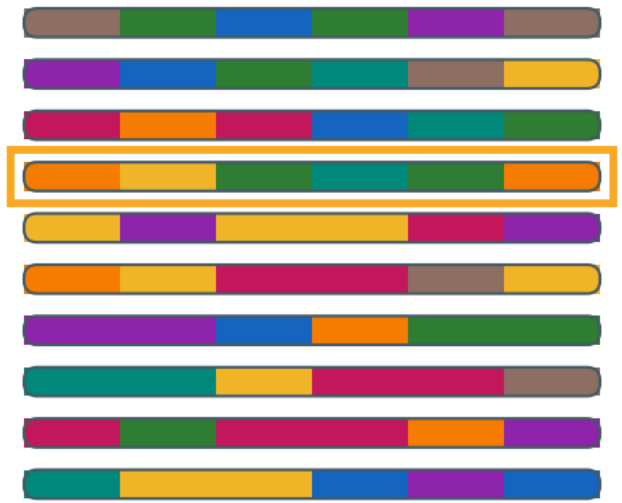
One seed reaches the island

Mainland: lots of variation



One seed reaches the island

Mainland: lots of variation

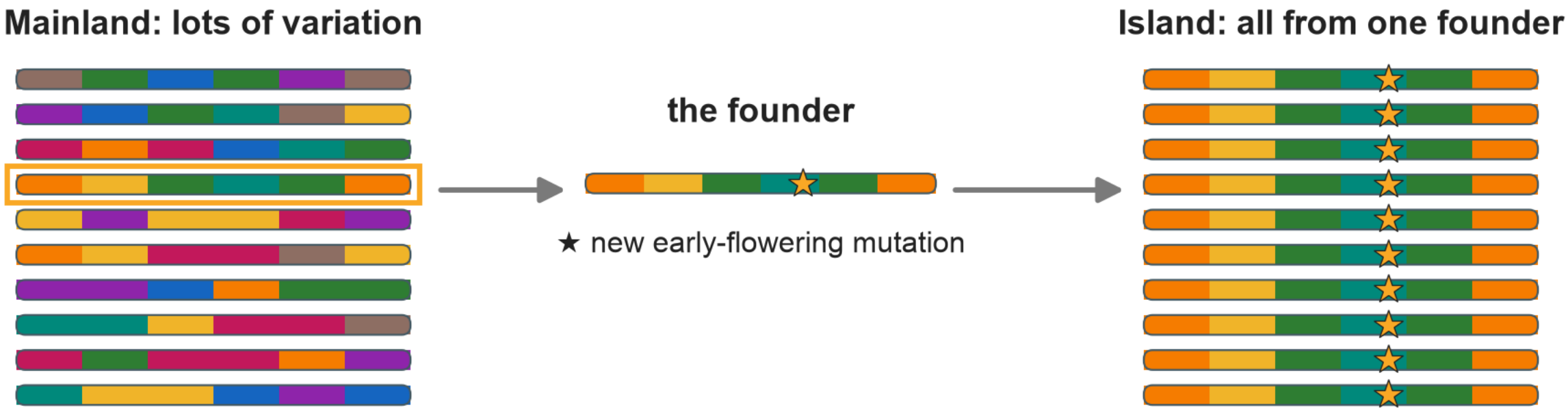


the founder

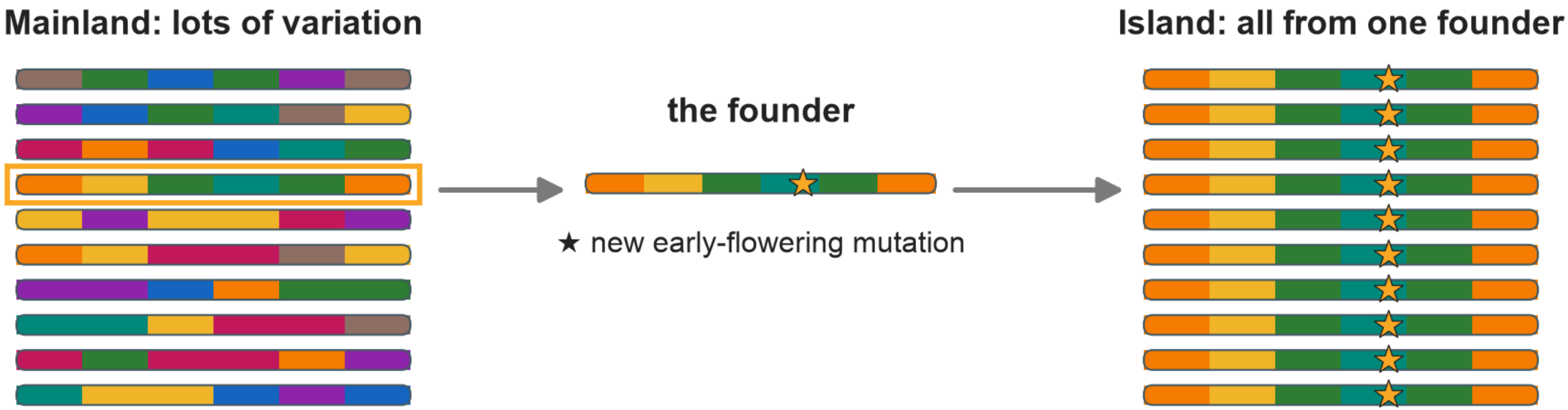


★ new early-flowering mutation

One seed reaches the island



One seed reaches the island

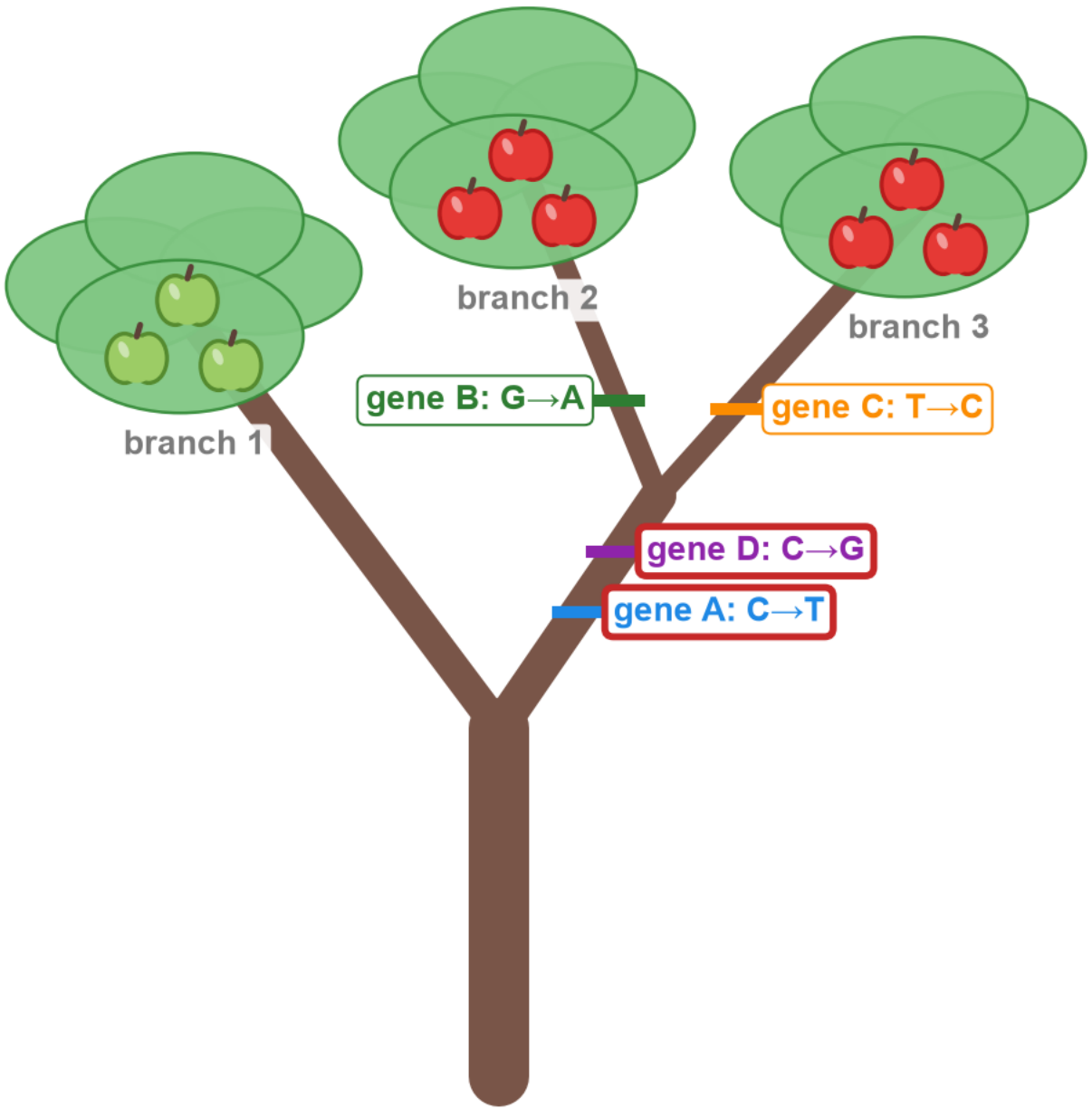


The early-flowering mutation and every neutral founder allele now travel together.

In a GWAS, all of them line up with "island" and with early flowering.

Ignore structure → false positives. Correct for it → the real gene can vanish too.

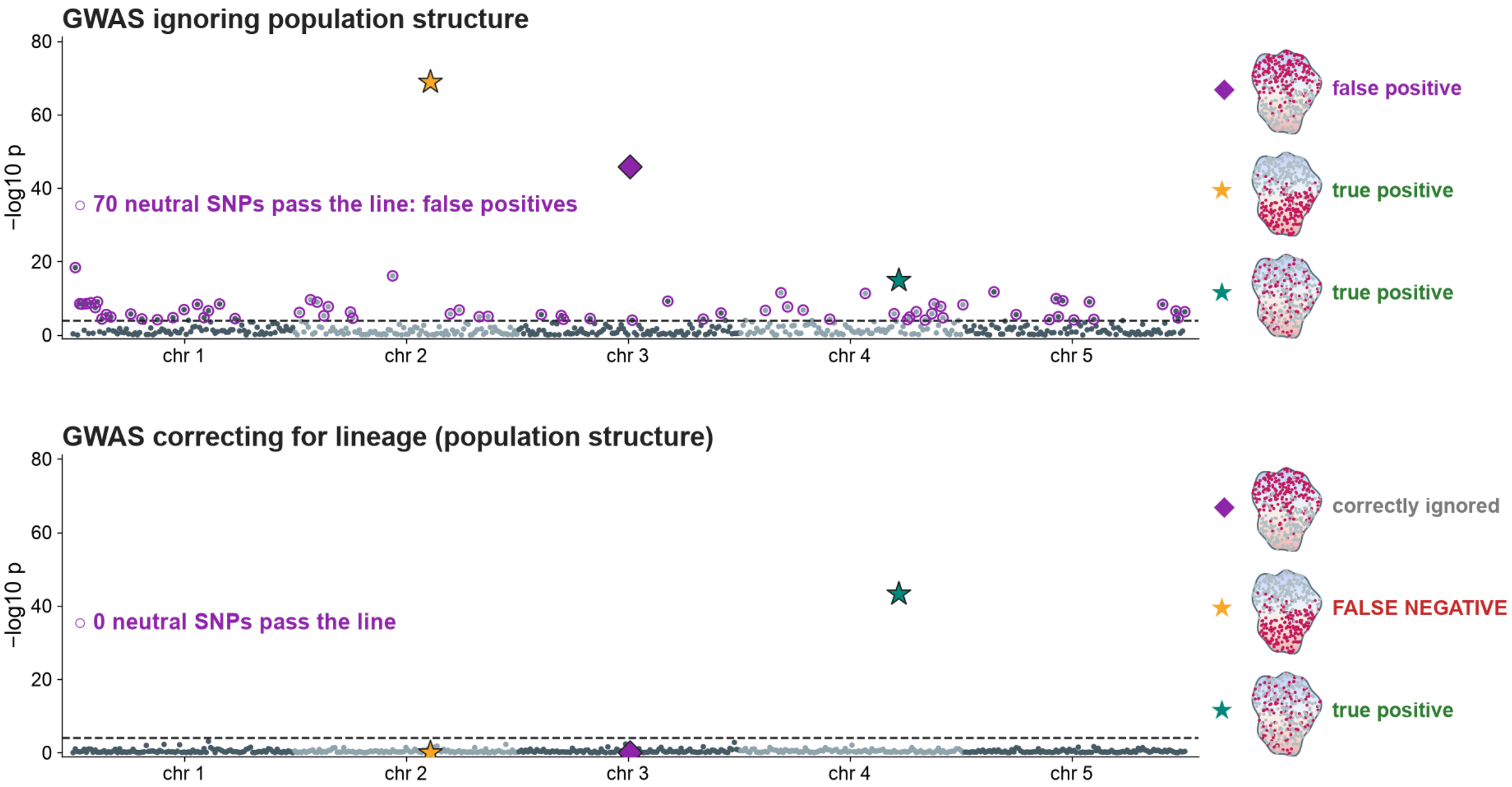
A or D? Shared ancestry = population structure



	gene A	gene B	gene C	gene D
branch 1	GACCTGA	TTAGCAA	CAGTACG	AGGCTTA
branch 2	GACTTGA	TTACAA	CAGTACG	AGGGTTA
branch 3	GACTTGA	TTAGCAA	CAGCACG	AGGGTTA

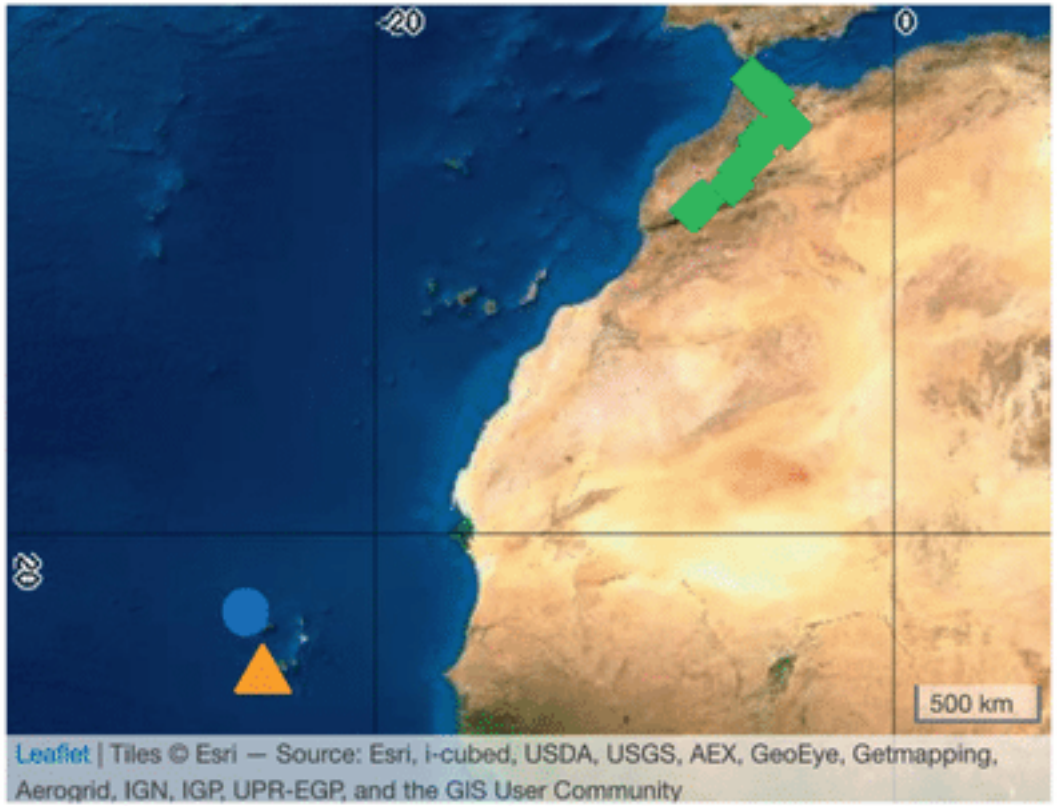
So does gene D: both changes happened on the shared branch. Shared history makes every passenger look causal: population structure

after Korte & Farlow 2013 · toy sequences

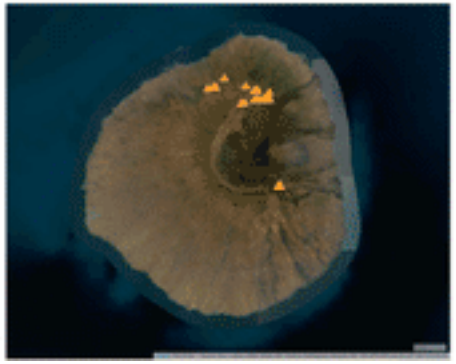

GWAS ignoring population structure
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◆ our neutral SNP ★ the two real flowering genes dashed = significance line lineage of each plant assigned from its genome

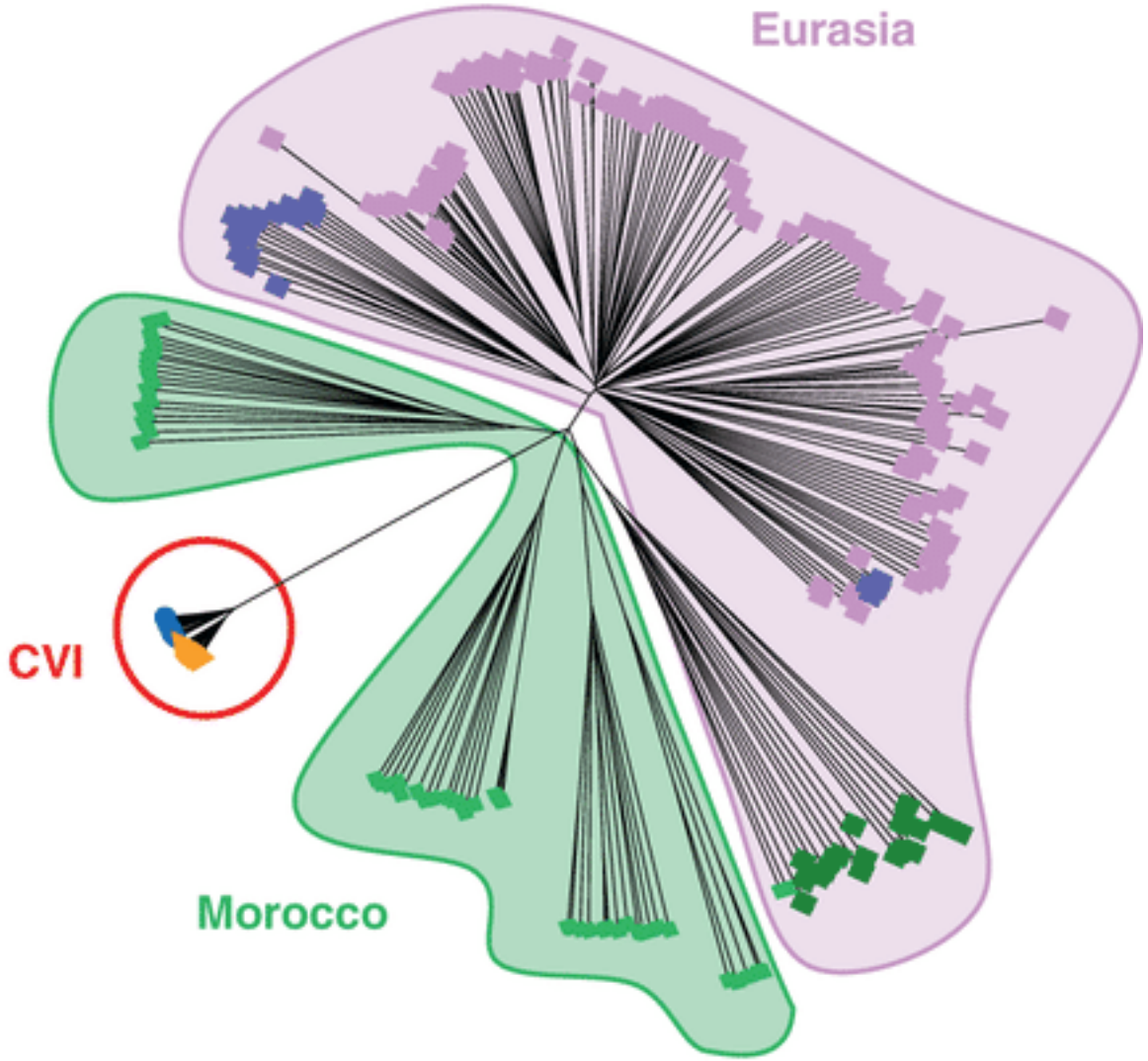
Know your system, pick the right experiments!

a





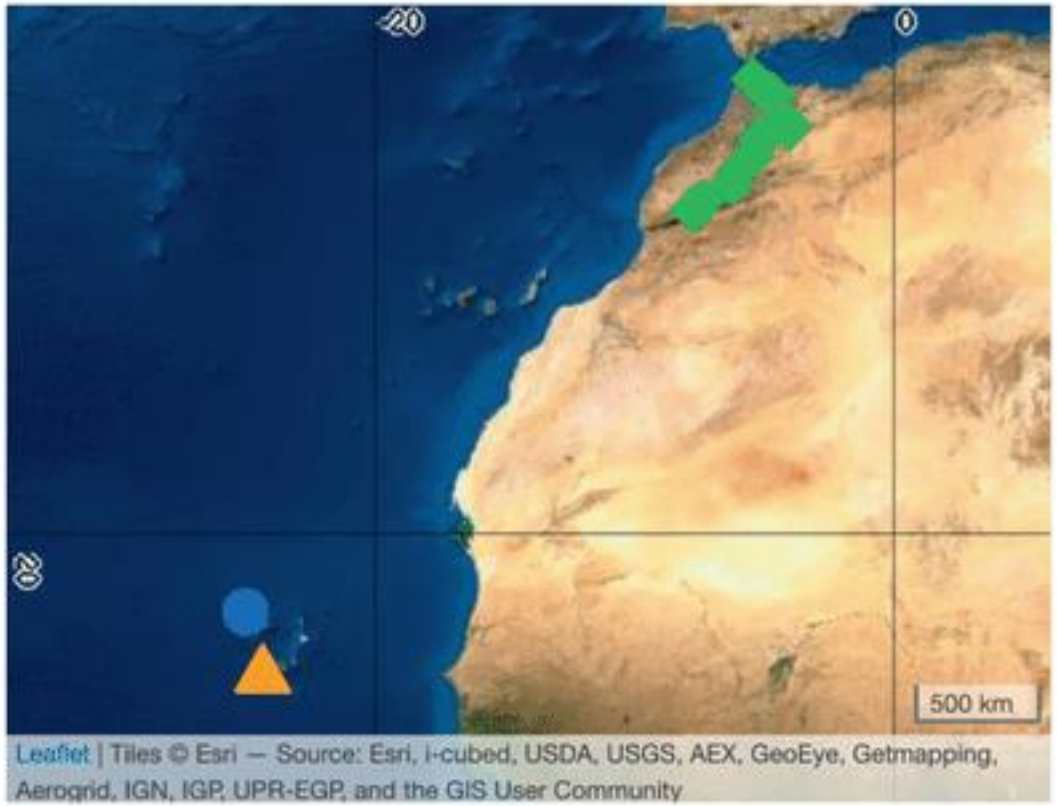




Fulgioni et al 2022

Variation in flowering time (days to bolting)

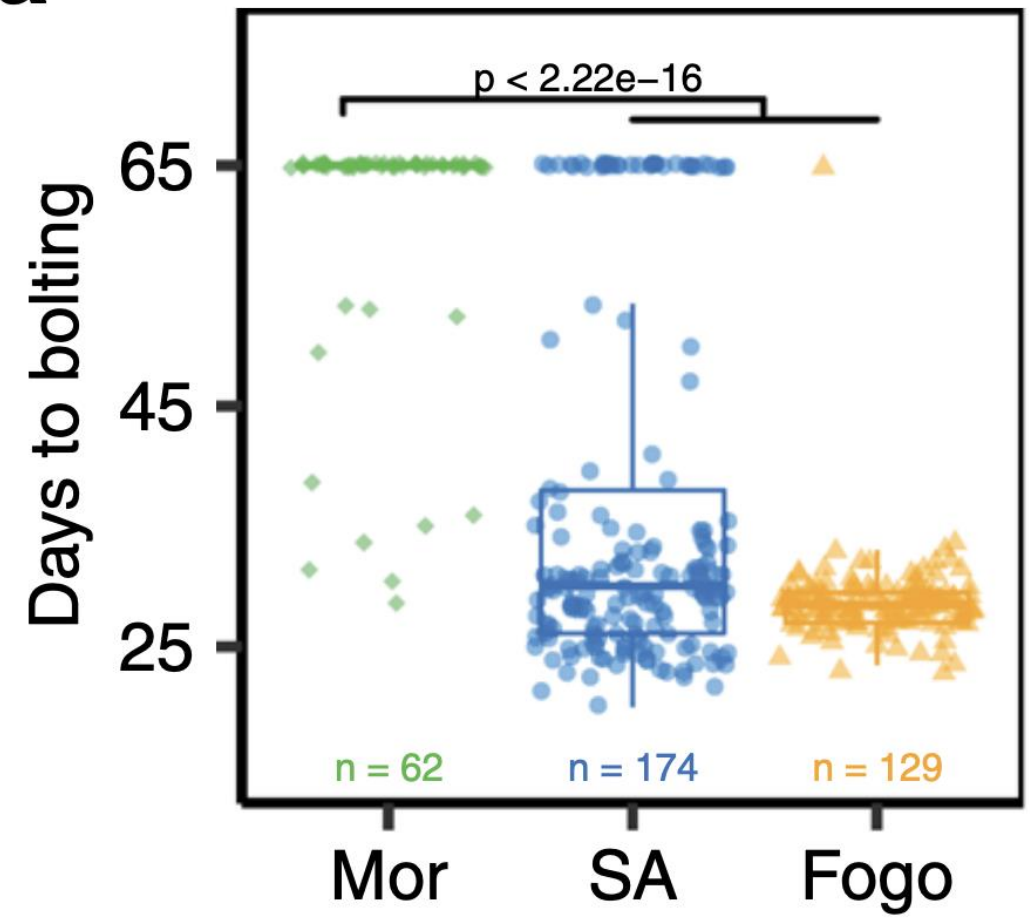
a





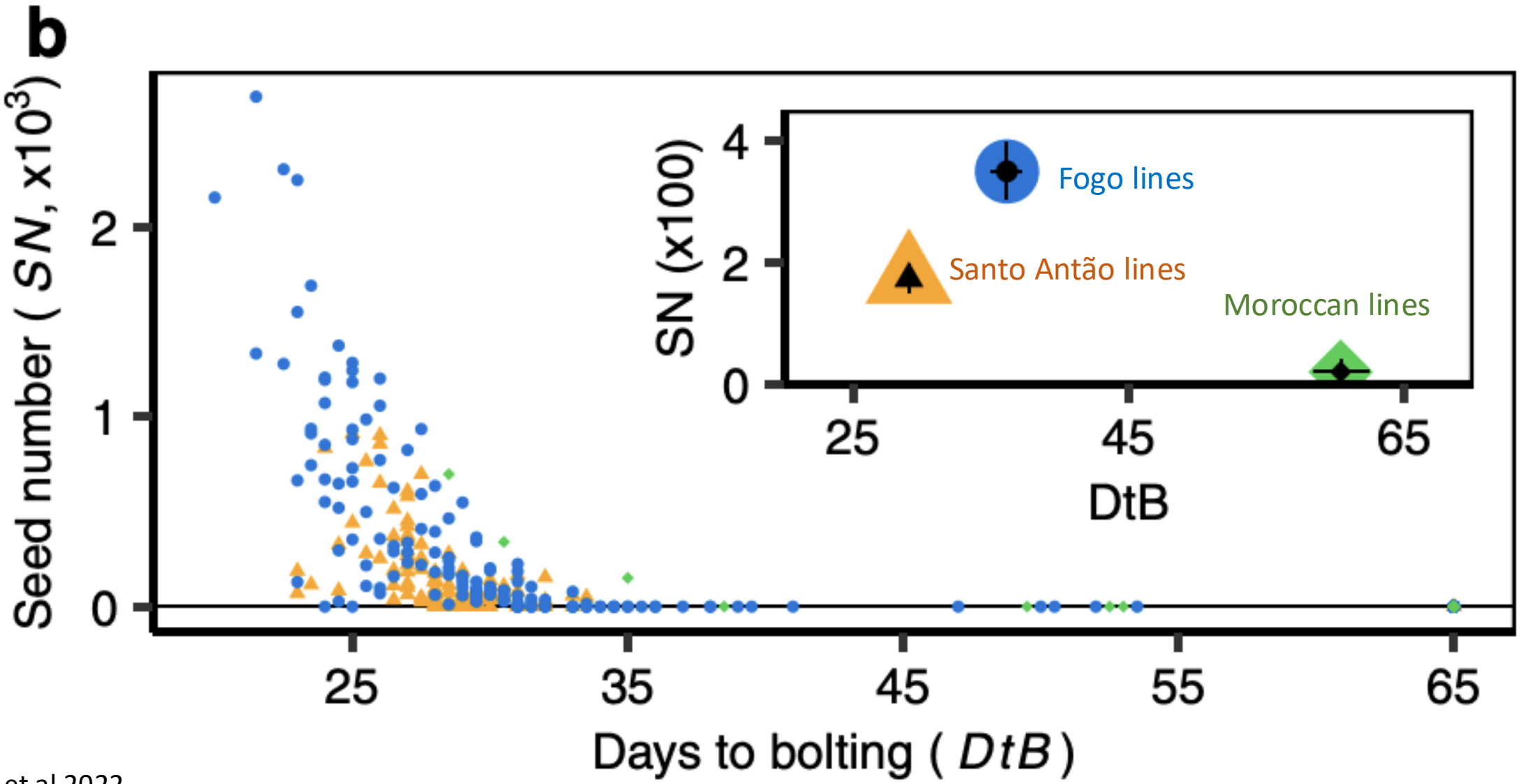


a

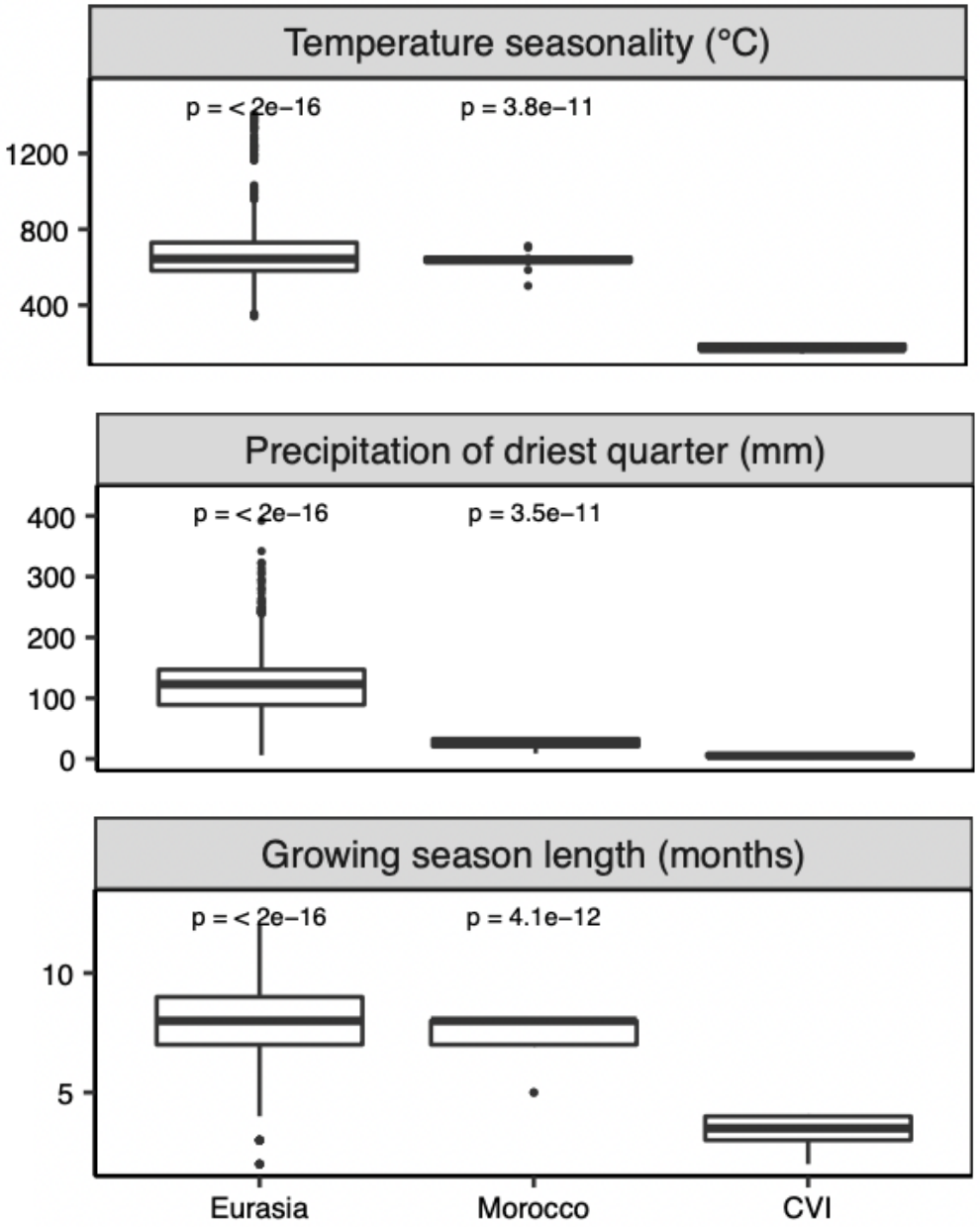


Fulgioni et al 2022

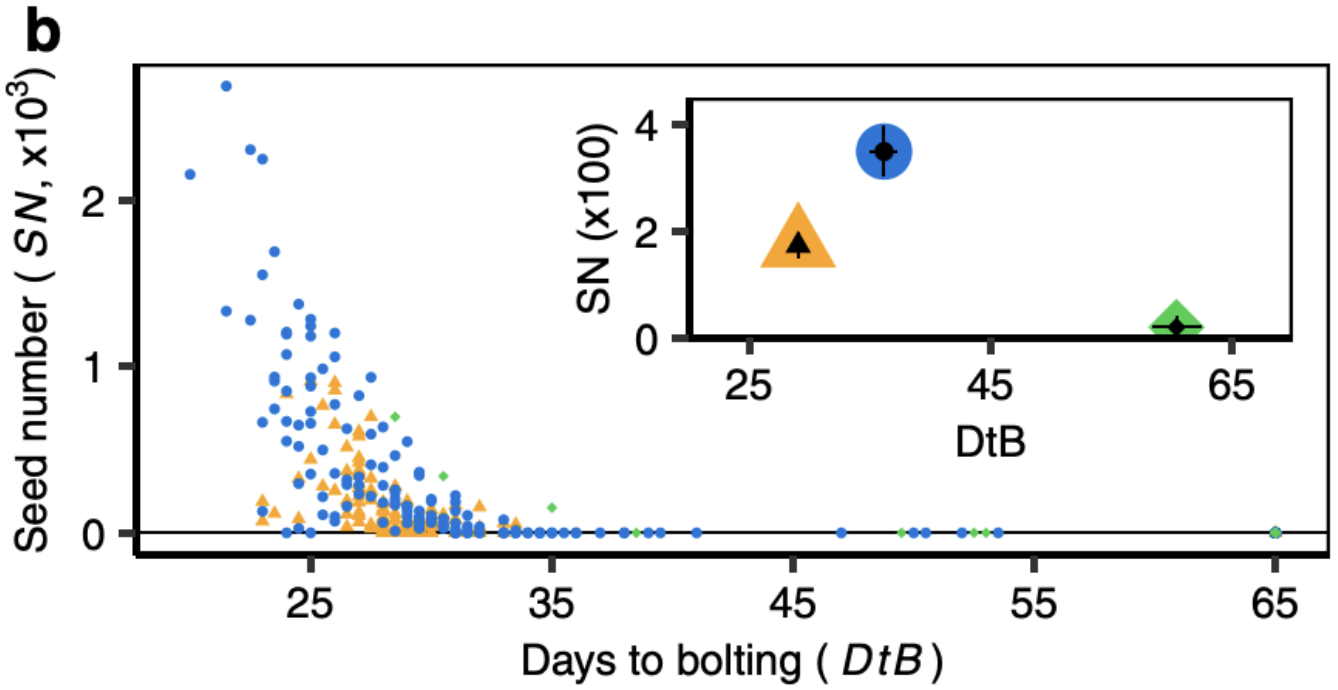
Flowering time was negatively correlated with seed # (fitness) when plants were grown in a CVI environment



Fulgioni et al 2022



Why would flowering early be "adaptive" (increase fitness) in a CVI environment?



Live poll

Text wall

Why would flowering EARLY increase fitness on the Cape Verde Islands?

Scan with your phone

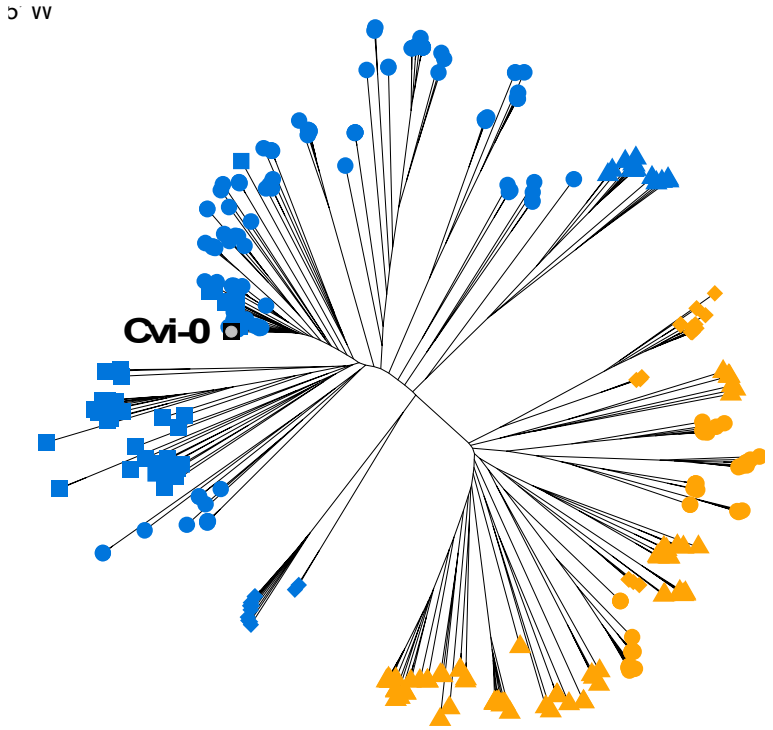
[Join on your phone or laptop: greymonroe.github.io/live-poll-test/submit.html?poll=pls152-ftg-cvi](greymonroe.github.io/live-poll-test/submit.html?poll=pls152-ftg-cvi)

[host screen]


a
Espongeiro
Lombo de Figueira
Cova de Paúl
Pico da Cruz
Google Earth
17º 10' N
17º 05' N
25º 05' W
25º W

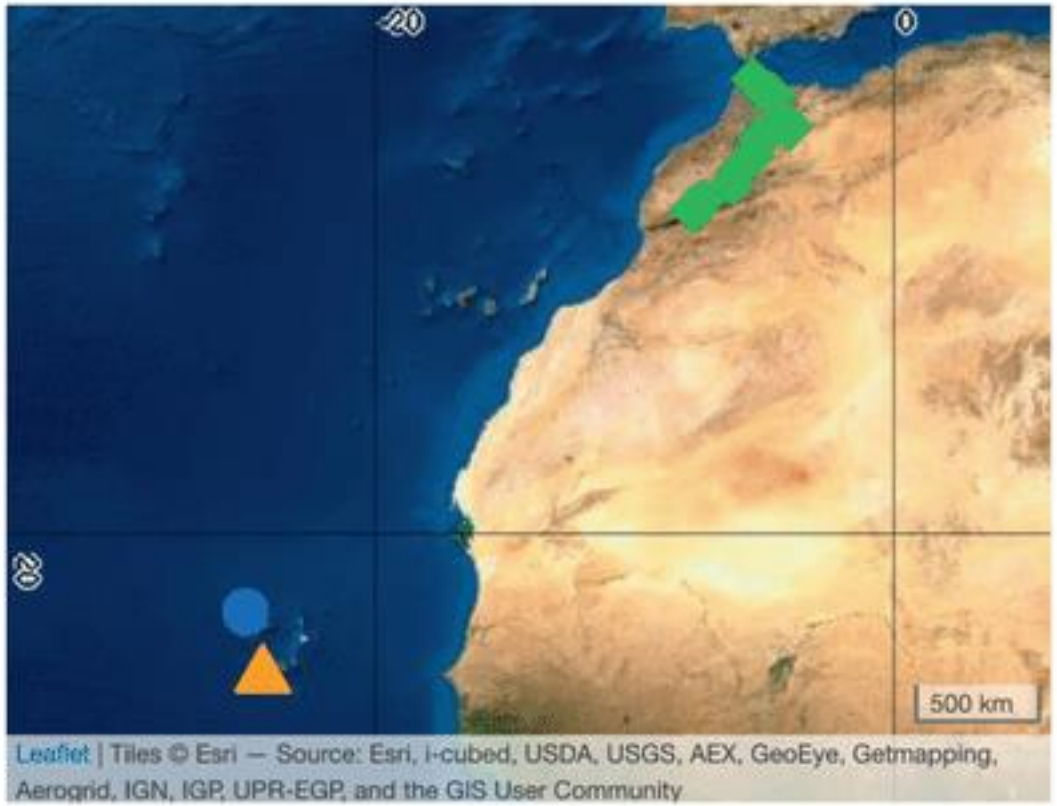

b
Inferno
Monte Velha
Lava
Google Earth
15º 10' N
14º 55' N
24º 25' W
24º 17' W


Cvi–0 (1001)
Cvi–0 (new)
SA–Lombo de Figuera
SA–Cova de Paúl
SA–Espongeiro
SA–Pico da Cruz
Fogo–Lava
Fogo–Ribeira Inferno
Fogo–Monte Velha


Cvi-0

Variation in flowering time (days to bolting)

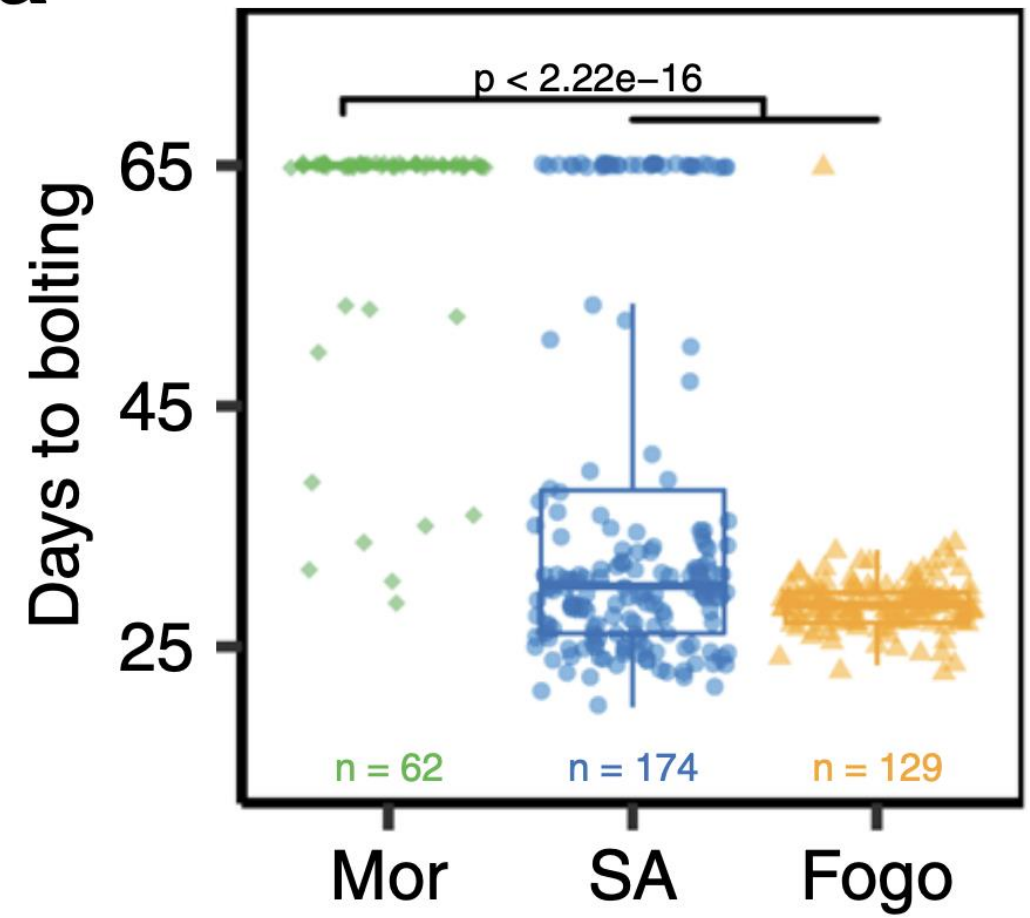
a







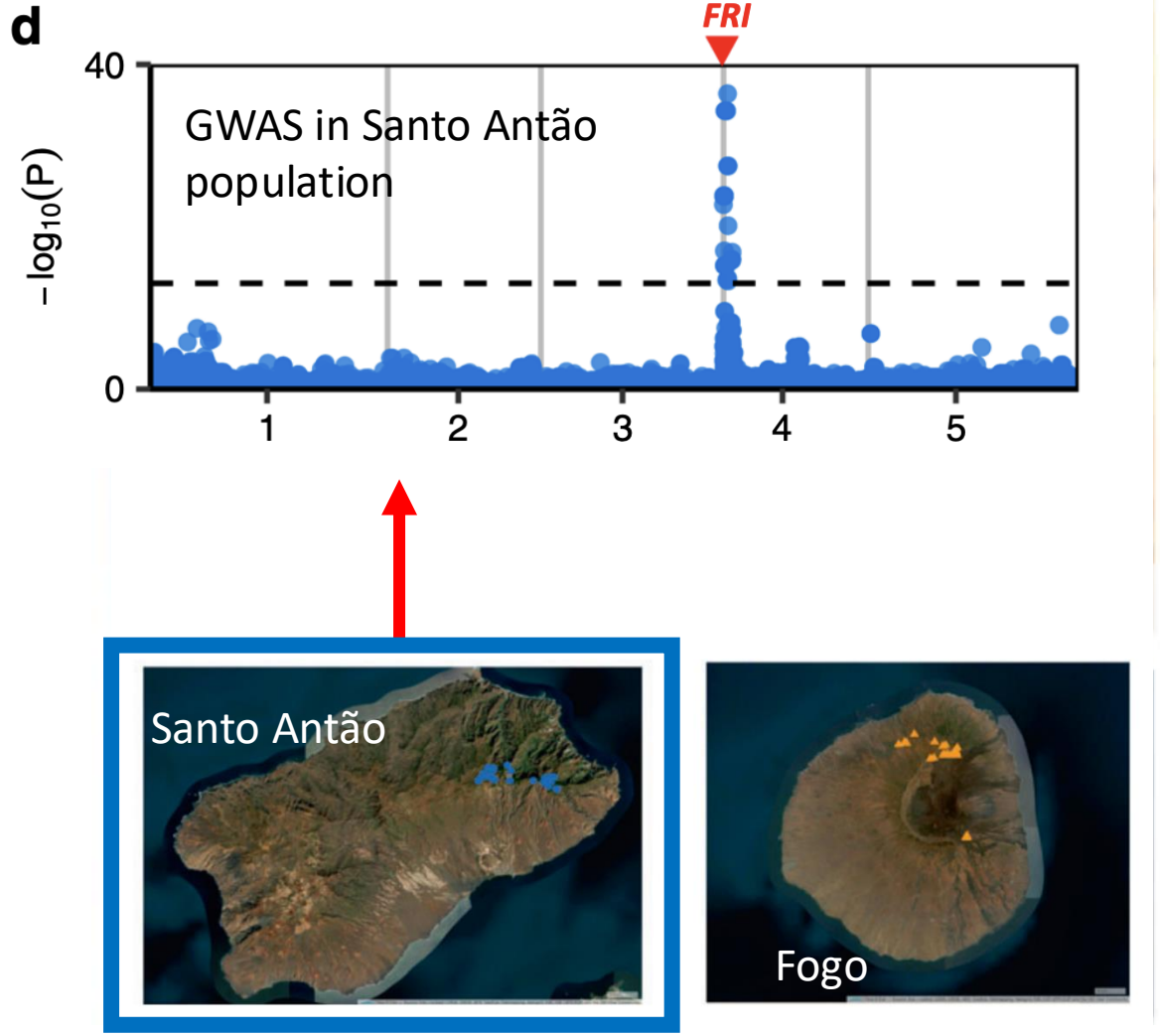
a

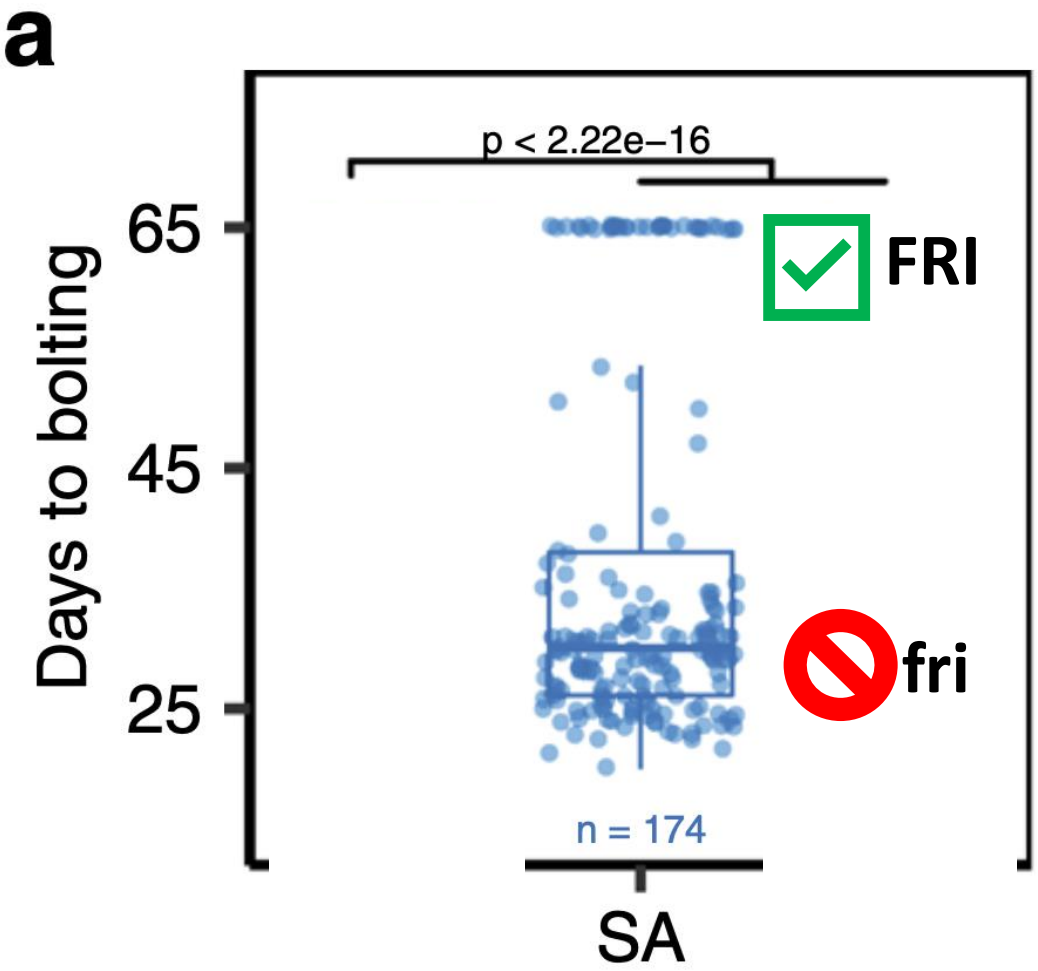


Fulgioni et al 2022

Santo Antão population is segregating for FRI stop codon mutation

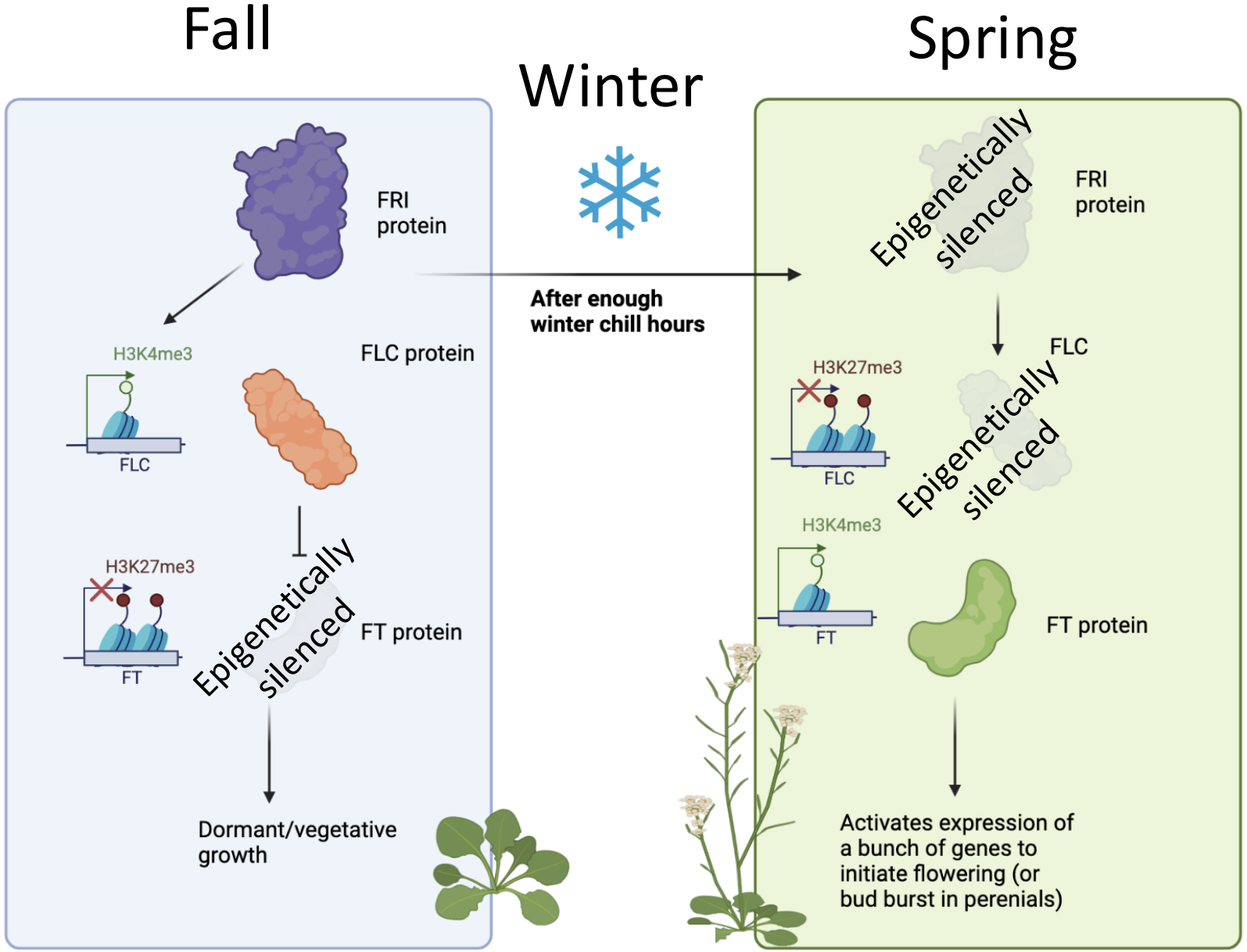
FRI **Premature stop mutation** in FRI gene that arose about 2.5 thousand years ago



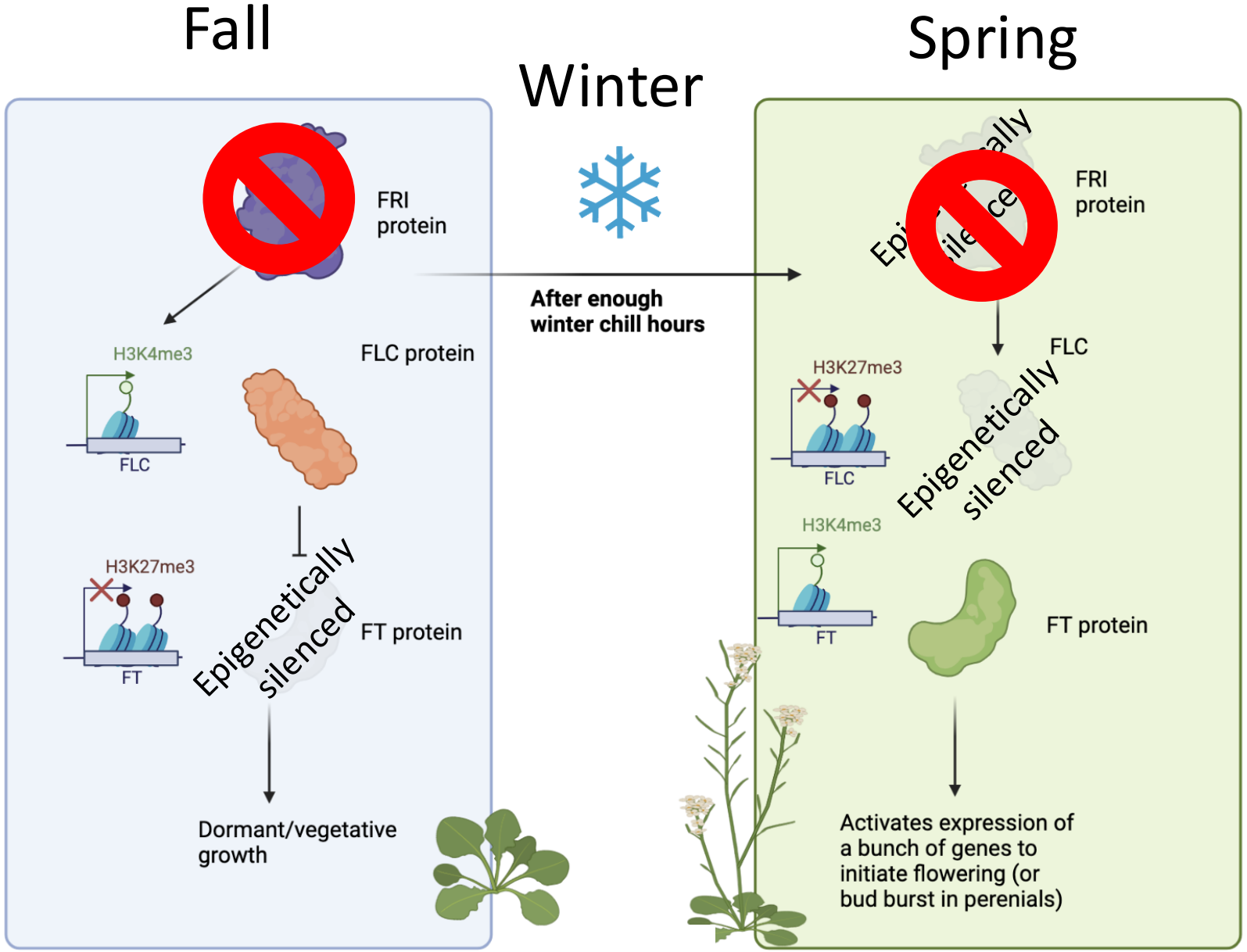


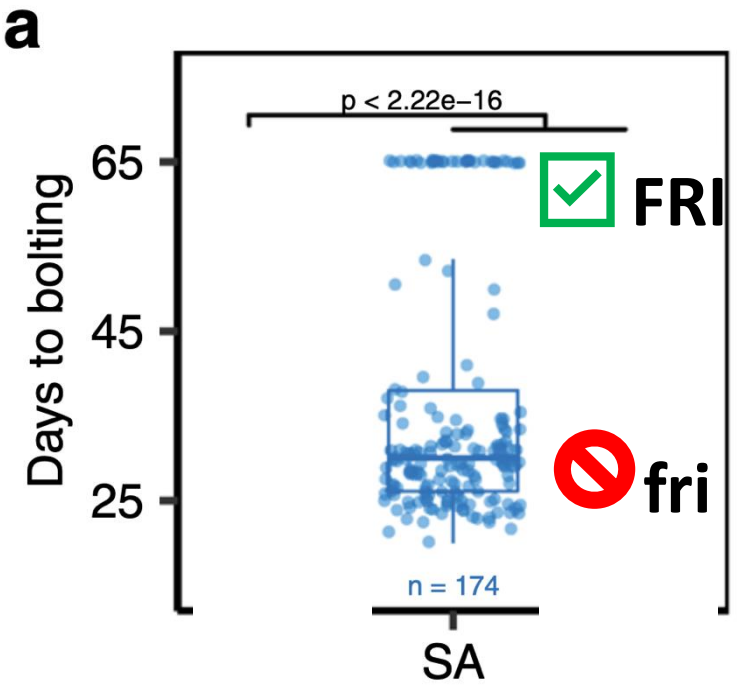
Fulgioni et al 2022

Vernalization control of flowering

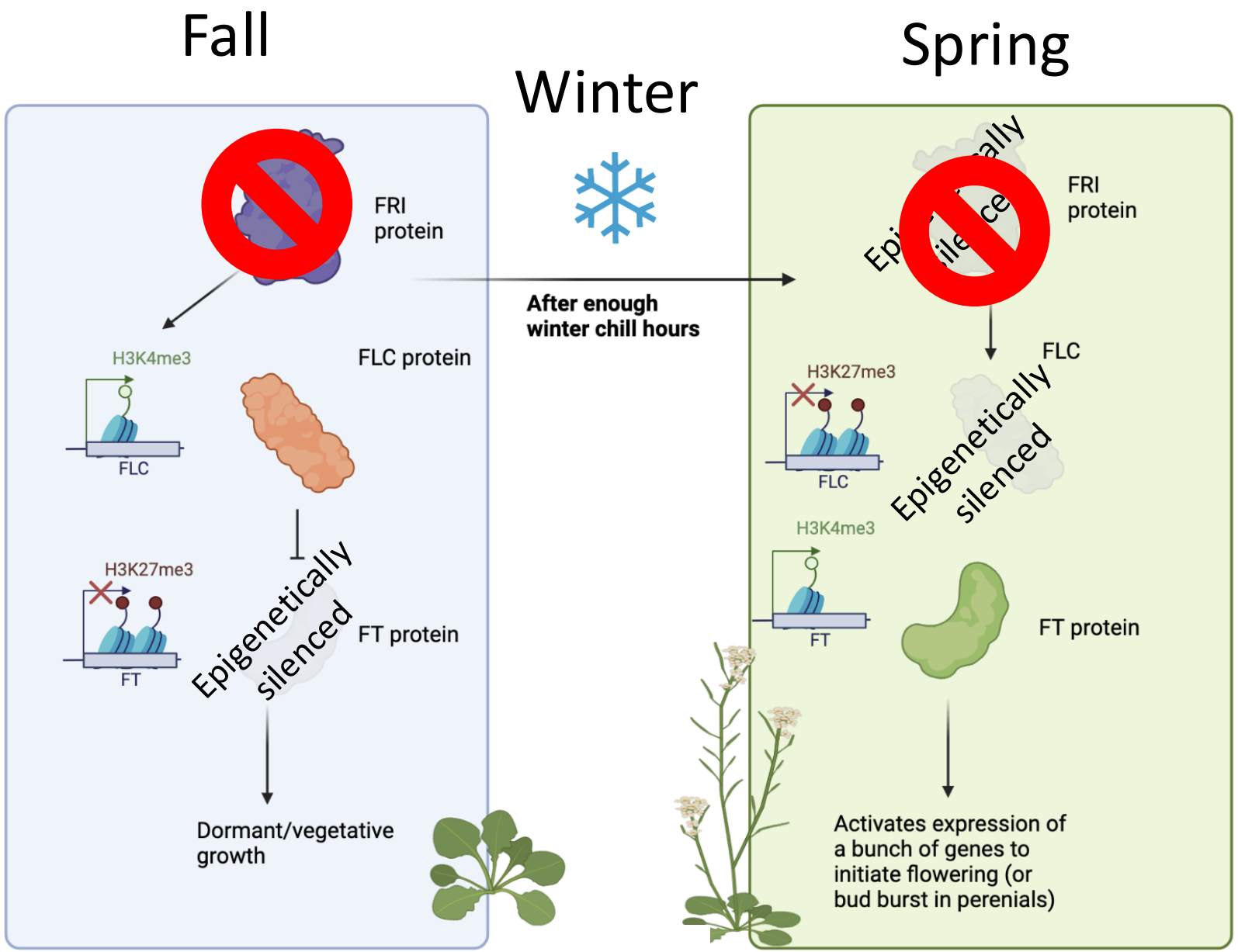


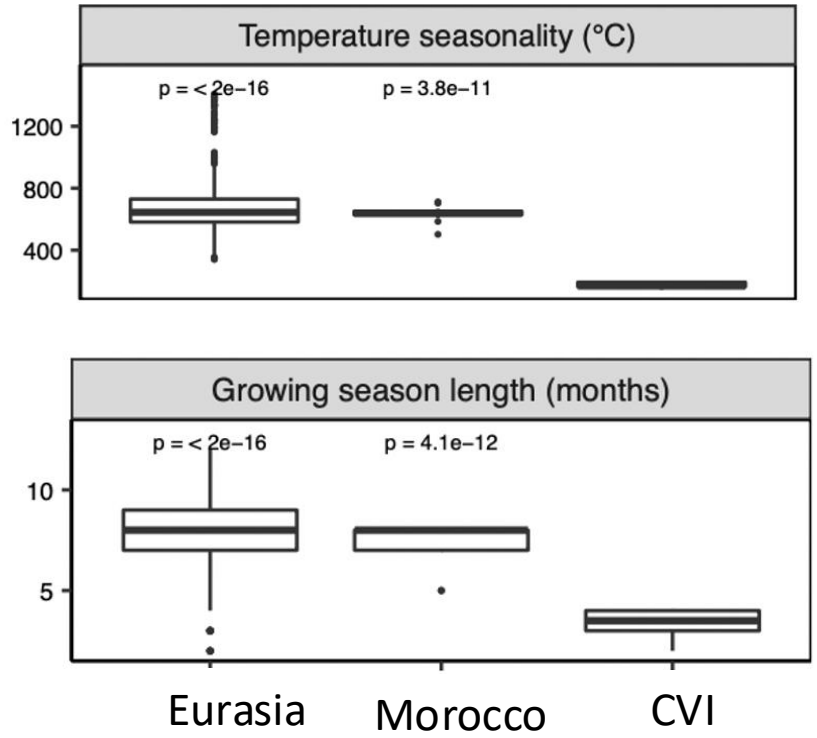
Vernalization control of flowering





Vernalization control of flowering

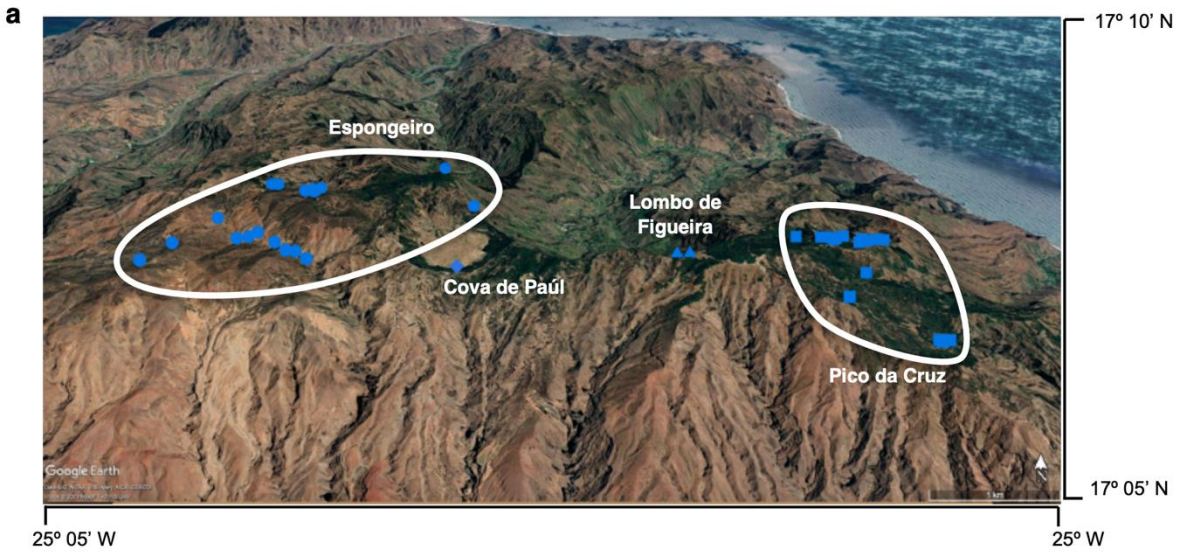


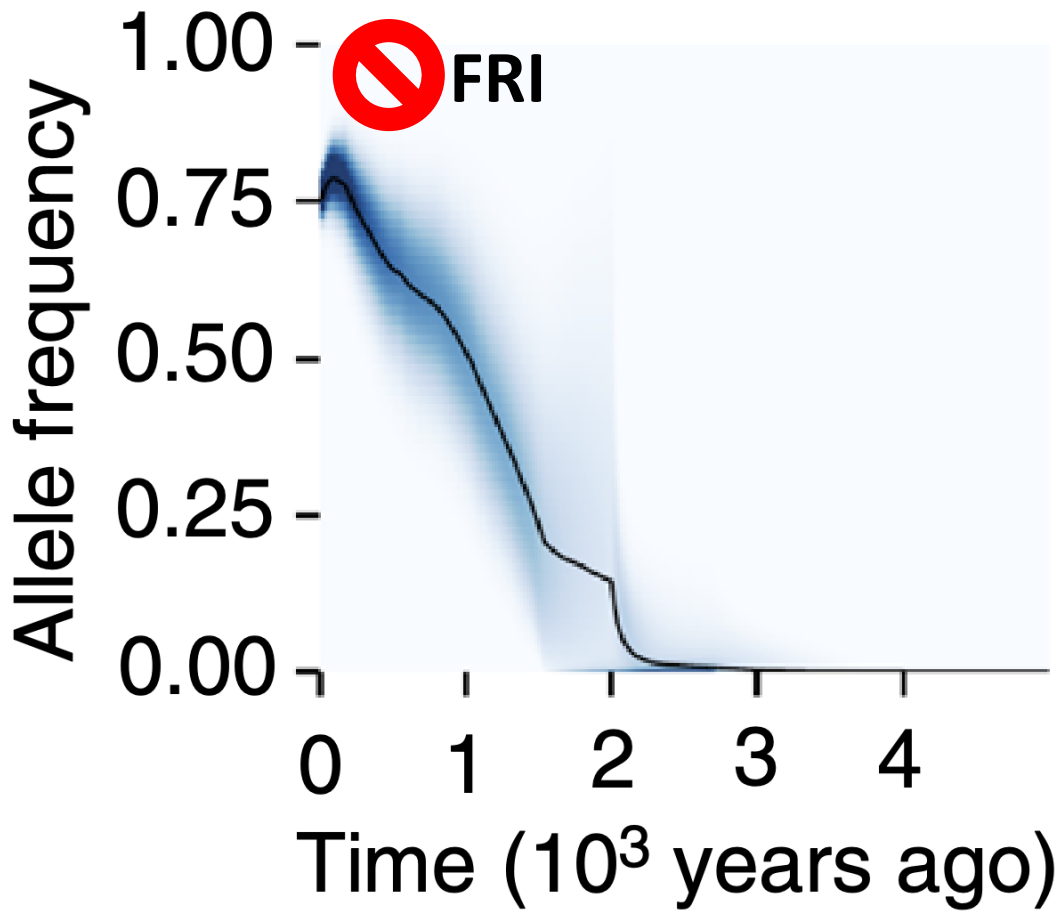


Why might having a functional FRI be deleterious (reduce fitness) in Cape Verde?

FRI loss-of-function allele has been rising in frequency since it arose 2.5K years ago

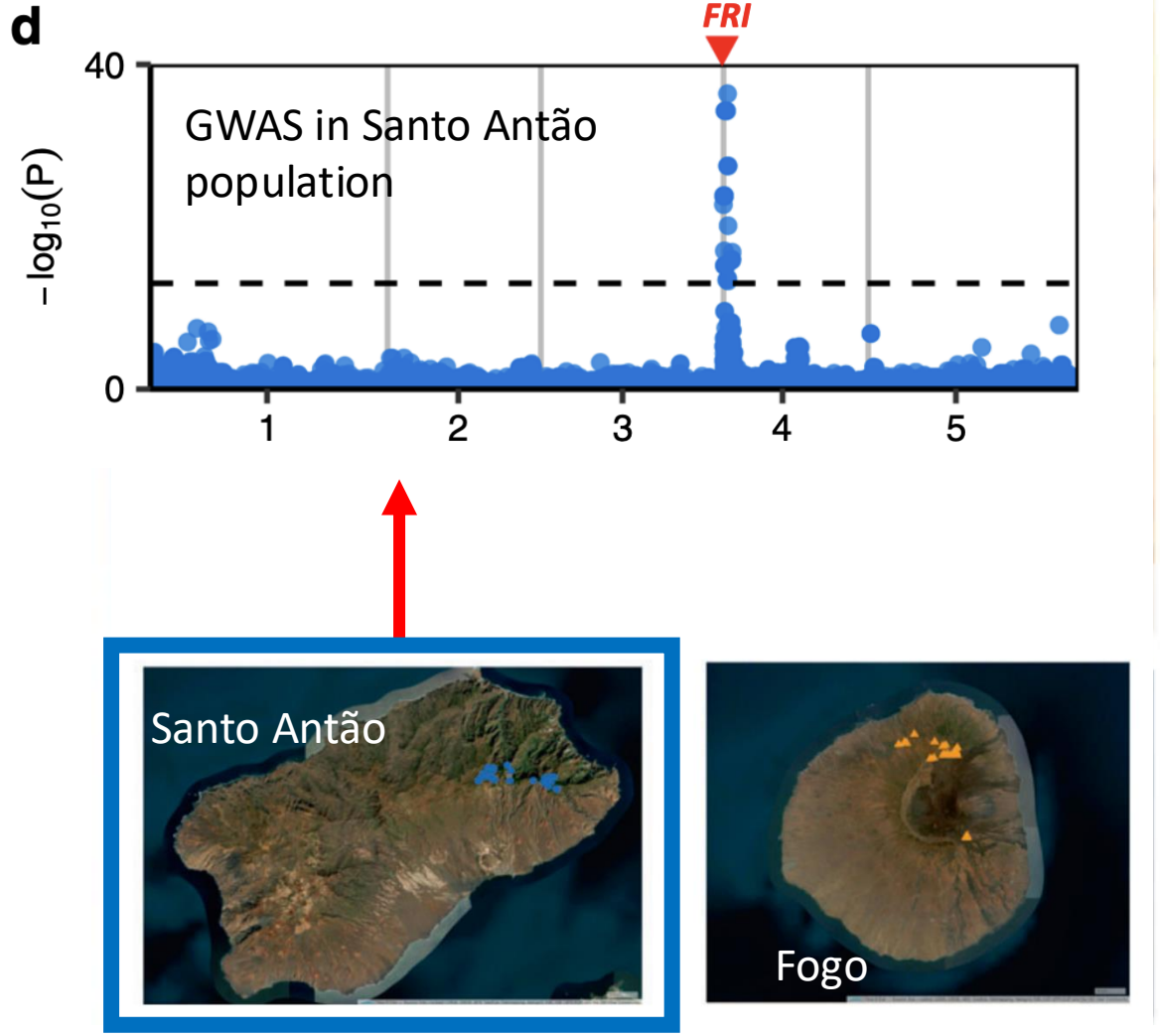


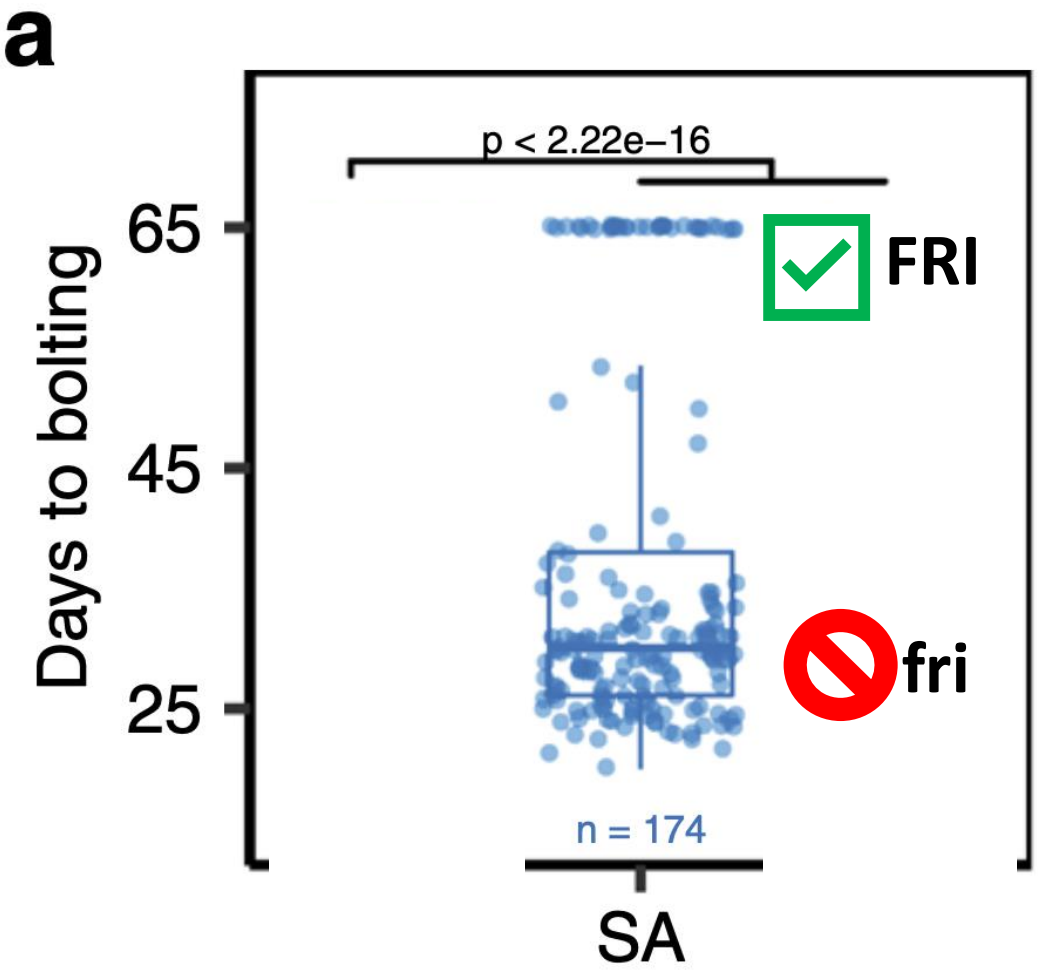




Santo Antão population is segregating for FRI stop codon mutation

FRI **Premature stop mutation** in FRI gene that arose about 2.5 thousand years ago

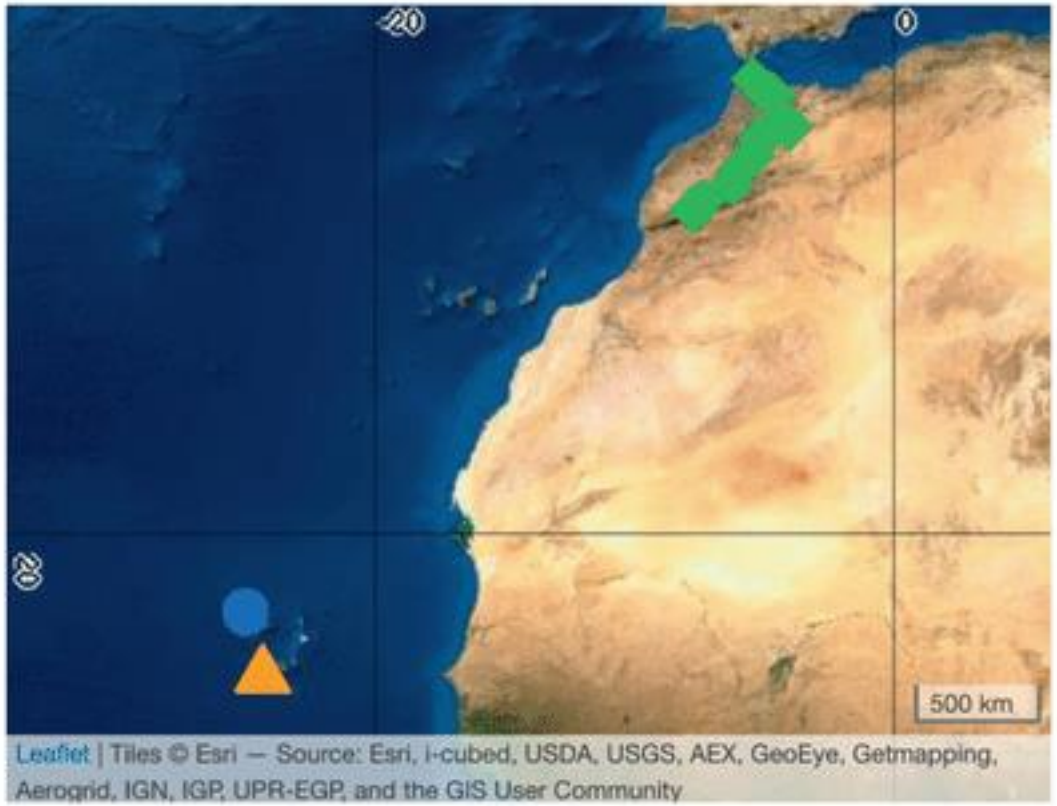




Fulgioni et al 2022

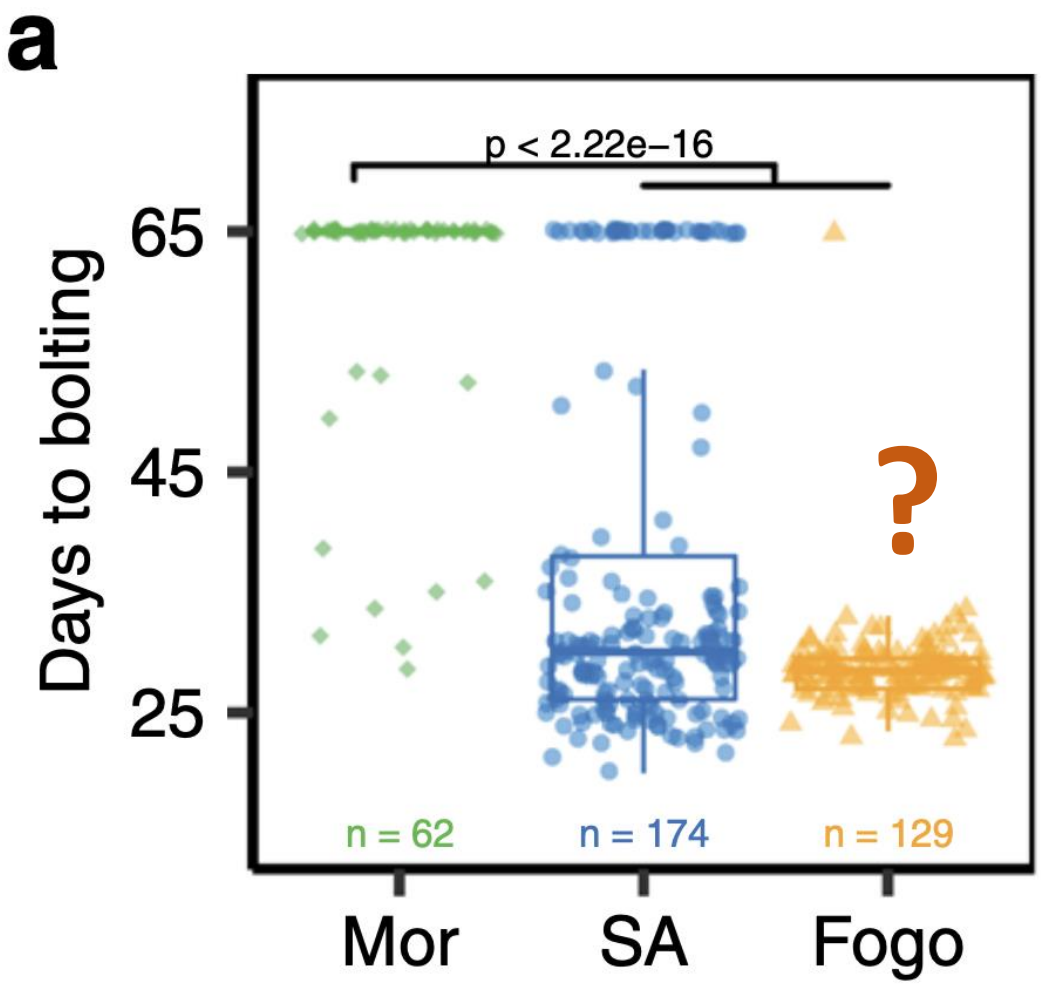
Variation in flowering time (days to bolting)

a





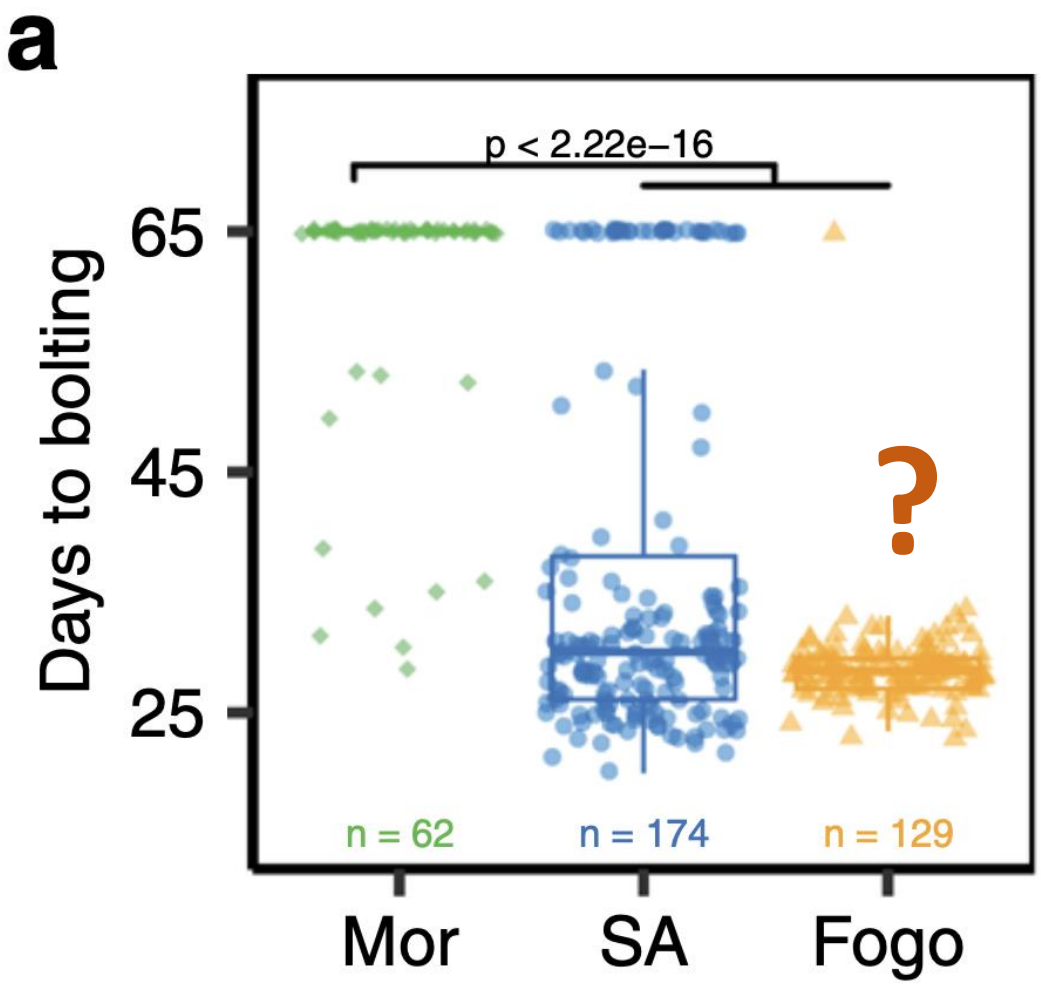




Fulgioni et al 2022

No obvious flowering time allele in Fogo Population

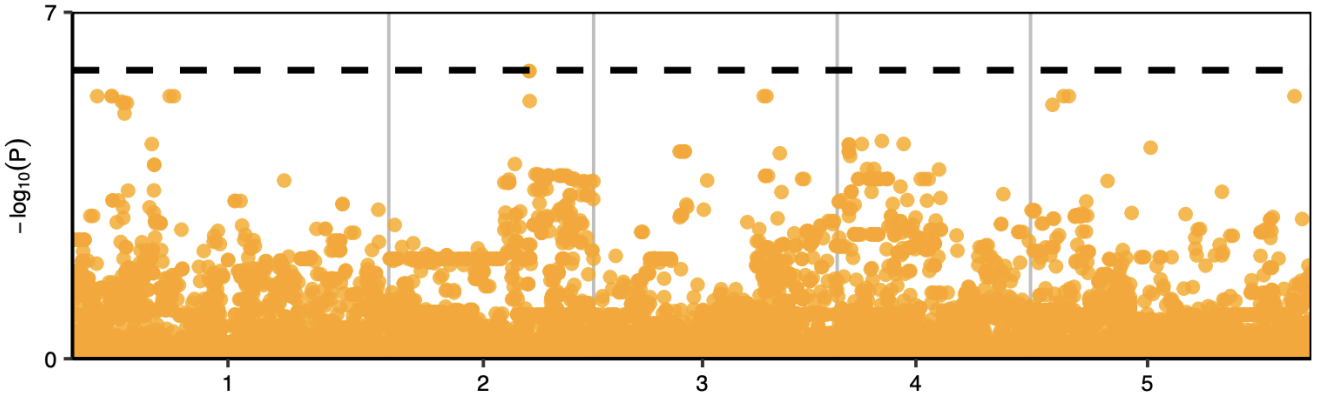




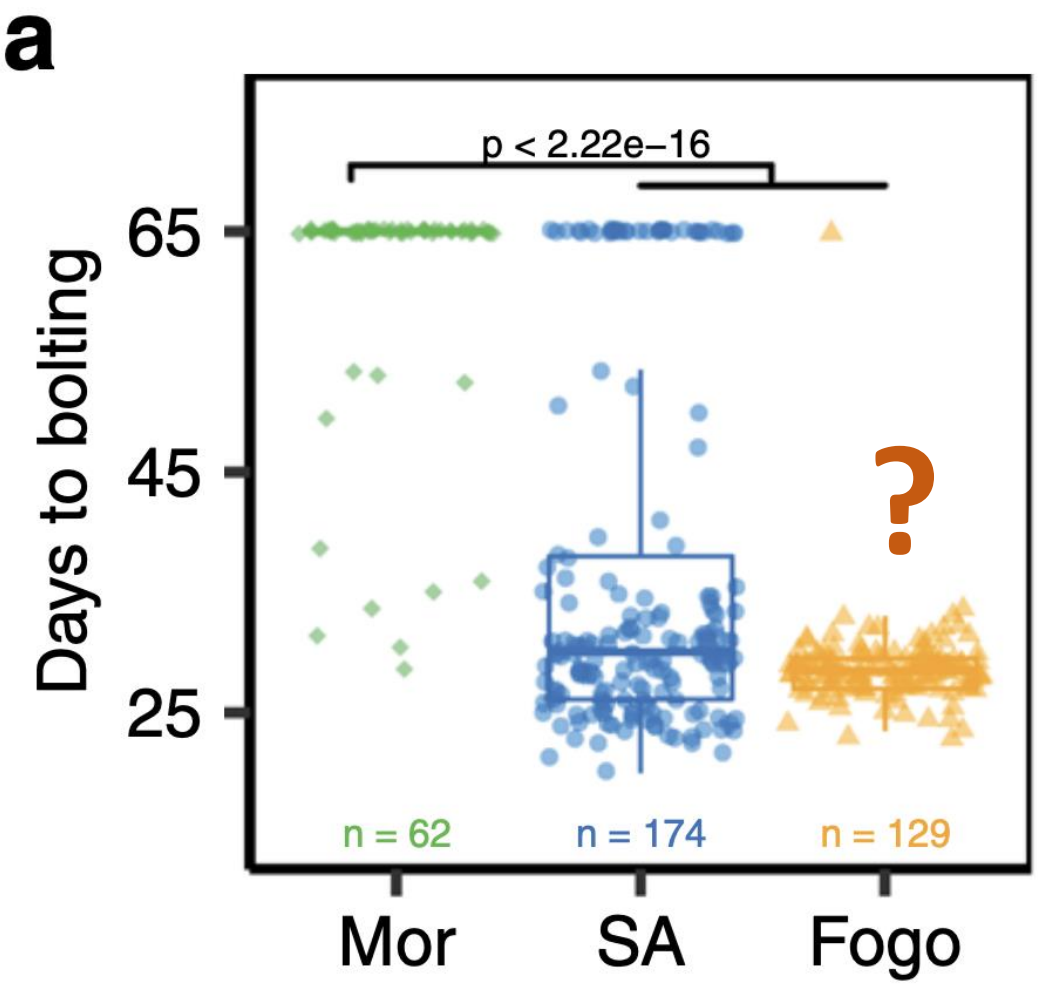
Fulgioni et al 2022

No obvious flowering time allele in Fogo Population

No significant hit ☹







Fulgioni et al 2022

Bulk segregant analysis: another way to ”map” genes

Early flowering parent

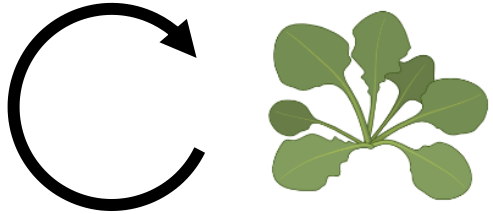
Late flowering parent



X

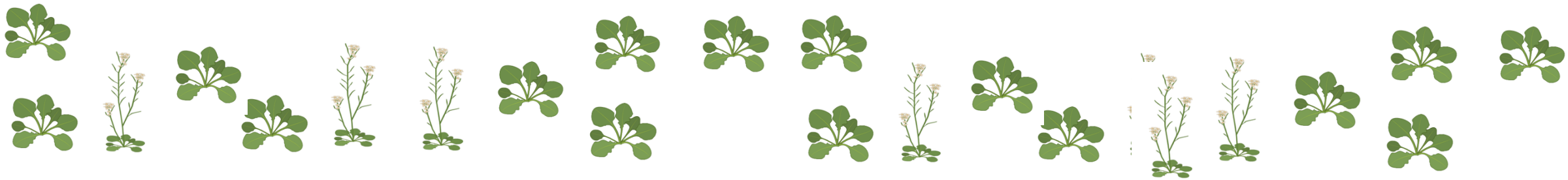


Sequence genomes of both parents



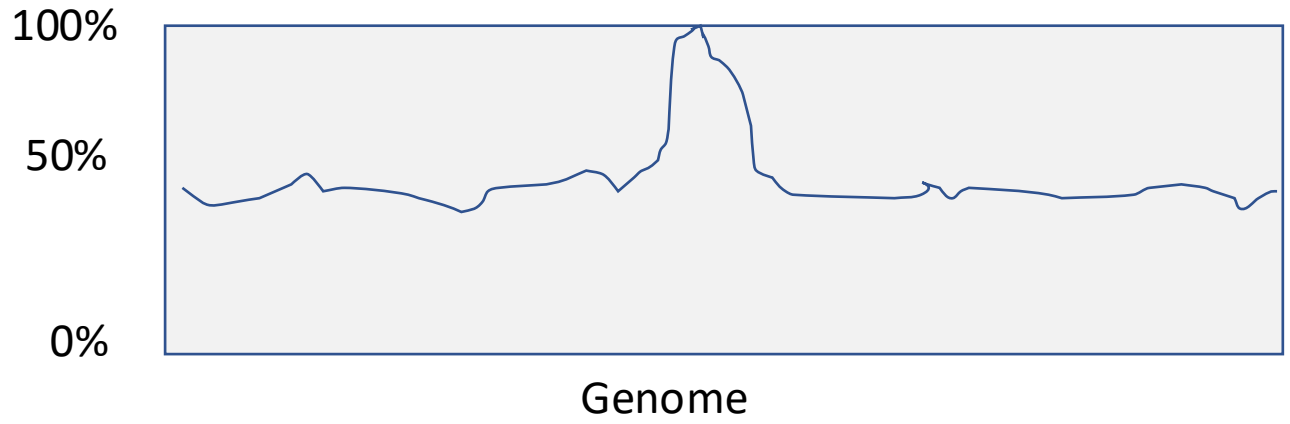
F1

F2



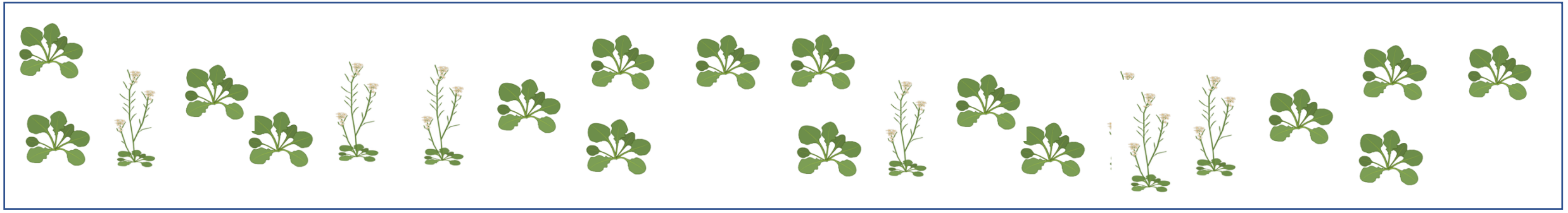
Bulk segregant analysis: another way to ”map” genes

Causal allele has to be at 100% frequency because it must be homozygous



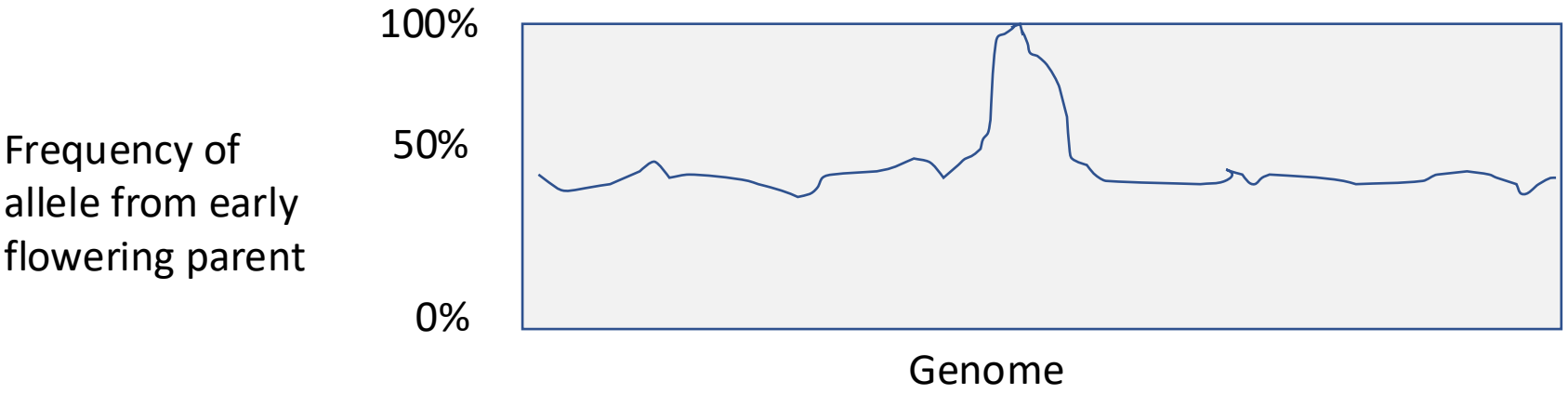
Pool all individuals with <u>one</u> of the phenotypes and sequence the genomes of all individuals

Which phenotype do you pick?

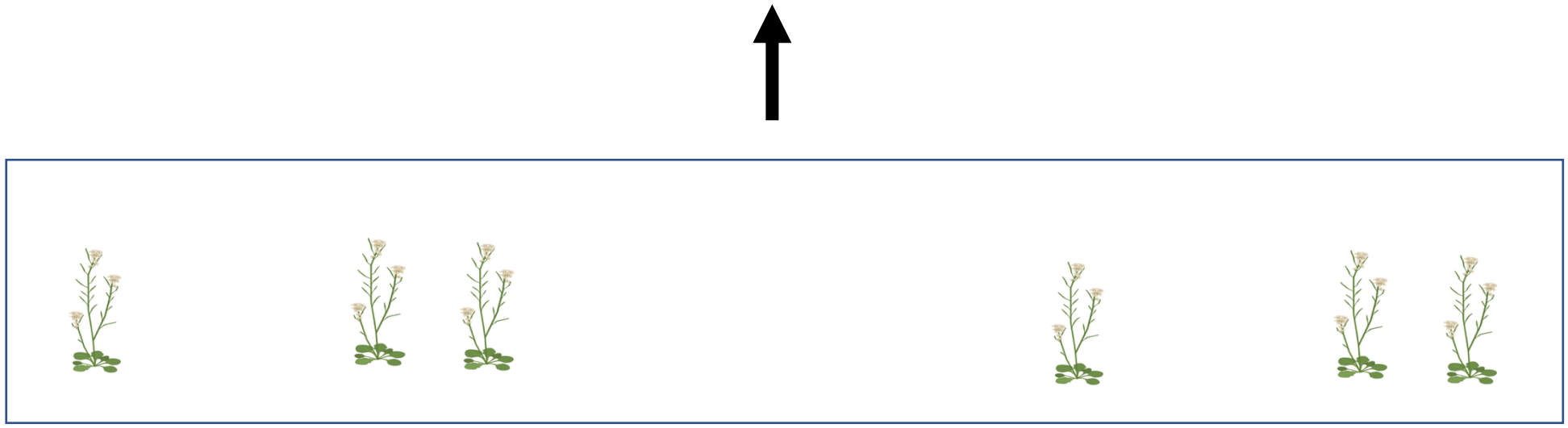


Bulk segregant analysis: another way to “map” genes

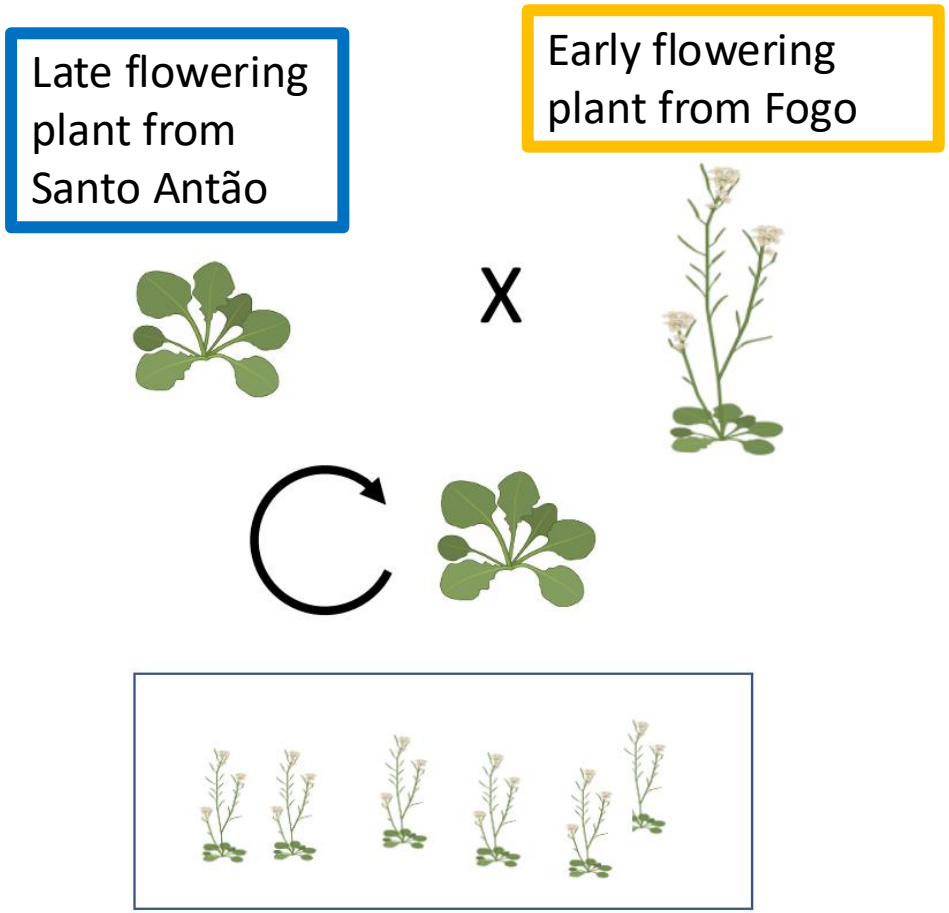
Causal allele has to be at 100% frequency because it must be homozygous (since it is recessive)



Pool all individuals and sequence the genomes of all individuals



Fogo population is “fixed” for a FLC stop codon mutation



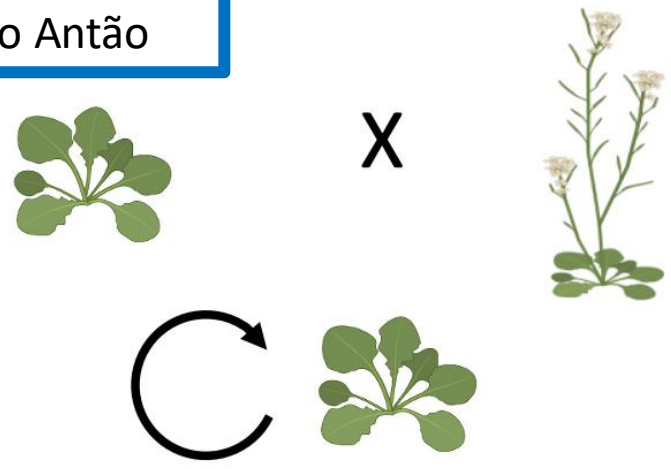
Sequence all individuals with recessive phenotype

Fulgioni et al 2022

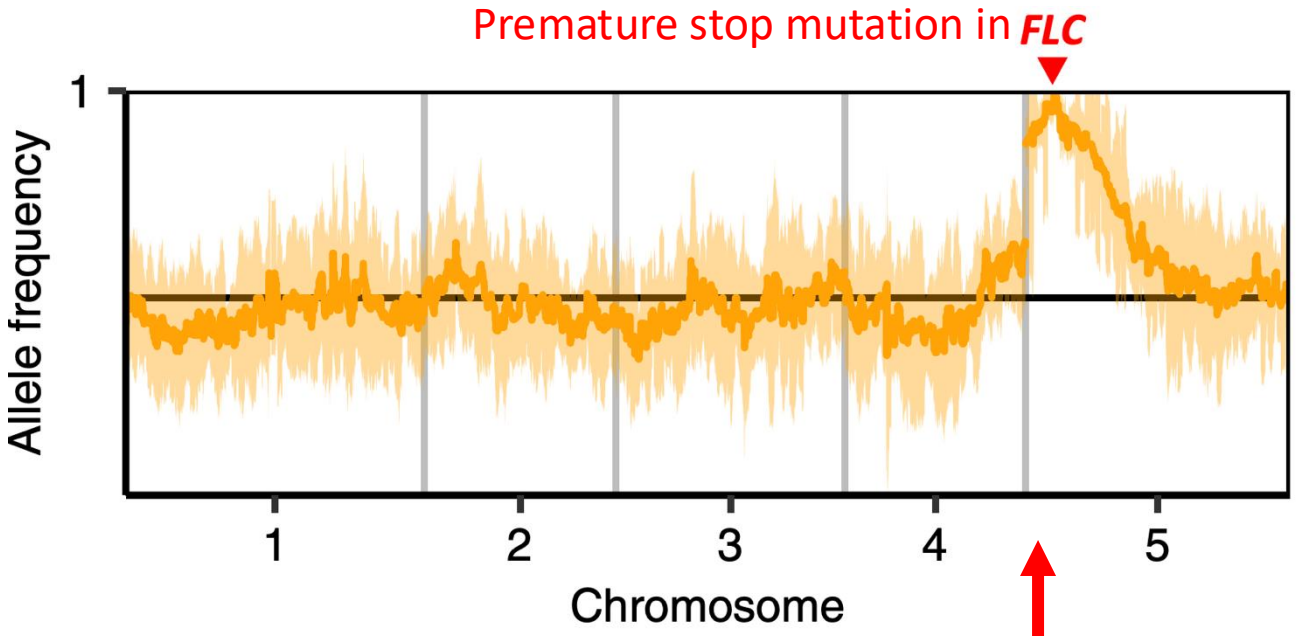
Fogo population is “fixed” for a FLC stop codon mutation

Late flowering plant from Santo Antão

Early flowering plant from Fogo



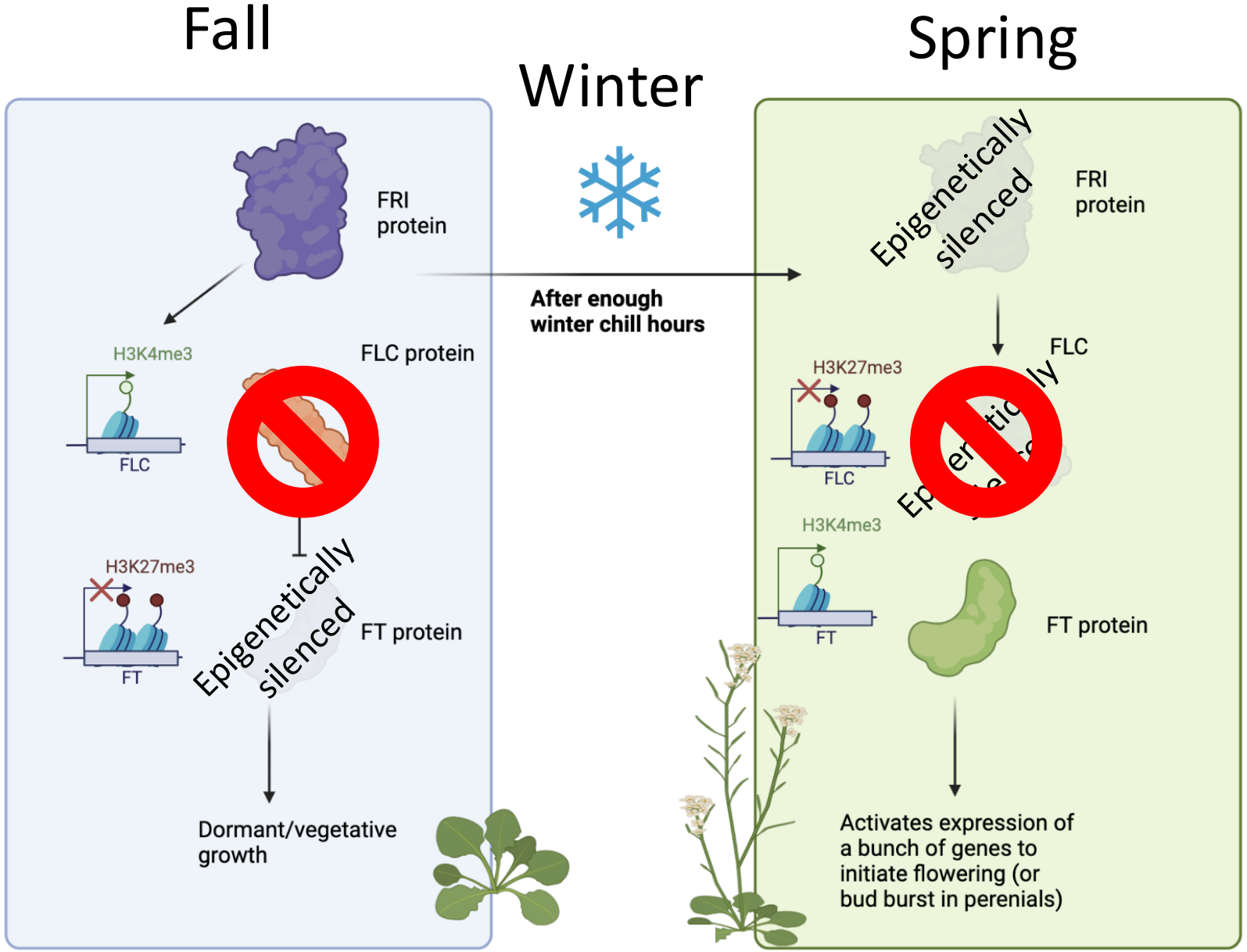
Sequence all individuals with recessive phenotype





Fulgioni et al 2022

Vernalization control of flowering



Loss of function mutations in FLC



Hypothesis: FLC (rather than FRI) knockout in Fogo causes early flowering
Predict the result of these experiments:

Cross 1

fri knockout line
***FLC* functional**

fri/fri
FLC/FLC



X

Early flowering plant from Fogo



Hypothesis

FRI/FRI
flc/flc

F1 phenotype?

late  Or  early

Cross 2

***FRI* functional**
flc knockout line

FRI/FRI
flc/flc



X

Early flowering plant from Fogo



Hypothesis

FRI/FRI
flc/flc

F1 phenotype?

late  Or  early

Live poll

Multiple choice

Hypothesis: a broken FLC (not FRI) makes Fogo plants flower early. Predict the F1: Cross 1 = fri knockout x Fogo plant. Cross 2 = flc knockout x Fogo plant.

Scan with your phone

[Join on your phone or laptop: greymonroe.github.io/live-poll-test/submit.html?poll=pls152-ftg-cross](greymonroe.github.io/live-poll-test/submit.html?poll=pls152-ftg-cross)

[host screen](greymonroe.github.io/live-poll-test/host.html?poll=pls152-ftg-cross)

“Complementation test” confirmed *flc* is causal

fri knockout line
***FLC* functional**

fri/fri
FLC/FLC



X

Early flowering
plant from Fogo



FRI/FRI
flc/flc

***FRI* functional**
flc knockout line

FRI/FRI
flc/flc



X

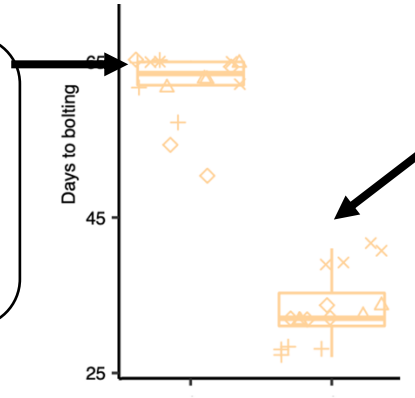
Early flowering
plant from Fogo



FRI/FRI
flc/flc



fri/FRI
FLC/flc

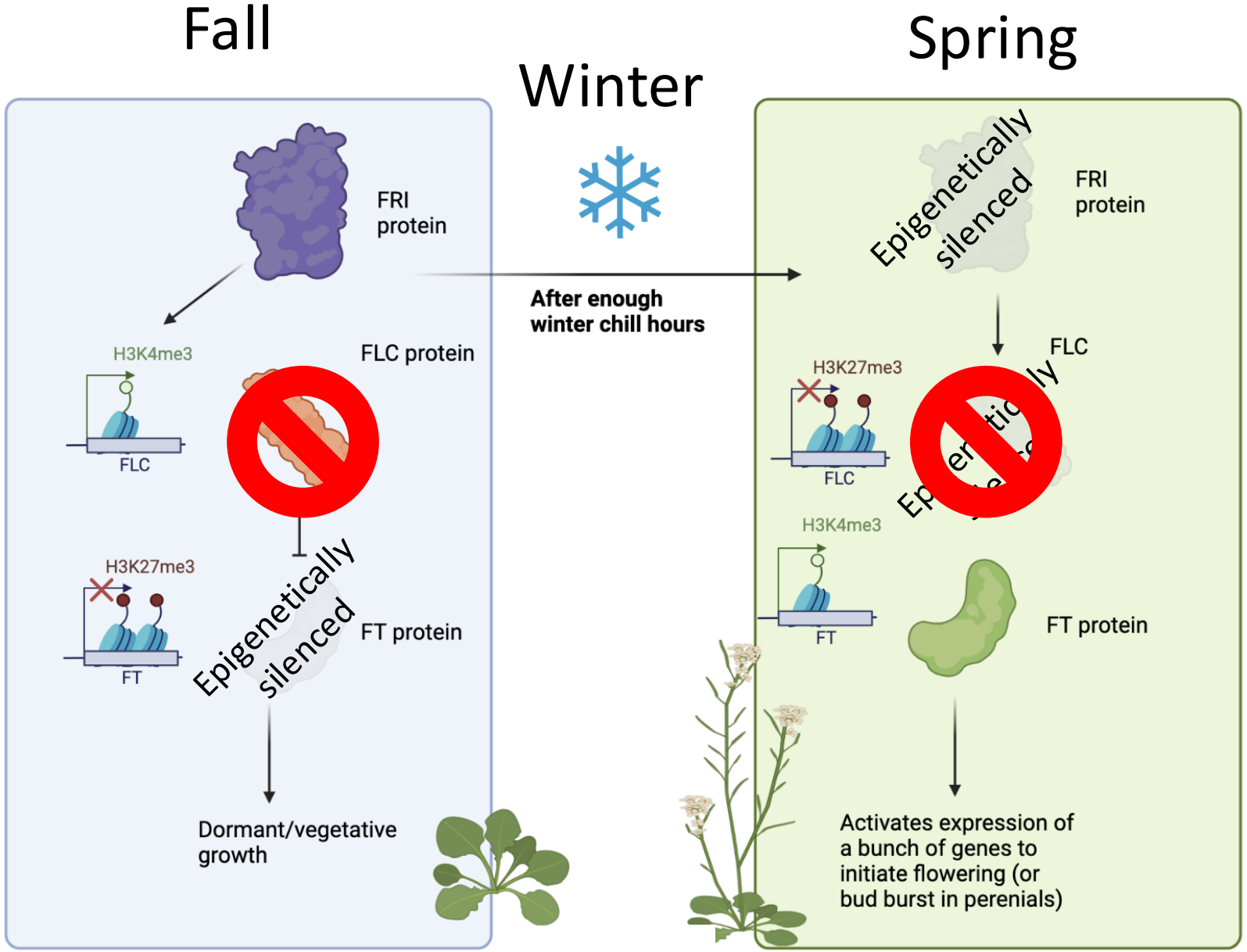


FRI/FRI
flc/flc



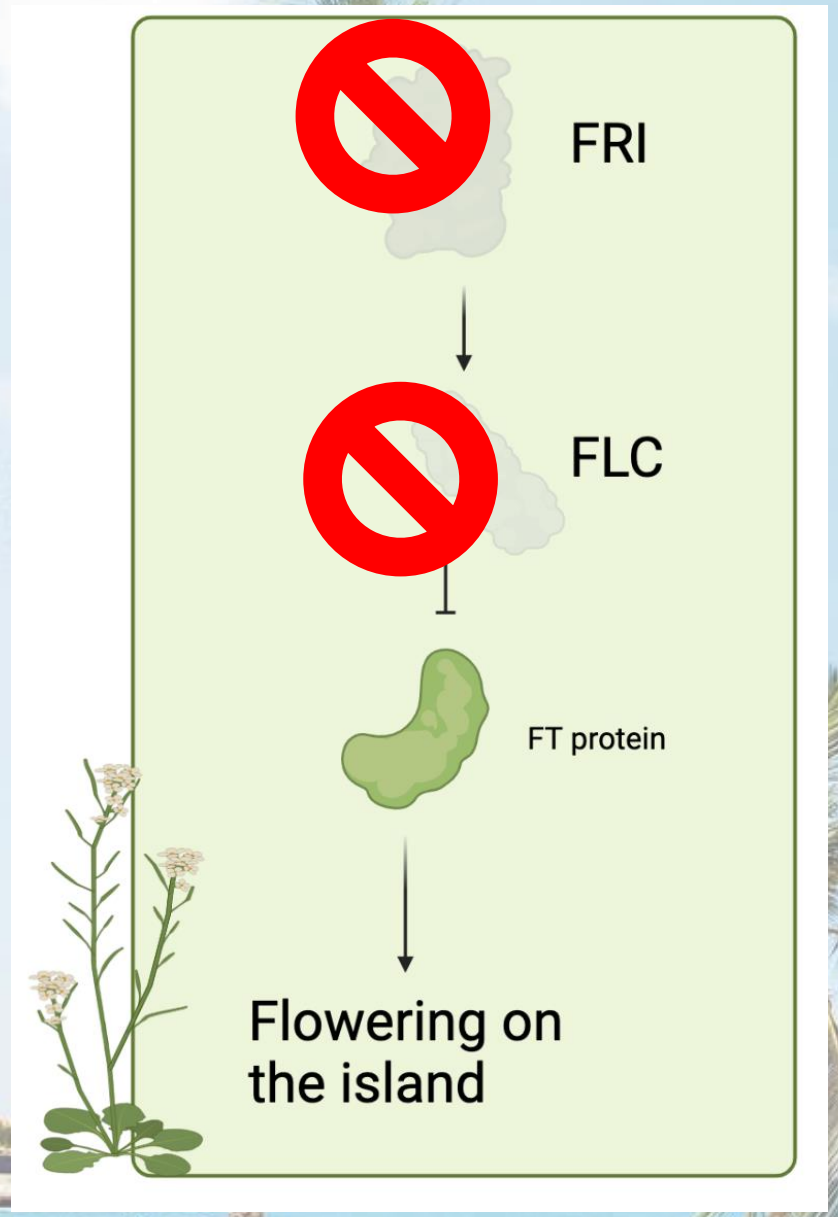
Fulgioni et al 2022

Vernalization control of flowering



Loss of function mutations in FLC

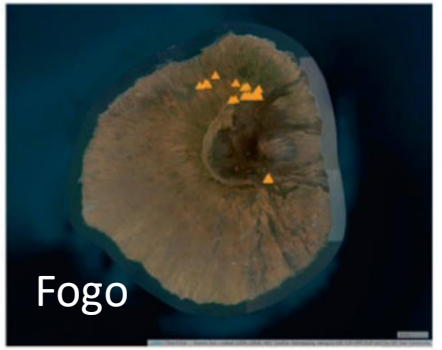


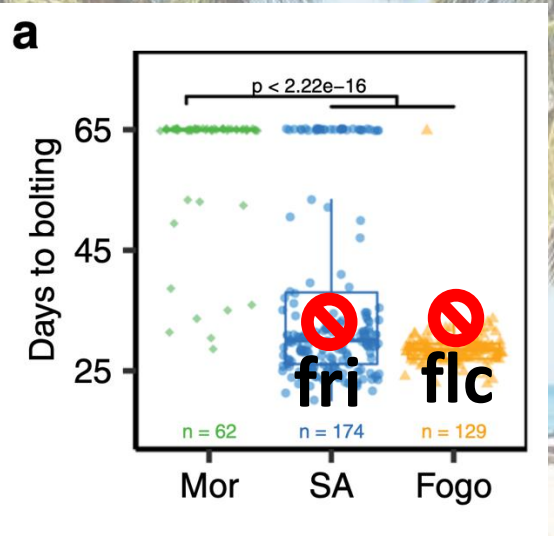


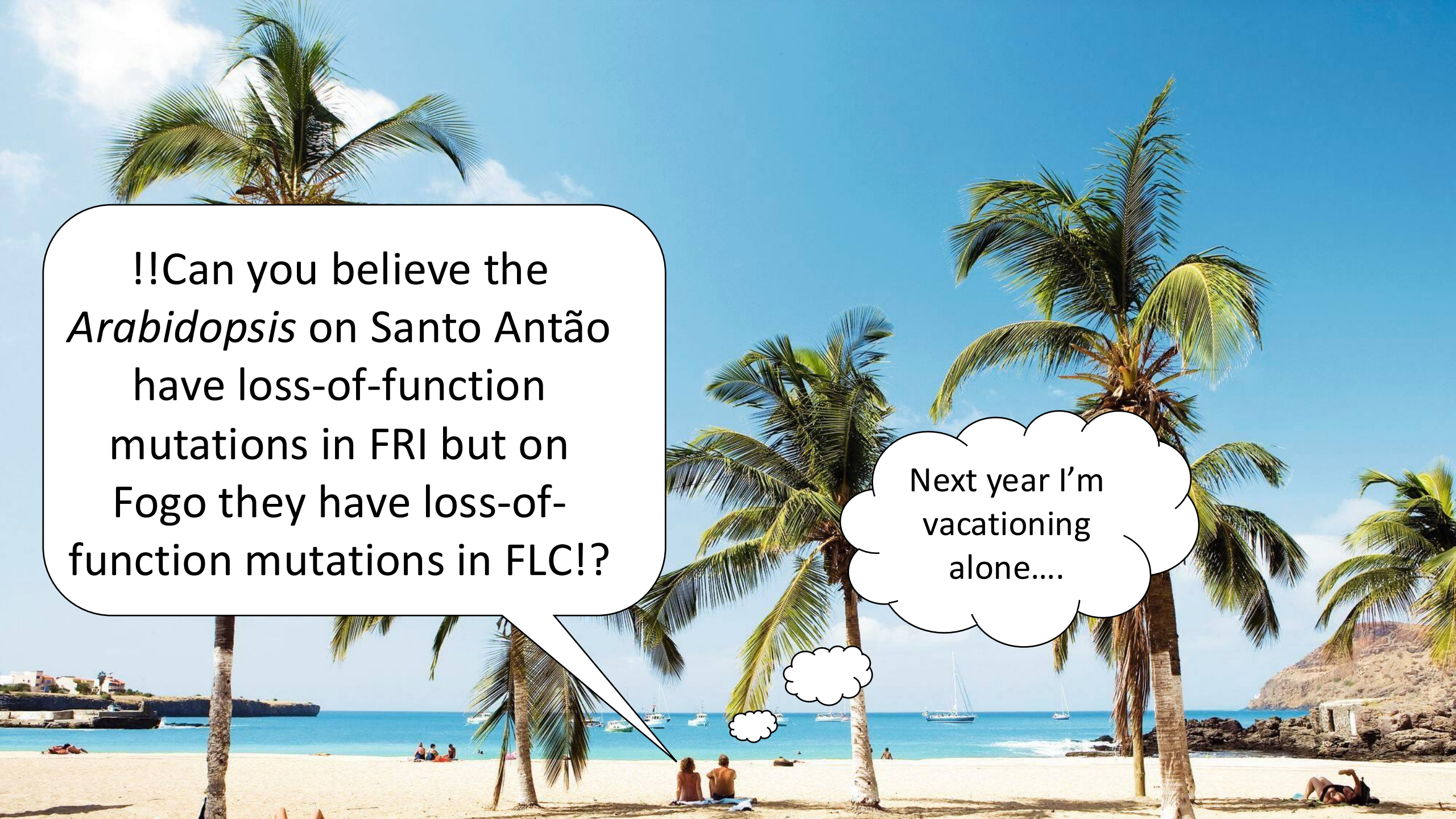
Loss of function mutations in FRI



Loss of function mutations in FLC






!!Can you believe the *Arabidopsis* on Santo Antão have loss-of-function mutations in FRI but on Fogo they have loss-of-function mutations in FLC!?
Next year I'm vacationing alone….

Take-home: why causal genes hide

Genetic heterogeneity	different genes, same trait
Allelic heterogeneity	many rare breaks in one gene: test 'any broken copy'
Population structure	ignore it: false positives · correct it: false negatives
Cape Verde	two islands, two broken genes (FRI, FLC), both flower early
Proof	complementation, not a peak